

## Highlights

### **Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction**

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- Scoring the same model at the fire, not region-wide, costs 0.143 AUC
- The cause is the negative pool, not prevalence: +0.155 against -0.000
- Under blocked CV, thermal predictors add +0.06 to +0.15 AUC in five regions
- Equalising the evaluation frame lifts mean transfer from 0.541 to 0.616
- None of 20 transferability diagnostics is shown to order transfer skill

# Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

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## Abstract

Pre-fire thermal dryness separates a fire year from a normal one, but such models are rarely tested outside the region where they were fitted. Six thermal predictors were added to a terrain, fuel and greenness baseline in five Mediterranean regions, with MCD64A1 labels and spatially blocked validation. The first result is **evaluation geometry**: with model, predictors and fitting held fixed, scoring on the burn scar and its 2 km collar rather than region-wide costs **0.143 ROC-AUC** [**+0.077**, **+0.208**]. A control isolates the cause as the negative pool's composition, not class balance. Applied between regions the same effect withdrew five of our own claims: the study areas enclose very unequal far fields, and equalising them to a 10 km collar, which drops no burned cells, lifts mean transfer from 0.541 to 0.616 and leaves no between-region sign reversal supported under our own criterion, though the LST anomaly still differs on a difference interval. Local skill nonetheless does not travel. The thermal block adds +0.045 to +0.148 within regions at 5 km blocking, but +0.022 [−0.032, +0.077] when a whole burn scar is withheld and +0.004 [−0.028, +0.036] across twenty transfer directions, with a sign

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that varies by pair. The static baseline transfers no better, at 0.537 against 0.541, and equalised transfer sits 0.155 below a matched within-region reference. Of twenty candidate diagnostics from five families, none was shown to order transfer, and the two that appeared to need target labels. Transfer skill has to be measured, not inferred from similarity.

*Keywords:* wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

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## 1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns [29]. The dominant pattern in fire susceptibility mapping [42, 14] is settled: geospatial predictors  
5 are assembled over a study region, paired with a historical burned-area record and fitted with a supervised classifier, most often a random forest [5], and the surface is published with a cross-validated AUC of 0.85 to 0.95. **That pattern does not report portability**, and this paper measures it.

### 1.1. Portability goes unmeasured

10 A susceptibility model’s reported skill is almost always a *within-region* estimate: held-out folds come from the same study area and season, often the same fire, and random folds over cells let spatial autocorrelation inflate it further. The problem is documented across ecological modelling [33, 31] and addressed by spatially blocked cross-validation [39, 25], but blocking does  
15 not make the estimate honest about a fire the model has not seen.

Because only that side of the ledger is reported, portability is never entered: a predictor block is adopted on the increment it delivers inside its

training footprint, and whether that survives a change of region is not asked, even though any regional product built from locally trained models implicitly promises generalisation beyond it. Meteorological fire-danger indices are known not to port cleanly in a Peruvian case study [32], and the two studies that test transfer systematically [8, 16] both report that it largely succeeds between similar regions — and both transfer models whose dominant predictors are *spatially stationary*. Whether a dynamic, season-specific class behaves the same way is this paper’s question.

### 1.2. Pre-fire thermal dryness is the natural test case

Pre-fire thermal dryness is the dynamic class most plausibly *expected* to transfer. Standard predictors are static or near-static over the timescale at which fire danger varies, so they explain poorly why one summer burned and the preceding one did not; what changes is the state of the surface, to which satellite thermal observation gives partial access (Section 2.2). The physics linking moisture stress to combustion is universal, so portability should be most expected here, which makes the class diagnostic: a loss cannot be dismissed as a peculiarity of a locally defined covariate. The expectation is sharpest for the internally normalised channels, which should be least exposed to absolute-temperature offsets between regions; **they transfer no better than the absolute ones** (Appendix A(f)). The expectation motivates the design and is not a finding of it — Section 4.4 reports the associations running the other way.

40 1.3. *Why the loss is invisible to the diagnostics in use*

The field’s instrument for anticipating transfer failure is predictor-space dissimilarity — the area of applicability and its relatives [23, 24, 18] — which asks whether a target’s predictor values fall inside the training envelope. That is a question about the marginal distribution of the predictors, and it cannot  
45 see a change in the *relationship* between predictors and response, because detecting one requires the labels being withheld. If the failure is conditional rather than marginal, those diagnostics look in the wrong place; Section 4.6 tests whether they do.

1.4. *Contributions*

50 Three findings carry this paper, stated here as claims and established in Section 4, which carries every interval.

**Contribution 1. Where a model is scored decides what it appears to know, and the effect is large enough to dissolve findings of our own.** That evaluation extent inflates AUC is established in species  
55 distribution modelling (Section 2.4); what is new is a magnitude under a controlled design. Holding model, predictors and fitting fixed and changing only which cells are scored costs **0.143 ROC-AUC**, the size of the increments this design measures for a predictor block, with the cause measured as the negative pool’s composition rather than class balance (Section 4.3). Ap-  
60 plied between regions it withdraws five claims of our own, including the sign reversal we had offered as the mechanism and the one arm that held place fixed (Section 4.4), and the practical consequence is a reporting standard (Section 5.7).

**Contribution 2. Local skill does not travel, and correcting the**  
65 **frame does not rescue it.** The thermal block is worth a substantial within-region increment in all five regions, every bootstrap interval above zero, and stays positive when the frame is equalised — but much of it is a property of interleaved holdout, and across twenty transfer directions the paired contribution spans zero on both frames under the primary resampling unit, 70 though much closer to the boundary once equalised and not under every admissible unit (Section 4.4), with a sign belonging to the pair rather than the block. Two controls bound this: the static baseline transfers no better on either frame, so the failure is not the thermal block’s peculiarity, and on matched frames and blocking equalised transfer still falls **0.155** short of the 75 within-region reference. The within-region half is not novel [1, 13]; the paired contrast against portability is.

**Contribution 3. The shortfall cannot be anticipated by any diagnostic we could run.** Of twenty candidates from five families, eighteen were not shown to order the matrix on the frames as drawn. Two conditional 80 variants did, but both need burned labels on both sides, both rest on a data-selected feature subset whose all-nine counterparts span zero, and **Contribution 1 removes even those** (Section 4.4). The point survives without any ranking: at the point estimates the pair with the highest burned-niche overlap is among the weakest in the matrix while the lowest transfers above 85 chance in both directions, contradicting an expectation the fire literature carries [8, 16] and agreeing with species distribution modelling [41, 34].

Two consequences follow, as supporting results: label-free alignment by standardisation and covariance alignment [37] does not repair transfer, in

what we believe is its first application to fire susceptibility; and because the  
90 residual is conditional, the resource that closes it is target labels, priced in  
Appendix A(u).

The leakage-audited, spatially blocked protocol is released with code,  
configuration and frozen outputs, so most of this can be re-run rather than  
taken on trust, the three exceptions being named in the declarations —  
95 which matters given evidence that wildfire transfer conclusions are sensitive  
to evaluation design [45]. A companion paper treats the observational layer  
beneath this one.

## 2. Related work

### *2.1. Fire susceptibility mapping, and what it reports*

100 The dominant pattern is described in Section 1; the figure it publishes, a  
cross-validated AUC of 0.85 to 0.95, is a within-region estimate. Comparable  
within-region results already exist for the landscapes studied here [1, 13].

### *2.2. Pre-fire thermal dryness*

Satellite thermal observation gives repeated access to surface state through  
105 fuel moisture content [47]. Chuvieco et al. [7] established the pairing of land  
surface temperature with a vegetation index as a live-fuel-moisture estima-  
tor for fire-danger rating, and the Temperature-Vegetation Dryness Index [35]  
formalises that feature space into an internally normalised measure, which is  
why it is often expected to travel better than raw temperature. That pre-  
110 fire thermal state carries information about subsequent fire is established  
[19, 21, 20], and Gelabert et al. [9] found dead fine fuel moisture and its

anomalies the most influential predictor of human-caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site evaluation. What has not been tested is whether such a classifier survives strict,  
115 label-free application to an unseen region.

### *2.3. Spatial validation and transferability*

Random cross-validation over spatially autocorrelated cells inflates skill estimates [33, 31], and spatially blocked designs are the standard remedy [39, 25], though they have themselves been argued to introduce pessimistic  
120 bias [43, 26, 6]. This paper blocks throughout and reports every result at three block sizes, so a reader can see which verdicts depend on the choice.

Blocking corrects the inflation random folds produce within a region. It does not speak to performance on a fire the model has not seen, which Section 4.3 measures on identical cells. For that, the field’s instrument is  
125 the area of applicability and related predictor-space dissimilarity measures [23, 24, 18], which ask whether a target’s predictor values fall inside the training data’s envelope. Species distribution modelling has examined transferability far more systematically than fire has [46], and two studies there report that geographic and environmental similarity do not reliably predict  
130 transfer success [41, 34]; transferability of satellite-derived empirical models is an active question in fire remote sensing itself [22].

The canonical taxonomy separates covariate shift, where the predictor distribution moves but the predictor-response relationship holds, from concept shift, where the relationship itself changes [27]. Covariate shift is in principle  
135 correctable without target labels, by per-region standardisation or covariance alignment such as CORAL [37], and domain adaptation has an established

remote-sensing literature [38, 30]. Concept shift is not: no realignment of inputs can repair a reversal in the sign of an association, because detecting one requires the labels being withheld. Shift decomposition in applied remote sensing is not itself new [12], but we are aware of no prior application of covariance alignment to fire susceptibility, fire occurrence or burned-area prediction.

#### 2.4. Cross-region generalisation of fire models

Few studies test fire-model transfer directly. Podschwit et al. [32] report, in a Peruvian case study, that meteorologically derived danger indices do not port cleanly between fire environments. WildfireGenome [16] runs a leave-one-county-out matrix across seven US counties and reports strong within-county performance with highly variable off-diagonal transfer, similar pairs transferring well and dissimilar pairs collapsing; its label is a principal-component composite of hazard *indicators* rather than observed burned area, and it applies no adaptation. Xu et al. [45] argue that wildfire transfer conclusions depend strongly on evaluation design, a caution we address by fixing the protocol in a project log before the diagnostics were computed and by reporting every sensitivity axis; that log is not a formal pre-registration, and Section 3.11 states what was fixed and when. Kondylatos et al. [15] provide Mesogeos, a 1 km Mediterranean datacube.

**Evaluation extent and AUC.** Species distribution modelling settled long ago that the area a model is evaluated over is not a neutral choice. Lobo et al. [17] make it the fifth and, by their own ranking, most important reason to distrust AUC comparatively: the extent of the modelled area governs how many easy absences enter the calculation, and therefore the score. VanDer-

Wal et al. [40] show the same lever on the calibration side, and Barve et al. [3] give the argument its general form as the accessible area. That literature is qualitative about magnitude, because magnitude is problem-specific, so  
 165 our Contribution 1 is a measurement inside that framework rather than a new phenomenon. The wildfire literature has largely not imported the lesson: region-wide figures are reported as though they described performance at the fire, and we know of no wildfire study that holds the model fixed and varies only the evaluation cells — which is why Section 5.7 treats the 0.143  
 170 as a reporting problem rather than a caveat.

**The nearest neighbour, and the contrast this paper draws.** Di-marco et al. [8] is the closest Mediterranean analogue: 500 m predictors harmonised across four countries, tree ensembles under spatial cross-validation, transfer tested both leave-one-country-out and as a full  $4 \times 4$  matrix, ev-  
 175 ery transfer exceeding AUC 0.80 and bioclimatically similar countries scoring higher, with no domain adaptation.<sup>1</sup> Much of the design is shared — Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation, an explicit transfer matrix — and two things differ. **The predictor class:** their model rests on attributes of a place, all  
 180 spatially stationary, ours on the state of a surface in one season. **The response variable:** theirs is human-driven ignition, ours burned area, which have different dominant controls. Read together the two results suggest the relationship between domain similarity and transfer success may be predictor-class dependent, with Vesk et al. [41] and Rousseau and Betts [34] aligning

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<sup>1</sup>Their Results text refers to "LST anomalies", inconsistent with their own Methods,

185 from species distribution modelling while Dimarco et al. and WildfireGenome  
run against it. We present that as a live disagreement; Section 5.6 states why  
our data do not settle it.

where no land surface temperature or TVDI variable appears and the only  
temperature predictor is a static ERA5-Land seasonal climatology [28]. We  
190 follow their Methods and note the discrepancy so the comparison is trans-  
parent.

### 3. Data and methods

#### 3.1. Study regions and temporal windows

Five Mediterranean wildfire regions are analysed (Fig. 1): Manavgat  
195 2021 and Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in  
Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangle in  
EPSG:4326, defined from place coverage rather than from a fire perimeter  
and deliberately not clipped to it, so that unburned cells around each fire  
form the negative class; none was tuned on burned prevalence, gate outcome  
200 or any model metric. **One choice was label-informed and is stated as  
such:** the North Evia box was extended after the legacy box proved atyp-  
ically high in burned prevalence, the extended geometry then defined from  
place anchors and the legacy variant kept as a sensitivity arm. Section 4.4  
shows this framing decision is consequential and Appendix C.5(x) treats it  
205 as the design lesson of the paper. A sixth region, Kozan 2023, is carried as  
a negative control and excluded by the gate of Section 3.3.

Each region has two non-overlapping windows. The **predictor window**  
closes the day before the **label window** opens, so no predictor observation

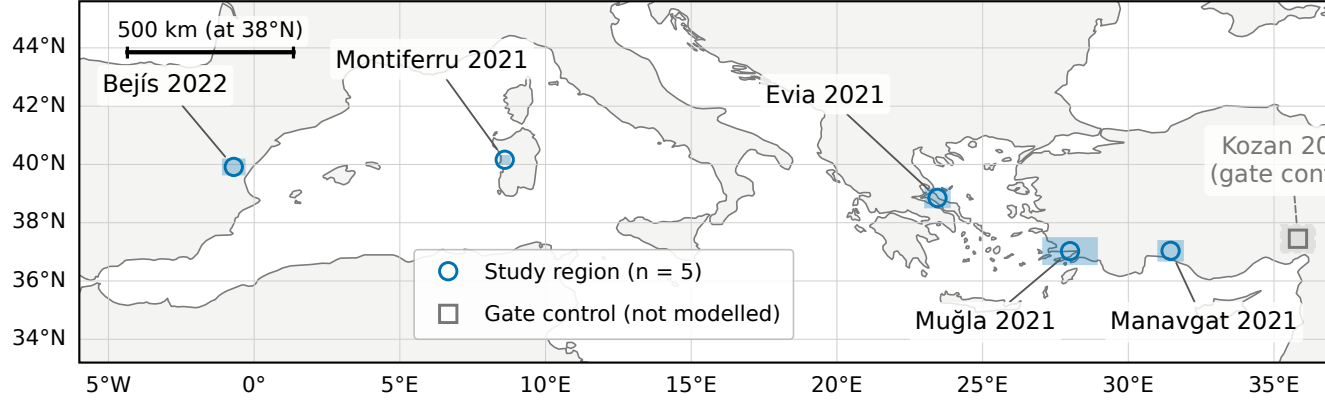


Figure 1: **Study regions and the gate negative control.** Locations of the five wildfire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table C1 to  $5 \times 10^{-3}$  degrees.

can post-date the first labelled burning. Lengths vary with the event: 57 to  
210 61 days for predictors, 35 to 59 for labels, counted inclusively. A four-year  
baseline of window-symmetric composites precedes each predictor window  
and supplies the climatological reference for the anomaly channels. Region  
identifiers, bounding boxes, window dates and baseline years are registered  
in `core/regions.py` and reproduced in Appendix C, Table C1.

### 215 3.2. Burned-area label and the $\sim 500$ m analysis grid

Labels come from MODIS MCD64A1 Collection 6.1 [10] retrieved through  
Google Earth Engine [11], whose omission and commission characteristics [4]  
bound every model here. The analysis grid is **reconstructed rather than  
native**: the pipeline’s 30 m EPSG:4326 reference grid is partitioned into  
220 17 x 17 blocks, giving a nominal 510 m cell that approximates rather than  
reproduces the MODIS cell and is square in degrees but not on the ground. A  
cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested  
against the label window; the label never affects eligibility for modelling. Two  
safeguards are recorded rather than assumed: cells burning inside a region’s  
225 own predictor window are removed, which ran for three of five regions, and  
burning in earlier years is screened for none. Appendix C.1 gives the full  
specification, including what follows from the grid’s shape.

### 3.3. Burned-landcover admissibility gate

Before any modelling each region passes a gate asking what fraction of  
230 its burned cells is dominated by natural vegetation, using ESA WorldCover  
classes on the same cells. A region is admitted at 0.50 with at least 30 burned

cells, and rejected as a cropland-dominated control at 0.50 cropland. Verdicts and the purpose of the gate are in Section 4.1, the full rule in Appendix C.1.

### 3.4. Predictor variables

235 Two feature sets are used. The **baseline** is terrain, fuel and green-  
ness: elevation, slope, dominant land cover and a predictor-window median  
vegetation-index composite, all static or near-static over the timescale at  
which fire danger varies. The **thermal** set adds six pre-fire channels: current  
land surface temperature, its anomaly against a four-year window-symmetric  
240 baseline at the same cell, the Temperature-Vegetation Dryness Index and its  
difference against that baseline, and two coordinate-informed products, a  
downscaled and a fused surface temperature. The two differenced channels  
are the ones constructed to isolate the dynamic anomaly. TVDI's wet and  
dry edges are percentiles of the values a given area and window happen to  
245 contain, so **it is not portable as a physical quantity independently of  
any concept shift** (Appendices C.4, C.5).

### 3.5. Cell aggregation, validity and analysis populations

Cell-level values are means over the  $\sim 510$  m cell from valid 30 m pixels  
only, with the valid fraction recorded; a cell is **valid\_for\_modeling** when  
250 it meets a 30 % valid-pixel floor and no thermal channel is missing. The  
**primary** population is natural vegetation, cells whose combined tree, shrub  
and grass fraction reaches 0.50, which excludes cropland from every burnable  
mask. The **secondary** population is all valid cells; the frozen export carries it  
for the within-region arm in two regions, reported as a sensitivity in Appendix  
255  $A(v)$ , and the transfer matrix is defined on the primary population only.

### 3.6. Classifier

The two feature sets of Section 3.4 are nested, the thermal set being the baseline plus the six thermal channels. Land cover is one-hot encoded and numeric features are standardised inside the fitting pipeline. The classifier is  
260 a random forest [5] with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced class weights and `random_state = 42`, identical for every region, population, feature set and transfer direction, so that no comparison here is confounded by a model choice.

### 3.7. Spatial-block cross-validation and bootstrap uncertainty

265 Cross-validation is spatially blocked. Blocks are squares of the analysis grid formed as `row_500m // B` by `col_500m // B`, and five folds are drawn over whole blocks, so no block is split between training and test. `B` is reported at 2, 10 and 20 cells, about 1, 5 and 10 km.

Uncertainty is a spatial-block bootstrap: blocks are resampled with re-  
270 placement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the interval. Because it is blocks that are resampled, what bounds an interval's reliability is the number of blocks carrying at least one burned cell, and those counts are reported alongside the intervals. Where a verdict rests on too few such blocks it is stated as indicative rather than as an interval.

### 275 3.8. Cross-region transfer protocol

For each ordered pair the model is fitted on the source population and applied to the target with **no target label used at any point**: no refitting, no threshold selection, no calibration. Both feature sets are transferred, so the thermal block's contribution is a paired difference within a direction

280 rather than a comparison across directions. All twenty ordered directions are computed.

### 3.9. Label-blind domain adaptation

Two label-free remedies are tested on every direction, in both feature sets.

**Region-wise z-score.** Each region’s numeric features are standardised  
 285 using its own statistics, source statistics from source data and target statistics from target data, never pooled. The classifier is refitted on the z-scored source and applied to the z-scored target. This removes first- and second-order marginal offsets.

**CORAL after region-wise z-score.** The source covariance is aligned  
 290 to the target’s by the standard whitening-recolouring map [37],  $\lambda = 10^{-5}$ . Critically **the transform is applied to the source only**; the target is left as it is, and the classifier is refitted on the aligned source. Neither variant sees a target label, and both are verified label-blind at run time.  $\lambda$  sensitivity was assessed over nine values on four of the twenty directions, moving transfer  
 295 AUC by at most 0.014; the canonical  $\lambda = 1$  lies outside that sweep (Appendix A(b)).

### 3.10. Transfer-gap decomposition and the concept-shift criterion

The transfer gap is decomposed as (adapted – raw) / (within – raw), signed and unclipped, with its interval from the same paired bootstrap; Sec-  
 300 tion 4.3 shows the remainder should not be read as a conditional residual, because much of it is incurred inside a single region (Appendices A(j), C.7).

The mechanism is diagnosed by **signed univariate association**: the raw ROC-AUC of each numeric predictor against **burned** is computed per

region and never folded to  $\max(\text{AUC}, 1 - \text{AUC})$ , so a value below 0.5 is a  
 305 direction rather than weakness. A reversal is called bootstrap-supported only  
 when the two regions' point estimates fall on opposite sides of 0.5 **and each**  
**region's own interval excludes 0.5**. That is stricter than requiring the  
 intervals to be disjoint, and a feature failing the second condition is recorded  
 as a point reversal only.

### 310 3.11. *Transferability diagnostics versus transfer*

Twenty candidate diagnostics from five families are computed for every  
 region pair and rank-correlated (Spearman) against observed raw transfer  
 AUC, under one bootstrap that resamples unordered region pairs with both  
 of their ordered directions travelling together. The families are marginal  
 315 predictor-space measures  $P(x)$ , including area-of-applicability dissimilarity,  
 climatic and geographic distance and a learned domain classifier; burned-  
 niche overlap  $P(x|y=1)$ ; regime distance  $P(y)$ ; and conditional direction  
 agreement  $P(y|x)$ . **No family-wise error control is claimed**, and the  
 number of variants per family is reported with the result. Appendix C.2  
 320 gives the full specification.

### 3.12. *Interventions*

**Pooled multi-region training.** Leave-one-region-out: the model is  
 trained on the pooled primary populations of the other four regions and  
 evaluated on the held-out region, with folds blocked as before. This asks  
 325 whether pooling recovers what single-source transfer loses.

**Removal of direction-reversing features.** The two predictors re-  
 versing with bootstrap support **on the frames as drawn** are **elevation\_**

**mean** and **lst\_anomaly\_mean**. Section 4.4 withdraws that support under an equalised frame, so the selection rule is frame-dependent and this arm is  
 330 reported as a measurement under the original protocol, not as a consequence of an established reversal. Both are dropped and everything refitted, within-region and across every direction, and each is dropped singly so the cost can be attributed. The two features are selected by the same reversal analysis the result is then read against, so **both quantities are post-selection**  
 335 **estimates** with no correction applied.

### *3.13. Controls on the transfer path*

Four evaluations establish what the transfer arms are measuring, all using the transfer protocol of Section 3.8 unchanged: a **within-region half-split** (modelled cells cut at the median of a grid axis, both axes and directions,  
 340 a split discarded when either half is single-class); **leave-one-scar-out** (each burned component of at least 50 cells, dilated by a 2 km buffer, withheld from training and used as the target); a **foreign-region evaluation** of the same held-out scar areas; and the **same blocked model restricted** to those areas, which isolates the evaluation region from the training regime. Buffers  
 345 of 2, 5 and 10 km were run, 2 km primary. The per-split and per-scar positive counts are unequal and bear on the interpretation (Appendix A(i)).

### *3.14. Leakage control and reproducibility*

An explicit forbidden-column set is enforced at every model fit as an assertion rather than a convention: coordinates and their normalised forms,  
 350 every burn-date and label-provenance column, and the agreement fraction are excluded from all feature sets. The natural-vegetation mask defines the

population and is never a predictor. All randomness uses seed 42 and the bootstrap 1000 replicates, with one qualification: the five diagnostic bootstraps behind Table 5 and Table B1 use per-measure offsets from that seed rather than the seed itself, so that independent measures do not share a re-sampling draw. Two of the twenty transfer verdicts are not stable across seeds and are identified in Section 4.5. Because the transfer analysis runs in an environment separate from the upstream pipeline’s, every within-region model was refitted against the frozen upstream output: the within-region comparisons agree exactly and the twenty transfer directions to within  $1.6 \times 10^{-7}$ , with the tolerance that applies if the library version is not pinned in Appendix C.5(vi). Headline results are repeated across two populations, three block sizes, the CORAL sweep, both feature sets and four classifier capacities, and where a conclusion depends on one of those choices **the dependence is reported rather than resolved by choosing the favourable setting** (Appendices A, C.6).

### *3.15. Same-geography event-to-event comparison (Muğla 2021 versus 2022)*

Muğla admits a comparison in which place is held fixed and the event varies: a second fire burned inside the identical AOI, on the identical grid, eleven months after the first. Signed univariate AUCs are computed for both arms under the same 10-cell bootstrap used elsewhere. **Season, year and population all differ**, since the 2022 arm is defined by removing the 2021 scar (Appendices A(m), C.3).

## 4. Results

### 375 4.1. Study regions and the admissibility gate

All five candidate regions pass the burned-landcover gate and the negative control fails it as intended: Kozan 2023 returns a natural-vegetation fraction of 0.017 against the 0.50 threshold and is excluded. The separation is not marginal, the five admitted regions carrying 0.723 to 0.991, so  
380 the threshold falls in an empty interval rather than between neighbouring cases. That is consistent with the gate separating natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish, **though one negative control cannot establish it**. North Evia is analysed on an extended AOI, which leaves the burned scar essentially unchanged  
385 while cutting TSG prevalence from 0.676 to 0.287; the effect on transfer is in Appendix A(a).

### 4.2. Within-region: the thermal increment replicates in five regions

Adding the six thermal predictors to the baseline raises spatially blocked out-of-fold ROC-AUC in every region, the increment's bootstrap interval  
390 excluding zero in all five at 1 km and at 5 km blocking (Fig. 2) — the two scales this design supports as intervals. At 10 km the point estimates hold, from +0.048 to +0.154, but rest on 6 to 33 positive-carrying blocks and are indicative.

*Table note (resampling units).* The bootstrap resamples spatial blocks,  
395 so an interval's reliability is bounded by the number of blocks carrying at least one burned cell, and those counts fall from 192 to 843 at 2 cells to **6 to 33** at 20 cells. An interval built on six such blocks has no meaningful

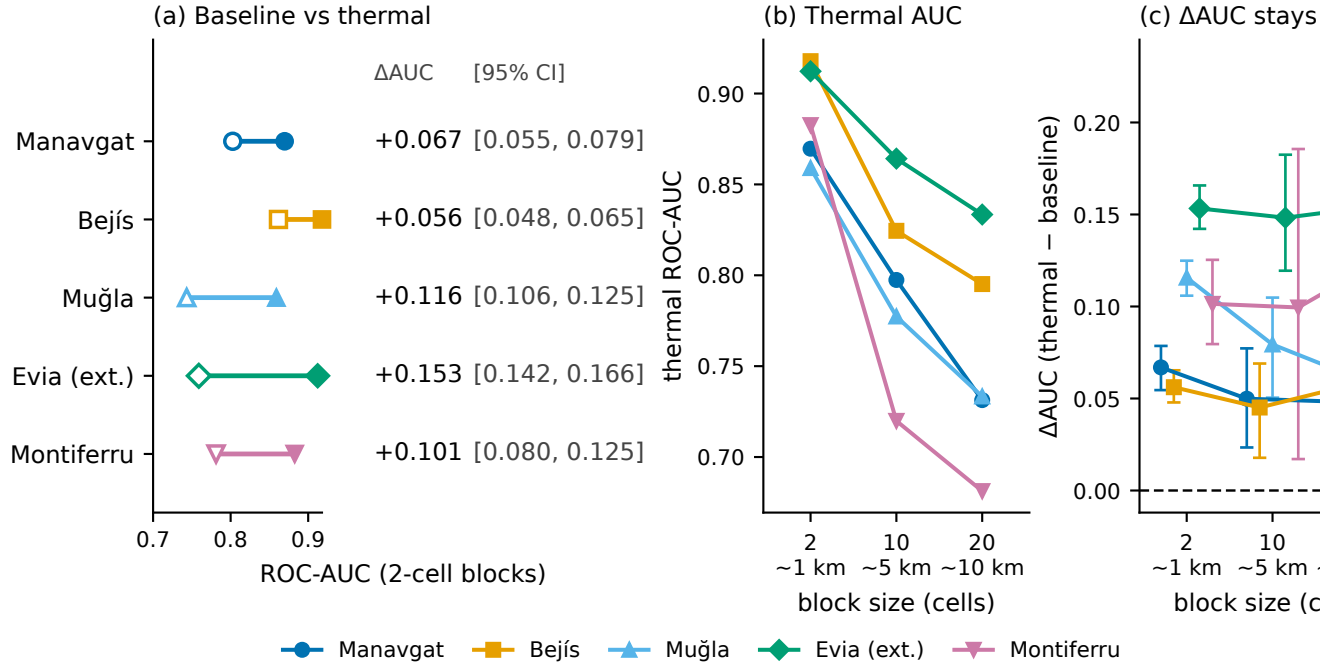


Figure 2: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell ( $\sim 1$  km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired  $\Delta AUC$  and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells,  $\approx 1, 5$  and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment  $\Delta AUC$  against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region-block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 1, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes** (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

Table 1: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.7); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells  $\approx$  1/5/10 km.

| Region                       | Block ( $\approx$ scale) | Baseline AUC | Thermal AUC | $\Delta$ AUC | $\Delta$ AUC 95% CI |
|------------------------------|--------------------------|--------------|-------------|--------------|---------------------|
| Manavgat<br>2021             | 2 ( $\sim$ 1 km)         | 0.803        | 0.870       | +0.067       | [+0.055, +0.079]    |
|                              | 10 ( $\sim$ 5 km)        | 0.748        | 0.797       | +0.050       | [+0.023, +0.077]    |
|                              | 20 ( $\sim$ 10 km)       | 0.683        | 0.731       | +0.048       | [+0.014, +0.085]    |
| Bejís<br>2022                | 2 ( $\sim$ 1 km)         | 0.862        | 0.918       | +0.056       | [+0.048, +0.065]    |
|                              | 10 ( $\sim$ 5 km)        | 0.779        | 0.825       | +0.045       | [+0.018, +0.069]    |
|                              | 20 ( $\sim$ 10 km)       | 0.739        | 0.795       | +0.057       | [+0.031, +0.090]    |
| Muğla<br>2021                | 2 ( $\sim$ 1 km)         | 0.743        | 0.859       | +0.116       | [+0.106, +0.125]    |
|                              | 10 ( $\sim$ 5 km)        | 0.698        | 0.777       | +0.079       | [+0.050, +0.105]    |
|                              | 20 ( $\sim$ 10 km)       | 0.673        | 0.733       | +0.061       | [+0.030, +0.094]    |
| North Evia<br>2021<br>(ext.) | 2 ( $\sim$ 1 km)         | 0.759        | 0.912       | +0.153       | [+0.142, +0.166]    |
|                              | 10 ( $\sim$ 5 km)        | 0.716        | 0.864       | +0.148       | [+0.119, +0.183]    |
|                              | 20 ( $\sim$ 10 km)       | 0.679        | 0.833       | +0.154       | [+0.124, +0.189]    |
| Montiferu<br>2021            | 2 ( $\sim$ 1 km)         | 0.781        | 0.883       | +0.101       | [+0.080, +0.125]    |
|                              | 10 ( $\sim$ 5 km)        | 0.620        | 0.720       | +0.100       | [+0.017, +0.186]    |
|                              | 20 ( $\sim$ 10 km)       | 0.555        | 0.681       | +0.126       | [+0.053, +0.228]    |

coverage, so **the 20-cell row should be read as indicative rather than as an interval**. The 10-cell row is the coarsest blocking this design supports properly, every region there having 16 to 70 positive-carrying blocks, and the increment holds at that scale in all five regions (Appendix A(c)).

This within-region result is not itself novel; it is reported because the transfer arms below are measured against it.

#### *4.3. What the evaluation frame is worth, within one region*

This section isolates the effect Section 4.4 applies between regions. It comes first because it is measured on one model in one region with no transfer involved, so nothing the transfer arms do can explain it (Appendix A(i)).

**Where the skill is lost, on a matched comparison.** Four evaluations are reported; the last three are scored on **identical cells**, so they differ only in what the model was trained on, and the first shows why an unmatched comparison misleads. The held-out unit is a burned connected component of at least 50 cells with all cells within 2 km of it. What makes that harder than a whole region is the composition of its negatives, not its burned fraction, and that is measured rather than argued. Scoring the same predictions on a random sample of region cells drawn at the scar area’s own burned fraction gives **0.782 against the region-wide 0.782**, so matching prevalence changes nothing, at  $-0.000 [-0.003, +0.002]$ ; scoring them on the scar area gives 0.627, a fall of **+0.155 [+0.093, +0.217]**. That swaps both pools at once, so each was swapped singly: replacing only the negatives costs **0.147 [0.104, 0.191]**, the whole of it, and replacing only the positives costs **0.002 [-0.052, +0.048]**, null on average though the per-scar effect runs from  $-0.12$  to  $+0.12$ . **The effect is the negative pool** — every negative

in a scar collar is fire-adjacent, sharing the terrain, land cover and synoptic conditions of the positives, whereas a region’s negatives include its easy far field — and ROC-AUC is in any case invariant to class balance at fixed class-conditional distributions (`pool_decomposition.json`).

Table 2: The four evaluations, scored on identical cells. Primary natural-vegetation population. Rows B, C and D are scored on the held-out scar area; row A is the whole region and is shown to make the mismatch visible. Means and Student  $t$  intervals are over the eight held-out scars.

| Evaluation                        | Model trained on                 | Scored on        | Mean AUC | 95 % CI        |
|-----------------------------------|----------------------------------|------------------|----------|----------------|
| A. Blocked cross-validation, 5 km | the region, scar included        | the whole region | 0.776    | [0.738, 0.814] |
| B. Same blocked model, restricted | the region, <b>scar included</b> | the scar area    | 0.634    | [0.552, 0.716] |
| C. Leave-one-scar-out             | the region, <b>scar withheld</b> | the scar area    | 0.552    | [0.501, 0.602] |
| D. Foreign region                 | another region, 306 to 2,802 km  | the scar area    | 0.555    | [0.495, 0.616] |

All four rows are means over the **same eight scars**, and Bejís has no row C because it is its region’s only component of any size, which is why the controls above are computed on nine and report 0.782 and 0.627 against this table’s 0.776 and 0.634. The intervals are Student  $t$  over the scars, this arm’s resampling unit rather than the spatial-block bootstrap used elsewhere. Four scars are in Muğla and two in Montiferru, and **row A is a region-level quantity repeated identically across a region’s scars, so its interval is pseudo-replicated and should not be read as coverage.** That repetition propagates into A – B and A – C; clustering by region

gives **+0.155** [**+0.082, +0.228**] and **+0.251** [**+0.093, +0.409**], both still excluding zero but the second 2.3 times wider, so **the scar-level intervals on these two differences overstate precision** (Appendix A(i)).

**The same model, scored two ways on the same region, differs by**  
440 **0.143 AUC.** Rows A and B are one model and one set of out-of-fold predictions; the only change is which cells they are scored on. That change alone costs **0.143** [**+0.077, +0.208**] of the 0.225 [**+0.157, +0.293**] fall from A to C, about two thirds — a ratio of two estimates, compatible with anything from roughly a third to nine tenths, so the point is the size of the numerator  
445 rather than the precision of the fraction. It is the size of the predictor-block increments this design itself measures, so a region-wide blocked figure is an upper bound on what the same model achieves where the fire actually is.

**Neither withholding the fire nor moving 2,800 km has a measurable cost.** B minus C, the fire-specific residual on identical cells, is **+0.082**  
450 [**−0.011, +0.175**]; C minus D, the effect of replacing a same-region model with one fitted 306 to 2,802 km away, is **−0.003** [**−0.075, +0.069**]. Both span zero and both arms sit close to chance, so **with eight scars, three of them starved of positives, this design cannot establish a fire-specific residual, only bound it at about 0.18.** It is not shown to be zero on an  
455 unseen fire; what this design establishes is that it is not established there. Nor does it establish the fire *event* as the unit, since the held-out patch is defined by the labels and its identity cannot be separated from its location. Appendix A(z) reports the spread behind row D and the sweeps showing the patch definition does not drive the result.

460 **The increment declines with the holdout, and is not established**

**once the fire is withheld.** Contribution 1 is about the paired thermal-minus-baseline difference, so the same evaluations were run on it: +0.056 to +0.153 under blocked cross-validation with every interval above zero, +0.027 on a within-region half-split, **+0.022** [**-0.032**, **+0.077**] under leave-one-scar-out, and +0.004 [-0.028, +0.036] across regions. The point estimate falls monotonically as the holdout hardens and the last two intervals span zero, so **the within-region increment is substantially a property of interleaved holdout.**

#### *4.4. The same effect between regions, applied to our own matrix*

Section 4.3's effect applies with equal force between regions. This test is reported before the transfer matrix because it changes what Section 4.5 and Appendix A(s) can claim; it withdraws nothing in Section 4.3, which is what it is built from. Sources are `aoi_frame_auc_frozen_mugla.csv` and `aoi_frame_transfer_frozen_mugla.csv`, which supersede the pre-correction files for the reason given below, with `collar_frame_bootstrap.csv` and `diagnostics_collar_frame.csv`; code is under `paper/code/` and the elaboration in Appendix A(w).

**The five areas of interest are not comparable frames.** Each is a rectangle drawn around a fire, and they differ by an order of magnitude in how much unburnt far field they enclose: the share of modelled cells beyond 10 km of any burned cell runs from **2.1 %** in Montiferru to **63.1 %** in Bejis (Appendix B, Table B6). That far field is not neutral — in Manavgat the median elevation of modelled cells rises from 472 m within 5 km of the fire to 1,273 m at 20 to 50 km, against 512 m for the burned cells themselves.

**Under an equalised frame the sign reversals do not survive.** Re-

stricting every region to cells within 10 km of any burned cell removes only  
 far-field negatives — every burned cell is at distance zero and is retained  
 at any radius — so the protection against a flattering radius is the sweep,  
 not the retention of positives. On that frame all five regions agree in sign  
 490 on elevation, LST and TVDI, and both bootstrap-supported elevation re-  
 versals of Table B3 disappear. Under this paper’s own criterion from Sec-  
 tion 3.10, **no between-region reversal remains bootstrap-supported:**  
 the two features straddling 0.5 at the point estimate are the internally differ-  
 enced channels, and in each at least one region’s interval covers 0.5 — Evia’s  
 495 by 0.003. The weaker difference instrument that Table B3’s note commits  
 this paper to does support four opposite-sided pairs on `lst_anomaly_mean`,  
 with three caveats that keep it weak. **The honest statement is that no**  
**between-region reversal meets the strict criterion under the collar,**  
**and that the LST anomaly differs between regions on the weaker**  
 500 **one.** Two scope limits belong with it, both in Appendix A(l) and A(w): the  
 interval criterion was evaluated at the 10 km collar only, though at the point  
 estimate the two radii agree exactly; and the collar reduces two regions’ cell  
 counts substantially, so part of the loss of support is a loss of power, against  
 which features supported in both regions *rise* from 1.20 to 3.40 per direction.  
 505 **The sign they now agree on is not the one the dryness framing**  
**predicts:** at the point estimates a hotter pre-fire surface burned *less* in every  
 region, and on mutual adjustment temperature survives where greenness does  
 not, so the absolute channels behave here as static land-surface descriptors  
 rather than as dryness. Within distance to the nearest burned cell the sign  
 510 crosses 0.5 in Manavgat and is attenuated to near-null in Montiferru, so in

those two it is largely the residual spatial gradient the collar was introduced to remove. The stratifications, the correlations and the reciprocal adjustment are in Appendix A(k).

**The same test determines what the diagnostics of Section 4.6 can establish.** Both candidates there with intervals excluding zero measure agreement in the sign of each predictor’s association between source and target — built from exactly the signed AUCs shown above to be frame artefacts, and correlated against transfer on the same unequal frames. Recomputing both sides under the collar:

Table 3: The diagnostics that ordered transfer, recomputed on an equalised frame. Spearman  $\rho$  against target ROC-AUC over the ordered directions in which each is defined. The first two rows are the two variants that cleared zero in Table B1; the third is the all-feature cosine, which did not and is shown for contrast. Both sides are recomputed here under one bootstrap setting, 1000 replicates, seed 42. Source `diagnostics_collar_frame.csv` and `collar_increment_and_cosine.csv`.

| Diagnostic                                  | Full frame                                 | 10 km collar                                                     |
|---------------------------------------------|--------------------------------------------|------------------------------------------------------------------|
| Sign-agreement fraction, supported features | $\rho = +0.86$ ( $p = 0.0001$ , $n = 14$ ) | <b>1.0 in all 18 directions, variance exactly 0 — degenerate</b> |
| Cosine, supported features                  | $\rho = +0.81$ ( $p = 0.0005$ , $n = 14$ ) | $\rho = -0.06$ ( $p = 0.82$ , $n = 18$ ), variance 0.00014       |
| Cosine, all nine features                   | $\rho = +0.50$ ( $p = 0.023$ , $n = 20$ )  | $\rho = +0.12$ ( $p = 0.61$ , $n = 20$ )                         |

The two fail differently and both fail: the agreement fraction has no

variance left once every pair agrees, and the supported cosine keeps a trace of variance but stops tracking transfer. The disagreements they were reading were the far fields. **Only these two were recomputed, so the other eighteen are unknown against the equalised transfer vector rather**  
525 **than shown to be null** (Appendix A(o)). This settles what Contribution 3 can claim: **no diagnostic tested here has been shown to order transfer**, and the two that appeared to were reading how the rectangles were drawn.

**Two further arms move with the frame.** The same-geography arm  
530 is the most extreme case in the cohort — a fixed study area is not a fixed evaluation frame, and its 2022 arm has 93.2 % of cells beyond 10 km of any burned cell against 55.3 % for 2021 — so under the collar its two arms fall on the same side of 0.5 on elevation, the feature whose reversal we had reported. The claim is scoped: that arm is reconstructed from the 2021  
535 predictors, valid for the year-invariant channels and **not** the seasonal ones, so **only elevation and slope carry a verdict, on both the reversal is removed, and on the thermal channels the arm is silent**; it rests on eleven positive-carrying blocks against a floor of sixteen (Appendices A(m), A(o)). The within-region increment, by contrast, **survives**, positive in all  
540 five regions at a mean of +0.077 against +0.086 as drawn.

**The transfer matrix moves as well.** Restricting source and target to the same collar:

**A data-provenance defect in this arm was found and corrected:**  
one region’s predictor file at the canonical path had come to differ from the  
545 one the frozen tables were computed on, so every arm here was re-run against

Table 4: Cross-region transfer under equalised evaluation frames. Primary natural-vegetation population, thermal model, twenty ordered directions per row. Above/below chance are point counts; the supported counts use a 10-cell ( $\approx 5$  km) spatial-block bootstrap on the target, 1000 replicates, seed 42; **Table B9 reports the same matrix under 2-cell ( $\approx 1$  km) blocking**, which is why its supported counts are the larger 12 and 6 (Section 4.5). Per-direction bounds are in `aoi_frame_transfer_frozen_mugla.csv`.

| Source frame | Target frame             | Mean target AUC | Above chance    | Below chance | Supported thermal delta  | Per-direction thermal delta |
|--------------|--------------------------|-----------------|-----------------|--------------|--------------------------|-----------------------------|
|              |                          |                 |                 |              | above<br>/<br>be-<br>low |                             |
| full         | full ( <b>Table B9</b> ) | 0.541           | 14 of 20        | <b>6</b>     | 9                        | +0.004                      |
|              |                          |                 |                 |              | /<br><b>4</b>            |                             |
| full         | 10 km                    | 0.576           | 17 of 20        | 3            | 11                       | +0.003                      |
|              |                          |                 |                 |              | /<br>1                   |                             |
| 10 km        | full                     | 0.570           | 17 of 20        | 3            | 11                       | +0.014                      |
|              |                          |                 |                 |              | /<br>3                   |                             |
| <b>10 km</b> | <b>10 km</b>             | <b>0.616</b>    | <b>19 of 20</b> | <b>1</b>     | <b>15</b>                | <b>+0.023</b>               |
|              |                          |                 |                 |              | /<br><b>1</b>            |                             |
| 5 km         | 5 km                     | 0.608           | 19 of 20        | 1            | 12                       | +0.014                      |
|              |                          |                 |                 |              | /<br>0                   |                             |

the frozen export. It moves forty of a hundred per-direction values by up to 0.022 and leaves **every headline quantity above unchanged to within 0.0012**, with the signed AUCs of that region’s two channels moving by at most 0.008 and no verdict changing (Appendix A(w)). The table reports the  
550 corrected values, and reproduces the frozen matrix at 0.541 against Table B9’s 0.541 and 14 of 20 exactly. **The baseline control must be restated on this frame**: the static baseline transfers at 0.593 against 0.616, a paired difference of +0.023 rather than +0.004, so the control holds in kind but the gap is about six times larger once frames are comparable, and about four  
555 times at the 5 km collar.

**What this settles, and what it leaves standing.** Five quantities reported below are properties of the frames rather than of the predictor-burning relationship, and are identified as such where they appear: the count of six anti-predictive directions, which becomes one; the sign reversal of ele-  
560 vation, LST and TVDI as a mechanism; the sign-agreement diagnostic; the same-geography arm; and the paired thermal contribution, +0.004 as drawn against +0.023 equalised. That last quantity carries the portability null, so it is given an interval on the frame this section argues for: **+0.023 [−0.004, +0.048]** under the pair-cluster resampling of Appendix A(o). It still spans  
565 zero, so the null survives the correction — but only just, and under the alternative admissible unit, clustering by target region, it does not (Appendix A(w)). What survives is the central negative result, and its size must be stated on a matched comparison: setting 0.616 against a within-region reference of about 0.87 would compare a collar number with a full-rectangle one  
570 at 1 km blocking. Recomputed on the same frame at 5 km blocking that

reference is 0.772, so the shortfall is **+0.155** [**+0.094**, **+0.217**], a Student  $t$  interval over the five target regions rather than the spatial-block bootstrap used elsewhere, and numerically almost identical to Section 4.3’s negative-pool effect over nine scars, which is a different quantity on a different unit.  
575 It is real, and 0.155 rather than the 0.25 an unmatched comparison implies. Appendix C.5(x) records the frame as a limitation of this cohort rather than of the method.

#### *4.5. Cross-region transfer, and what label-free adaptation does to it*

**Everything in this section is computed on the frames as drawn**  
580 **and should be read against Section 4.4**, which has shown the frames are not comparable and that the within-region reference used here is itself frame-dependent. The as-drawn matrix is reported because it is what the pre-registered protocol yields; per-direction values are in Appendix B, Table B9, and the supporting arms in Appendix A(x).

**Raw transfer is heterogeneous and includes anti-predictive di-**  
**rections.** Target AUC spans 0.326 to 0.686. The support counts depend on the blocking scale, so both are reported: at 2-cell ( $\approx 1$  km) blocking twelve of twenty directions are above chance with interval support and six below; at the 10-cell ( $\approx 5$  km) blocking this design defends, nine are above and four  
590 below with seven uncertain, the counts Table 4 uses. Of the below-chance directions only Manavgat to Bejís survives frame equalisation, and two verdicts are not stable across bootstrap seeds (`transfer_ci_blocksize.csv`). Even the best raw transfer sits far below that target’s own within-region 0.870, the raw deficit running 0.184 to 0.592 against a reference Section 4.3  
595 shows is not matched to a transfer evaluation.

In precision terms it is worse than the ROC figures suggest. A susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity. PR-AUC averages **0.156 against a no-skill baseline of 0.136**, and **six of twenty directions fall below their own baseline** at the point estimate, five of them with intervals entirely below it; the exception is Bejis to Manavgat, whose interval covers its baseline. Only one direction exceeds twice its baseline. These are frame-as-drawn quantities and the PR arm was not recomputed on the collar (Appendix B, Table B4).

**The static baseline does not transfer either**, at a mean of **0.537** against **0.541** for the thermal model, so the failure is not specific to the dynamic block: a baseline that does not travel, plus pre-fire thermal state, gives a model that does not travel. Section 4.4 restates this control on the equalised frame, where it is 0.593 against 0.616.

**The thermal block’s paired contribution is +0.004, and the mean is not the informative statistic.** The individual contributions span **−0.148 to +0.133**, twelve positive and eight negative, so the spread is thirty times the mean and the sign belongs to the pair rather than to the block; a mean near zero records cancellation, not consistent absence of effect. The directions are not independent, since each region appears in eight of the twenty, so the interval depends on the resampling unit — **all four units the design permits give the same answer**, from  $[-0.027, +0.034]$  to  $[-0.037, +0.046]$ , and none propagates within-direction sampling variability. The leave-one-region-out jackknife shows how little the mean is anchored: **dropping Evia alone reverses its sign.**

**Label-blind adaptation compresses the matrix toward chance**

**rather than repairing it.** Under region-wise z-scoring the twenty directions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half the raw spread, and fourteen of twenty move closer to chance — which helps the directions that failed and harms those that worked. The committed-  
625 in-advance CORAL arm averages **0.552**; taking whichever method scores better per direction gives 0.556, but that selection uses the target labels the protocol forbids, so it is **an oracle upper bound rather than an achievable result**. Even the oracle only reaches the reference a model can reach on an unseen scar (Section 4.3), so alignment is regressing the matrix  
630 onto that reference rather than failing beneath it. What it cannot do is exceed the reference, and a sign reversal is not a distribution mismatch that realigning inputs would repair. Appendix A(x) reports the recovery fractions, including the seven directions with *negative* recovery.

4.6. *Transferability diagnostics: what appears to order transfer, and why it*  
635 *does not*

Twenty candidate diagnostics from five families were each rank-correlated with raw thermal transfer over the twenty ordered directions, under one common pair-based bootstrap.

**Only two diagnostics have intervals excluding zero, and both are**  
640 **conditional:** the sign-agreement fraction over interval-supported features at  $\rho = +0.84$  [+0.58, +0.88], and its cosine variant at +0.81. No marginal measure was shown to order the matrix, including area-of-applicability dissimilarity, climatic distance and geographic distance, and neither were the niche-overlap and regime families. The learned domain classifier is at ceiling,  
645 separating source from target at  $\text{AUC} \geq 0.96$  for every pair, so it always

Table 5: Transferability diagnostics versus raw thermal transfer, by family. Spearman  $\rho$  against raw transfer AUC with pair-based bootstrap 95 % CIs. Exp. is the sign expected if the diagnostic orders transfer. The member named is the one with the largest absolute correlation in its family, which is not always in the expected direction. All twenty individual diagnostics are in Appendix B, Table B1.

| Family                                          | Diagnostics | Exp. | Largest correlation in family                                                                              | Sign as expected | Any CI excluding 0 |
|-------------------------------------------------|-------------|------|------------------------------------------------------------------------------------------------------------|------------------|--------------------|
| <b>P(y x)</b><br><b>condi-</b><br><b>tional</b> | 6           | +    | <b>+0.84</b><br>[ <b>+0.58</b> ,<br><b>+0.88</b> ]<br>agree-<br>ment<br>fraction,<br>supported<br>features | yes              | <b>yes, 2 of 6</b> |
| P(x y=1)<br>niche<br>overlap                    | 5           | +    | +0.24<br>[−0.45,<br>+0.74]<br>Schoener’s<br>D, 1-D<br>mean                                                 | yes              | no                 |
| P(x)<br>marginal                                | 6           | −    | −0.32<br>[−0.78,<br>+0.33]<br>domain-<br>classifier<br>AUC                                                 | yes              | no                 |
| P(y)<br>regime<br>structure                     | 2           | −    | +0.29<br>[−0.38,<br>+0.74] <sup>log</sup><br><sub>34</sub><br>effective-<br>N<br>distance                  | <b>no</b>        | no                 |
| geographic                                      | 1           | −    | −0.24<br>[−0.84,                                                                                           | yes              | no                 |

succeeds and carries no ordering information. Geographic separation is a single diagnostic here, centroid distance over the twelve directions for which the pipeline author’s export defines it, at  $\rho = -0.24 [-0.84, +0.73]$ : in the expected direction and spanning zero.

650 Four limits are stated with the result. **Size:** the index’s tie structure caps the achievable Spearman at +0.861, so the observed +0.840 sits on that ceiling, and its exact one-sided permutation p of 0.0060 — 240 of the  $8! = 40,320$  relabellings of the **eight** pairs on which the index is defined — is the smallest that structure can produce, against a Bonferroni threshold of  
655 0.05/19 = 0.0026. **No outcome of this diagnostic could have cleared family-wise correction. Labels:** signed associations need burned labels in both regions, so the family that appears to work is not available before deployment while the family that fails is. **Selection,** the sharpest: the two rows that clear zero are the *supported-feature* variants, whose subset is chosen  
660 by whether two regions’ bootstrap intervals happen to be disjoint — a data-dependent selection on the same data, with no correction; their unselected counterparts over all nine features are  $\rho = +0.50 [-0.17, +0.83]$  and  $+0.18 [-0.40, +0.72]$ , both spanning zero, so **the result lives in the selection step. Frame,** established in Section 4.4, removes it entirely. Recomputing  
665 every row on the common twelve directions reproduces the published values to  $4.8 \times 10^{-5}$  and leaves the ordering unchanged, so it is not an artefact of unequal samples (Appendix B). Everything here is what the original protocol yields; the conclusion that survives is that **no diagnostic tested here was shown to order transfer once the frames are comparable.**

670 *4.7. Further arms*

Six further arms bear on the findings above without changing them: the contrast pair (A(s)), the two interventions (A(n)), the sensitivity summary (A(v)), the same-geography two-event arm (A(m)), the distance curve (A(t)) and the label-budget curve (A(u)). Each is stated there with its own limits.

675 The two bearing directly on Sections 5.4 and 5.7 are the interventions, where a local cost of  $-0.081$  is measured against a transfer return of  $+0.014$   $[-0.017, +0.045]$  whose interval spans zero, and the label budget, where thirty-two labelled 5 km blocks recover 85 to 89 % of the target's matched ceiling in three of six directions and 30 to 57 % in the rest — 7 to 20 % of the target's

680 population, drawn from the event being predicted, which is not a resource available before that event burns.

## 5. Discussion

### *5.1. Reading the three findings together*

Section 1.4 states the three findings and Section 4 establishes them; this

685 section argues from them. One relation between them makes the paper cohere: the first is not a caveat attached to the other two, it is the instrument that sets their size. Applied to our own matrix it withdrew five claims, and what it left standing is a shortfall in transferred skill, not a reversed relationship.

### *5.2. Why the thermal increment is real but local*

690 The within-region increment and the transfer failure are measured at different separations, and Section 4.3 shows most of the difference is already

present inside a single region. The increment is local in a specific sense: it holds where held-out cells are interleaved with training cells and most of it  
695 is gone once they are not, before the fire or the region changes. It is not an artefact to be explained away, surviving every sensitivity arm of Appendix A(a)–(h), but it is established under interleaved validation and not beyond it — and even there the window-closure arm **weakens monotonically in Evia**, so what holds everywhere is survival, not improvement.

700 The natural objection is that Mediterranean regions are simply different systems, so a predictor meaning one thing in one place and another elsewhere is a comparison of two systems rather than instability. **We designed the two-Muğla-events arm to answer that objection and it does not answer it:** Section 4.4 shows it is the most extreme frame artefact in the  
705 cohort. Three further confounds were never resolved in any case: season and year, which **cannot be resolved in this study area**, an unfixed population, and a positive-block count below this design’s own floor (Appendix C.5(ii)). With one fire per region everywhere else and that arm withdrawn, **this cohort provides no evidence that the transfer shortfall is regional rather than event-specific.** Appendix A(t) adds a length scale for  
710 the within-region decay but cannot turn it into an attribution either.

One conclusion survives from the other direction: the static baseline transfers no better than the dynamic block, so whatever the shortfall is, it is not the thermal block’s peculiarity.

### 715 5.3. *Why the diagnostics fail, and what the conditional one is not*

A diagnostic built on distance in predictor space asks whether the target’s predictor values look like the training data’s, which is not the question that

matters when the failure is conditional: a target region can sit well inside the training envelope while the relationship between those predictors and burning points the other way. Manavgat to Muğla is exactly that case, with 0.875 of target cells inside the full-frame area of applicability and transfer among the weakest in the matrix **on either frame**, while the least similar pair transfers better (Appendix A(s)). High overlap does not buy transfer. **The qualifier is not decorative:** at the 5 km blocking this paper otherwise defends neither contrasted direction carries a verdict, so this is a statement about point estimates. The domain classifier is at ceiling for every pair, so separability carries no ordering information either.

The two conditional variants that did clear zero are not a remedy. Section 4.6 states four limits with them: a tie-structure ceiling and a family-wise threshold **no outcome could have cleared**, a label requirement that makes it a mechanism diagnosis rather than a screen, a feature subset selected on the same data, and the frame, which removes the result altogether — the index is built from signed associations Section 4.4 shows to be frame artefacts. So the practical conclusion is not "use this index instead", nor even "it works but needs labels": **none of the twenty candidates was shown to order transfer**, and the two that appeared to were reading how the rectangles were drawn — nulls on ten effective pairs, not shown to order transfer rather than shown incapable of it.

#### 5.4. *What the two interventions do and do not show*

Both interventions show the same shape. Pooling four regions never beats the best single-source transfer for any target and stays well below the within-region ceiling (Appendix A(n)), so it does not manufacture the missing condi-

tional information. Removing the reversing predictors costs  $-0.081$  of within-region skill with interval support in every region, and returns  $+0.014$   $[-0.017,$   
745  $+0.045]$  in mean transfer.

That pair of numbers is easy to read as an exchange, and it is not one. Both arms are null on the portability axis — the thermal block’s own contribution is  $+0.004$ , removal returns  $+0.014$ , both intervals spanning zero — so they measure a local cost and no compensating gain. That is not a con-  
750 servation law and not a rate at which local skill can be sold for portability; no such rate is estimated here.

### *5.5. The regime hypothesis, reported as it happened*

A regime-structure explanation was stated in advance and the data confirmed the null: the regime-distance correlation has the wrong sign at the  
755 point estimate, the most regime-similar pair fails in both directions, and the most regime-different pair transfers above chance. One mundane explanation can be set aside: subsampling Muğla, much the largest population, to Manavgat’s cell count and separately to its positive count leaves its transfer behaviour inside the subsampling range in both roles. The error was in the  
760 hypothesised grouping, not in the data, and with ten pairs this cannot refute regime typology [2] in general.

### *5.6. The empirical contrast with Dimarco et al.*

Dimarco et al. [8] transfer successfully across a comparable Mediterranean design and we do not, and the two results are not in conflict. Their pre-  
765 dictors are attributes of a place, ours the state of a surface in one season,

which suggests the relation between domain similarity and transfer success is predictor-class dependent.

That reading is one of at least two, and we cannot separate them here: their response variable is human-driven **ignition**, dominated by access and activity, while ours is burned **area**, dominated by spread, so a predictor-class explanation and a response-variable explanation are confounded in this comparison. Our own data speak against a simple predictor-class reading in any case: the static baseline transfers at a mean of 0.537 here, so within this cohort the place-attribute class does not travel either.

## 5.7. Implications

In precision-recall terms, which is how a susceptibility surface is used, transferred models average a PR-AUC of 0.156 against a no-skill baseline of 0.136 **on the frames as drawn**, six of twenty falling below their own baseline (Section 4.5). Whatever the ROC figures suggest, a model moved to a region it was not fitted in does not usefully rank burned cells there.

The paper supports one concrete change in reporting: alongside a spatially blocked within-region figure, report skill on a held-out burn scar and its surroundings. On these five regions the two differ by about 0.14 ROC-AUC on the same model, the size of the effect such papers usually claim, so **a blocked figure alone should be read as an upper bound**. Transfer skill likewise has to be *measured* rather than inferred from similarity, since the diagnostics used for that inference did not order it here, and where a model must be moved the resource that closes the gap is target labels (Appendix A(u)): a real answer, but not a cheap one.

The results also bound what unsupervised alignment can be asked to do.

Even the oracle selection, which uses the target labels the protocol forbids, only reaches the reference a model can reach on an unseen scar (Section 4.5). Alignment is not failing far below an achievable target; it is regressing the matrix onto it, at the cost of the directions that already worked. A sign reversal is not a distribution mismatch that realigning inputs repairs.

### 5.8. Limitations

Eleven limitations are stated in full in Appendix C.5; four bind the conclusions above. **The frames are not comparable and this cohort cannot fully repair it** (Section 4.4): the collar radius is itself a choice — 5 km and 10 km give 0.608 against 0.616 — and the frames were fixed upstream, so we can restrict them but not extend them. Any future cohort should fix the frame by an explicit accessible-area rule [3] before any predictor is computed; that is the main design lesson of this paper. **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology and the shortfall cannot be attributed to region rather than event (Section 5.2). **The diagnostic correlations rest on an effective sample of ten region pairs**, so Section 4.6’s successes and failures both read at that power. And **the interval-support counts are less stable than the point estimates behind them**: the published split turns on a lower bound of  $-0.00045$ . The remaining seven, from the absent meteorological covariates to the single classifier family (Appendix A(h)), bear on scope rather than on the conclusions above.

## 6. Conclusions

Where a fire model is scored decides what it appears to know. Holding  
815 model, predictors and fitting fixed and changing only which cells are scored,  
moving from a whole study region to the burn scar and its 2 km collar  
costs **0.143 ROC-AUC**, with a control measuring the cause as the negative  
pool’s composition rather than class balance. That is the size of the predictor-  
block increments this design itself measures, so a region-wide validation figure  
820 should be read as an upper bound on what a model achieves where fire  
actually occurs, and reported as one.

That correction is not only other people’s problem. Applied to our own  
five-region matrix it withdraws five of our claims. The study areas enclose  
very unequal far fields, and equalising them to a 10 km collar lifts mean  
825 transfer from 0.541 to 0.616, reduces the directions below chance, and leaves  
no sign reversal supported between regions under our own criterion. The  
same correction applies to the one arm that held place fixed, two fires in one  
study area eleven months apart, which we had wrongly exempted because  
its geography was constant while its evaluation frame was not: its elevation  
830 reversal does not survive equalisation either, and that arm can speak only  
for the year-invariant channels. What had looked like a reversed predictor-  
burning relationship was mostly a statement about how five rectangles were  
drawn.

What survives is the central result. Pre-fire thermal dryness adds a real  
835 and repeatable within-region increment in all five regions, surviving spatial  
blocking at about 5 km and a predictor window closed up to two weeks before  
the first labelled burning. It does not travel: on matched frames and blocking,

equalised transfer of 0.616 falls 0.155 short of the within-region reference; the paired cross-region contribution spans zero on both frames under the  
840 resampling unit we treat as primary, though not under every admissible one (Section 4.4), with a sign that varies by pair; and the static baseline transfers no better on either frame, so this is not a peculiarity of thermal predictors. The shortfall also cannot be anticipated: none of twenty diagnostics from five families was shown to order the matrix, and the two that appeared to  
845 require target labels on both sides and stop ordering transfer once the frames are comparable. Label-free alignment compresses most directions towards chance rather than repairing them, and supervised recalibration works but is not cheap.

Two limits belong with all of this. The agreed thermal direction is the  
850 opposite of the one the dryness framing predicts — hotter pre-fire surfaces burned less in every region, and holding greenness does not remove it while holding temperature reverses greenness in two regions — so on this cohort the absolute channels behave as static land-surface descriptors rather than as a dryness index. And with one fire per region, and the single arm that  
855 held place fixed now withdrawn, this design measures a transfer shortfall but cannot attribute it to region rather than to event.

Two practical consequences follow. Report local skill and portability separately, since a predictor block worth a large within-region increment may contribute nothing distinguishable from zero across regions. And fix the  
860 evaluation frame by an explicit accessible-area rule before any predictor is computed, because a frame chosen for convenience can manufacture both a transfer failure and a mechanism to explain it. Three extensions are left

for future work: temporal transfer, meteorological covariates, and physically normalised dryness variables.

## 865 **Appendix A. Appendix A. Sensitivity analyses**

Each arm below varies one design choice and leaves everything else fixed. None changes a conclusion in the main text. They are reported so that a reader can see which choices were tested, and what each was worth. Arm (f) is the exception in one respect: it is not a robustness check but a test of a  
870 claim the introduction makes, and the claim is not upheld.

### *Appendix A.1. A(a)–A(h). The eight sensitivity arms*

Each arm is labelled by its letter below, and is cited elsewhere in the paper as Appendix A(a) through A(h).

**(a) Evia AOI and prevalence.** The raw transfer arms were repeated  
875 with the legacy, high-prevalence Evia box. Every qualitative conclusion is unchanged. Thermal raw transfer AUCs move by up to 0.07, the largest being Evia to Bejís at 0.378 to 0.448, and no direction changes side of the chance line.

**(b) CORAL regularisation, including the canonical  $\lambda = 1$ .** Over  
880 nine  $\lambda$  values from 0 to  $10^{-1}$  on four directions, CORAL transfer AUC moves by at most 0.014 within any direction and 0.008 within the thermal family. The canonical  $\lambda = 1$  of the cited method lies outside that sweep and was computed separately on the same four directions. It changes the thermal mean from 0.519 to 0.522, a shift of +0.003 with a per-direction range of  
885  $-0.005$  to  $+0.016$ , and none of those four directions crosses the chance line.

The  $\lambda = 1$  values are 0.510, 0.444, 0.559 and 0.575 against 0.508, 0.445, 0.564 and 0.559 at  $\lambda = 0.1$ .

**Those four directions are Bejís–Muğla and Manavgat–Muğla, and they exclude the pair that moves.** An earlier version of this appendix generalised from them to the whole matrix, which was an error: the Manavgat–Bejís pair is not in the sweep and it is the pair whose CORAL result is  $\lambda$ -dependent. A separate frozen artefact, `experiments/cross_region/step10/coral_lambda_sensitivity.csv`, carries it. Bejís to Manavgat gives 0.5571 [0.5292, 0.5857] at  $\lambda = 10^{-5}$ , 0.5644 at  $\lambda = 10^{-3}$  and 0.5473 at  $\lambda = 0.1$ , all with intervals entirely above 0.5, but **0.4850 [0.4496, 0.5197] at  $\lambda = 1$** , which crosses the chance line and loses its interval support. Manavgat to Bejís falls likewise, from 0.5108 to 0.4776 [0.4501, 0.5013].

The correct statement is therefore narrower. CORAL-dependent conclusions are insensitive to  $\lambda$  over three orders of magnitude below 0.1; at the cited method’s canonical  $\lambda = 1$  the one direction whose adapted interval sits entirely above chance loses that status.  $\lambda = 10^{-5}$  is the minimally regularised choice and matches the reference implementation, but the dependence is real and is reported rather than absorbed.

**(c) Blocking scale.** Recomputing the transfer quantities at 10-cell ( $\approx 5$  km) blocking from the frozen per-cell predictions widens the intervals and moves the verdict counts, from ten positive, seven negative and three uncertain at 1 km to six, four and ten at 5 km. Coarser blocking therefore removes support from seven verdicts and adds none. Across five seeds the counts run 5 to 6 positive, 3 to 4 negative and 10 to 11 uncertain, so they are not exact.

The point estimates are unchanged. That is an identity rather than a

result, and it should not be offered as robustness. The blocking scale is the bootstrap *resampling unit*, and each point estimate is computed once over all target cells, so no choice of block size could have moved one. The comparison does establish two things. Every verdict that changes, changes  
915 towards "no verdict". And no direction crosses the chance line under the widened intervals. The counts are the fragile part of this paper. The sign pattern is not thereby shown to be robust. It is simply not tested by this variation.

**(d) Predictor-window closure.** The predictor window was closed 7  
920 and 14 days earlier in all five regions. The thermal contribution stays positive and bootstrap-supported everywhere. The direction of the change is region-specific. It strengthens in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla, from 0.115 to 0.128. It is flat in Montiferru. It weakens monotonically in Evia, from 0.156 to 0.149 to 0.135. What holds in every region is survival,  
925 not improvement. That is the claim carried in Section 5.2.

**(e) Quality screening of the coarse thermal input.** Two of the five regions' MODIS inputs are quality-screened and three are not. The split follows export date rather than design, and it induces an elevation-correlated change at the input, at  $r = +0.615$  in Manavgat. Manavgat's  
930 entire downstream chain was rebuilt from a quality-screened input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. Elevation is identical in both arms, at 0.374, the population is unchanged, and the within-region increment moves from  $[+0.055, +0.079]$  to  $[+0.054, +0.077]$ .  
935 The reason is structural. Elevation is a DEM variable the screening cannot

touch, and fusion falls back on the MODIS-derived surface across only 2.14 percentage points of coverage. Details are in `paper/modis_qc_downstream_propagation.md`.

(f) **Normalised against absolute dryness channels.** Section 1.2  
940 argues that an internally normalised index should be less exposed to absolute-temperature offsets between regions than raw land surface temperature. The thermal block contains both kinds, so the argument can be tested directly as a feature-set contrast.

One thing should be said before the result, because it makes the outcome  
945 less surprising than it might otherwise appear. The normalised channels are the two that hold their direction in the same-geography two-event comparison of Appendix A(m), where the absolute channels move. That is a statement about one region across two fires. **Across regions it does not hold:** `1st_anomaly_mean` is itself one of the two predictors whose reversal  
950 is bootstrap-supported on the frames as drawn, reversing between Bejís and Evia (Section 4.4 withdraws that support under an equalised frame), which is why Appendix A(n) drops it alongside elevation. Being internally normalised protects a channel against the offset between two seasons in one place. It does not, on this evidence, protect it against a change of place.

955 Three feature sets were run over all twenty directions and all five within-region folds, with the classifier, population, folds and bootstrap held fixed. The harness aborts unless its reference configuration lands on the frozen exports. It did exactly: the maximum absolute difference between the reference arm and the frozen transfer AUCs is 0.000000 across all twenty directions.

| Feature set                             | Mean transfer AUC | Directions > 0.5 | Mean<br>within-region<br>AUC |
|-----------------------------------------|-------------------|------------------|------------------------------|
| baseline only                           | 0.5371            | 16               | 0.7896                       |
| baseline +<br>normalised<br>anomalies   | 0.5442            | 15               | 0.8551                       |
| baseline +<br>absolute surface<br>state | 0.5479            | 13               | 0.8564                       |
| all ten features<br>(reference)         | 0.5414            | 14               | 0.8883                       |

**The advantage is not found.** Per direction, the normalised set minus the absolute set has a mean of  $-0.0037$ . It is positive in 12 of 20 directions and runs from  $-0.099$  to  $+0.094$ , so the spread is an order of magnitude larger than the difference. The normalised channels transfer very slightly worse at the mean, and nothing here supports a design rule favouring anomaly-referenced dryness for portability.

Within region the two sub-blocks are indistinguishable, at 0.8551 and 0.8564, and each recovers most of the gap between the baseline’s 0.7896 and the full model’s 0.8883. What separates them in sign stability does not become transferable skill.

This arm strengthens the negative finding rather than softening it. Four differently constituted feature sets were tried. None clears a mean of 0.548 across regions, and all differ sharply within them. The baseline arm also gives an independent confirmation of the control reported in Section 4.5: its

975 mean transfer of 0.5371 was recomputed here from the modelling datasets  
and matches the 0.537 read from the frozen per-direction export. Details are  
in `paper/anomaly_only_transfer.md`.

The comparison is between two sub-blocks of one thermal set on one  
cohort. It does not test normalised dryness indices in general, and it does  
980 not test a normalisation fitted against a pooled multi-region reference rather  
than each region’s own baseline years.

(g) **The coordinate-informed channels.** `downscaled_1st_mean` and  
`fused_1st_mean` come from a per-region downscaling model whose own in-  
puts include coordinates, which Section 3.14 names as the one route by which  
985 a coordinate-derived surface re-enters a feature set that excludes coordinates.  
A coordinate-smoothed surface is by construction locally informative and  
non-portable, so if the within-region increment depended on it, that incre-  
ment would be this paper’s own result in miniature rather than a finding  
about thermal dryness. The arm was therefore run here rather than de-  
990 ferred.

Both channels were dropped and everything refitted, over all twenty direc-  
tions and all five within-region folds. The reference configuration reproduces  
the frozen exports exactly, at 0.000000 against both the transfer and the  
within-region references.

| Region          | Increment, full set | Increment without the two<br>channels | Retained    |
|-----------------|---------------------|---------------------------------------|-------------|
| Manavgat 2021   | +0.067              | +0.063                                | 94 %        |
| Bejís 2022      | +0.056              | +0.046                                | 82 %        |
| Muğla 2021      | +0.116              | +0.097                                | 84 %        |
| North Evia 2021 | +0.153              | +0.145                                | 94 %        |
| Montiferru 2021 | +0.101              | +0.105                                | 103 %       |
| <b>Mean</b>     | <b>+0.099</b>       | <b>+0.091</b>                         | <b>92 %</b> |

**The increment does not depend on them.** It stays positive in every region and retains 82 % to 103 % of its size, rising slightly in one region. Mean transfer is likewise unchanged, at 0.545 without the two channels against 0.541 with them and 0.537 for the baseline alone. The local skill this paper reports is therefore not an artefact of a coordinate-smoothed surface.

This arm was computed independently here and reproduces the companion paper’s figure for the same quantity, 82 % to 103 %, from a separately written harness.

**(h) Model capacity.** Every number in this paper comes from one random forest with unlimited depth and `min_samples_leaf = 3`. That is the configuration most able to encode local structure and least able to extrapolate, so a reader may reasonably ask whether the transfer failure is a property of the predictors or of the estimator. Holding the model fixed is right for the internal comparisons and does not license the external claim, so three further estimators were run over the same twenty directions and the same five within-region folds.

| Estimator                                                            | Within-region AUC | Within increment | Transfer AUC | Transfer increment | Above chance |
|----------------------------------------------------------------------|-------------------|------------------|--------------|--------------------|--------------|
| Random<br>for-<br>est,<br>depth<br>un-<br>lim-<br>ited,<br>leaf<br>3 | 0.888             | +0.099           | 0.541        | +0.004             | 14 of 20     |
| Random<br>for-<br>est,<br>depth<br>6,<br>leaf<br>50                  | 0.829             | +0.055           | <b>0.556</b> | −0.007             | 14 of 20     |
| Random<br>for-<br>est,<br>leaf<br>200                                | 0.805             | +0.047           | <b>0.550</b> | −0.021             | 14 of 20     |
| Penalised<br>lo-<br>gis-<br>tic<br>re-<br>gres-<br>sion              | 0.741             | +0.045           | <b>0.510</b> | −0.024             | 14 of 20     |

**Regularisation does not rescue transfer.** All four estimators land

between 0.510 and 0.556, all four put exactly fourteen of twenty directions  
1015 above chance, and the linear model, the one built to extrapolate, transfers  
worst. Within-region skill falls as capacity is reduced, from 0.888 to 0.741,  
which is what regularisation is expected to cost; transfer does not rise to  
meet it. The thermal block’s contribution to transfer is +0.004 under the  
canonical forest and **negative** under all three regularised alternatives, so the  
1020 block does not become portable when the model is made simpler.

The transfer failure is therefore a property of the predictors on this cohort,  
not of an unregularised forest. Detail in `paper/model_capacity.json`.

#### *Appendix A.2. A(i). The four evaluations of Section 4.3, in full*

Section 4.3 reports four evaluations of the same models as a ladder. The  
1025 per-split and per-scar detail is here, so that the ladder can be checked without  
leaving the manuscript.

**Table A1. Within-region half-split, every split.** Source and target  
positive counts are given because they are unequal, which is the principal  
limit on this arm: a straight cut does not produce two exchangeable halves.  
1030 Two splits are unusable because one half of Manavgat contains no burned  
cells.

| Region             | Axis        | Direction   | Source positives | Target positives | Thermal AUC | Baseline AUC |
|--------------------|-------------|-------------|------------------|------------------|-------------|--------------|
| manavgat 2021      | east-west   | low to high | 700              | 84               | 0.695       | 0.661        |
| manavgat 2021      | east-west   | high to low | 84               | 700              | 0.557       | 0.580        |
| manavgat 2021      | north-south | low to high | 0                | 784              | —           | —            |
| manavgat 2021      | north-south | high to low | 784              | 0                | —           | —            |
| bejis 2022         | east-west   | low to high | 640              | 460              | 0.750       | 0.784        |
| bejis 2022         | east-west   | high to low | 460              | 640              | 0.578       | 0.560        |
| bejis 2022         | north-south | low to high | 357              | 743              | 0.713       | 0.662        |
| bejis 2022         | north-south | high to low | 743              | 357              | 0.603       | 0.613        |
| mugla 2021         | east-west   | low to high | 1,449            | 1,462            | 0.448       | 0.378        |
| mugla 2021         | east-west   | high to low | 1,462            | 1,449            | 0.636       | 0.628        |
| mugla 2021         | north-south | low to high | 774              | 2,137            | 0.533       | 0.503        |
| mugla 2021         | north-south | high to low | 2,137            | 774              | 0.294       | 0.549        |
| evia 2021 extended | east-west   | low to high | 2,052            | 612              | 0.589       | 0.506        |
| evia 2021 extended | east-west   | high to low | 612              | 2,052            | 0.645       | 0.482        |
| evia 2021 extended | north-south | low to high | 2,564            | 100              | 0.649       | 0.513        |
| evia 2021 extended | north-south | high to low | 100              | 2,564            | 0.533       | 0.495        |
| montiferru 2021    | east-west   | low to high | 410              | 129              | 0.475       | 0.461        |
| montiferru 2021    | east-west   | high to low | 129              | 410              | 0.507       | 0.513        |
| montiferru 2021    | north-south | low to high | 438              | 101              | 0.583       | 0.511        |
| montiferru 2021    | north-south | high to low | 101              | 438              | 0.536       | 0.447        |

**Table A2. Leave-one-scar-out at a 2 km buffer, every scar.** Muğla

1035 and Montiferru are the only regions containing more than one burned component of at least 50 cells. Muğla is the only region where holding one out still leaves the source model properly trained across every arm; Montiferru’s two components are very unequal, so one of its arms retains 472 source positives and the other 97. Muğla’s four arms mean 0.579; the four arms in the other  
1040 regions mean 0.525, and the pooled figure of 0.552 averages the two.

| Region             | Component | Source            | Target positives | Target cells | AUC   |
|--------------------|-----------|-------------------|------------------|--------------|-------|
|                    |           | positives<br>left |                  |              |       |
| evia 2021 extended | 1         | 11                | 2,653            | 3,059        | 0.465 |
| manavgat 2021      | 1         | 88                | 696              | 1,135        | 0.592 |
| montiferru 2021    | 1         | 97                | 442              | 758          | 0.584 |
| montiferru 2021    | 5         | 472               | 67               | 195          | 0.458 |
| mugla 2021         | 6         | 1,997             | 914              | 1,266        | 0.595 |
| mugla 2021         | 1         | 2,173             | 738              | 1,244        | 0.561 |
| mugla 2021         | 10        | 2,272             | 639              | 940          | 0.616 |
| mugla 2021         | 8         | 2,363             | 548              | 954          | 0.542 |

The evaluation populations of the two arms are not comparable with each other or with the transfer targets. A held-out scar with its 2 km collar contains only fire-adjacent negatives, against a whole target region’s inclusion of its easy far field; the burned fractions, 34 to 87 % against 3.8 to 28.7 %, are a symptom of that rather than the cause, since ROC-AUC is invariant to class balance at fixed class-conditional distributions. The scar arm therefore asks for discrimination against the nearest and most similar negatives only, while a transfer arm includes the whole easy far field. Section 4.3 states the consequence: the last three rows of the ladder are not distinguishable by this design.

**Table A3. The foreign-region arm, decomposed by source.** Each held-out scar area is scored with a model fitted on each of the other four regions. Table 2’s row D is the mean over the **eight** scars that carry a row C, that is 32 of the 36 combinations below; the nine-scar mean quoted in this appendix is 0.559 against row D’s 0.555.

| Target region   | Scar | Mean over sources | Min   | Max   | Spread |
|-----------------|------|-------------------|-------|-------|--------|
| Bejís 2022      | 1    | 0.589             | 0.521 | 0.631 | 0.110  |
| North Evia 2021 | 1    | 0.590             | 0.374 | 0.704 | 0.329  |
| Manavgat 2021   | 1    | 0.468             | 0.407 | 0.563 | 0.156  |
| Montiferru 2021 | 1    | 0.504             | 0.457 | 0.546 | 0.089  |
| Montiferru 2021 | 5    | 0.490             | 0.399 | 0.656 | 0.257  |
| Muğla 2021      | 1    | 0.518             | 0.425 | 0.608 | 0.183  |
| Muğla 2021      | 6    | 0.568             | 0.533 | 0.597 | 0.064  |
| Muğla 2021      | 8    | 0.632             | 0.570 | 0.724 | 0.155  |
| Muğla 2021      | 10   | 0.671             | 0.579 | 0.716 | 0.137  |

Over all 36 combinations here the mean is 0.559, the range 0.374 to 0.724, and ten fall below chance. Restricted to the 32 combinations behind row D the mean is 0.555 and the same ten fall below chance. The mean spread across the four sources for a single scar is 0.164 over the nine scars and 0.171 over the eight. Averaging over sources is what makes row D comparable with row C, which is fitted on one region; it is not a claim that the choice of foreign source is immaterial, and Section 4.3 states the distinction.

**Table A4. Prevalence is not the cause of the evaluation-area effect.** The same fitted model and the same out-of-fold predictions are scored three ways: on the whole region, on a random sample of region cells drawn at the scar area’s own burned fraction, and on the scar area. Twenty draws per scar, seed 42. Nine scars, since this control needs no leave-one-scar-out arm and Bejís therefore qualifies.

| Target region   | Scar | Burned fraction | A whole region | A'           | B scar area  |
|-----------------|------|-----------------|----------------|--------------|--------------|
|                 |      |                 |                | prevalence-  |              |
|                 |      |                 |                | matched      |              |
| Manavgat 2021   | 1    | 0.61            | 0.797          | 0.796        | 0.589        |
| Bejís 2022      | 1    | 0.66            | 0.824          | 0.825        | 0.575        |
| Muğla 2021      | 6    | 0.72            | 0.777          | 0.772        | 0.564        |
| Muğla 2021      | 1    | 0.59            | 0.777          | 0.779        | 0.654        |
| Muğla 2021      | 10   | 0.68            | 0.777          | 0.779        | 0.673        |
| Muğla 2021      | 8    | 0.57            | 0.777          | 0.782        | 0.761        |
| North Evia 2021 | 1    | 0.87            | 0.864          | 0.866        | 0.745        |
| Montiferru 2021 | 1    | 0.58            | 0.720          | 0.717        | 0.622        |
| Montiferru 2021 | 5    | 0.34            | 0.720          | 0.723        | 0.461        |
| <b>Mean</b>     |      |                 | <b>0.782</b>   | <b>0.782</b> | <b>0.627</b> |

A minus A', the effect of prevalence alone, is **−0.000** [**−0.003**, **+0.002**]. A' minus B, the effect of replacing the region's negatives with fire-adjacent ones, is **+0.155** [**+0.093**, **+0.217**]. The evaluation-area effect is entirely  
1075 the negative pool.

### *Appendix A.3. A(j). The transfer-gap decomposition, in full*

Section 4.5 states the two figures this table carries — at most 34 % of the gap recovered in the directions that started below chance, and seven direc-  
1080 tions with negative recovery. The table is kept here rather than in the body because Section 4.3 shows the within-region reference in its denominator is not matched to a transfer evaluation, and Section 4.4 shows its raw column is frame-dependent, so the fractions should be read as within-protocol quantities.

**Table A5. Transfer-gap decomposition (four-AOI set, 12 direc-**  
1085 **tions).** Within = target's within-region thermal AUC; best adapted = the

better of z-score/CORAL; recovered fraction = (adapted - raw)/(within  
 - raw), signed and unclipped, with paired bootstrap CI (1000 replicates).  
 Montiferru directions are not part of this decomposition (per-pair absolute  
 decompositions exist without fraction CIs). The status column asks whether  
 1090 the *adapted* value clears chance and uses the 2-cell adapted intervals of Ta-  
 ble B9. The adapted arms were not recomputed at the coarser blocking of  
 Appendix A(c), which covers the raw arm and the paired delta only.

| Direction      | Within | Raw   | Best adapted<br>(method) | Recovered<br>fraction [CI] | Status                              |
|----------------|--------|-------|--------------------------|----------------------------|-------------------------------------|
| Manavgat→Bejís | 0.918  | 0.326 | 0.511<br>(CORAL)         | +0.31 [+0.28,<br>+0.34]    | recovery,<br>chance not<br>excluded |
| Bejís→Manavgat | 0.870  | 0.444 | 0.555<br>(CORAL)         | +0.26 [+0.18,<br>+0.34]    | recovery<br>above chance            |
| Muğla→Manavgat | 0.870  | 0.401 | 0.560<br>(CORAL)         | +0.34 [+0.28,<br>+0.40]    | recovery<br>above chance            |
| Bejís→Evia     | 0.912  | 0.383 | 0.532<br>(z-score)       | +0.28 [+0.23,<br>+0.32]    | recovery<br>above chance            |
| Evia→Bejís     | 0.918  | 0.448 | 0.549<br>(z-score)       | +0.22 [+0.16,<br>+0.27]    | recovery<br>above chance            |
| Manavgat→Muğla | 0.859  | 0.470 | 0.443<br>(CORAL)         | −0.07 [−0.12,<br>−0.03]    | <b>negative</b><br><b>recovery</b>  |
| Muğla→Bejís    | 0.918  | 0.583 | 0.560<br>(CORAL)         | −0.07 [−0.16,<br>+0.01]    | <b>negative</b><br><b>recovery</b>  |
| Manavgat→Evia  | 0.912  | 0.613 | 0.542<br>(z-score)       | −0.23 [−0.32,<br>−0.14]    | <b>negative</b><br><b>recovery</b>  |
| Evia→Manavgat  | 0.870  | 0.686 | 0.527<br>(CORAL)         | −0.86 [−1.18,<br>−0.60]    | <b>negative</b><br><b>recovery</b>  |
| Evia→Muğla     | 0.859  | 0.577 | 0.530<br>(CORAL)         | −0.17 [−0.22,<br>−0.11]    | <b>negative</b><br><b>recovery</b>  |
| Muğla→Evia     | 0.912  | 0.653 | 0.563<br>(CORAL)         | −0.35 [−0.43,<br>−0.27]    | <b>negative</b><br><b>recovery</b>  |
| Bejís→Muğla    | 0.859  | 0.618 | 0.518<br>(z-score)       | −0.42 [−0.51,<br>−0.34]    | <b>negative</b><br><b>recovery</b>  |

#### Appendix A.4. $A(k)$ . The thermal sign, stratified

1095

Section 4.4 reports that all five regions agree on a negative association between pre-fire surface temperature and burning, and that neither of the

two obvious confounders explains it. The per-region values are here. Signed AUC against **burned** within the 10 km collar; stratified columns pool within-stratum concordance over deciles of the named variable, weighting each stratum by its positive-negative pair count. Source `matched_frame_gap.csv`, recomputable by `paper/code/verify_matched_gap.py`.

| Region     | LST raw | LST within elevation | LST within NDVI | LST within distance | NDVI raw | NDVI within LST |
|------------|---------|----------------------|-----------------|---------------------|----------|-----------------|
| Manavgat   | 0.386   | 0.402                | 0.440           | <b>0.505</b>        | 0.621    | 0.542           |
| Bejís      | 0.405   | 0.509                | 0.486           | 0.350               | 0.618    | 0.574           |
| Muğla      | 0.332   | 0.363                | 0.404           | 0.367               | 0.652    | 0.547           |
| Evia       | 0.286   | 0.327                | 0.279           | 0.319               | 0.663    | <b>0.380</b>    |
| Montiferru | 0.376   | 0.392                | 0.368           | <b>0.485</b>        | 0.582    | <b>0.405</b>    |

Three readings follow. The LST sign survives stratification within elevation in four of five regions and within greenness in all five, despite  $r(\text{LST}, \text{NDVI})$  reaching  $-0.92$ , so it is neither a lapse-rate proxy nor an inverse-greenness proxy. Within distance to the nearest burned cell it crosses 0.5 in Manavgat, at 0.505, and is attenuated to near-null in Montiferru, at 0.485 from an unstratified 0.376 — the two regions where the collar leaves the least residual gradient — so in those two the agreed sign is largely a weaker version of the same spatial effect the collar was introduced to remove, entirely so in Manavgat. And the reciprocal adjustment runs one way only: NDVI reverses in two regions once temperature is held, while LST reverses in none once greenness is held.

#### Appendix A.5. $A(l)$ . The LST anomaly under the difference instrument

Section 4.4 reports that no reversal meets this paper’s strict criterion once evaluation frames are equalised, and that the LST anomaly nonetheless differs between regions on the weaker instrument Table B3’s note commits

the paper to. These are the four pairs that have opposite-sided point estimates and a difference interval excluding zero, all on `lst_anomaly_mean`. Signed AUC within the 10 km collar; the two regions are bootstrapped independently under the 10-cell spatial-block scheme and differenced, 1000 replicates, seed 42. Source `matched_frame_gap.csv`, recomputable by `paper/code/verify_matched_gap.py`.

**Table A6. Between-region differences in the signed LST-anomaly association, 10 km collar.**

| Pair               | AUC A | AUC B | Difference | 95 % CI          |
|--------------------|-------|-------|------------|------------------|
| Bejís vs Evia      | 0.392 | 0.584 | −0.191     | [−0.295, −0.083] |
| Evia vs Montiferru | 0.584 | 0.400 | +0.184     | [+0.012, +0.321] |
| Manavgat vs Evia   | 0.462 | 0.584 | −0.122     | [−0.225, −0.020] |
| Bejís vs Muğla     | 0.392 | 0.507 | −0.115     | [−0.219, −0.002] |

Nine of the ninety feature-by-pair differences clear zero, against about 4.5 expected at nominal 5 % under the null, and no multiplicity correction is applied; four of the nine are this feature and three of those involve Evia. The nine features are effectively two to three dimensions (Section 4.4), so the ninety comparisons are not independent either. This is reported as a weaker result than a reversal, not as a restored one.

#### *Appendix A.6. $A(m)$ . The same-geography event pair*

Appendix A(m) states the result and its withdrawal; the design, the two structural asymmetries that have no analogue in the twenty-direction matrix, and the direction of the bias they impose are here.

This arm was designed to hold place fixed and vary only the fire, which would have separated regional concept shift from everything that differs be-

1140 tween study areas. Muğla burned twice, in 2021 and again eleven months later, on the same grid and through the same processing chain, and the 2022 arm is the 2021 population with the 2021 scar removed: 41,730 rows / 2,911 burned for 2021 against 38,790 rows / 331 burned for 2022. **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm,**  
1145 and eleven is below the sixteen this design sets as its own floor, so the 2022 intervals are read as indicative exactly as the 20-cell row of Table 1 is. Full per-feature values are in Appendix B, Table B5.

**On the frame as drawn, elevation reverses with bootstrap support.** In 2021 higher ground burned preferentially, at 0.611 [0.532, 0.690];  
1150 in 2022 lower ground did, at 0.296 [0.230, 0.355]. The intervals are disjoint and the difference is  $-0.317$  [ $-0.414$ ,  $-0.220$ ]: the same predictor, region and grid, and an association pointing the opposite way in two fires eleven months apart. The four absolute thermal channels change side as well.

**Section 4.4 has already withdrawn it.** The 2022 arm is one compact  
1155 scar inside the whole Muğla box, so 93.2 % of its cells lie beyond 10 km of any burned cell against 55.3 % for the 2021 arm — the most extreme far field in the cohort. Under the same 10 km collar the 2021 figure barely moves, 0.611 to 0.606 and still supported, while the 2022 figure moves from 0.297 to 0.565, onto the same side of 0.5 as 2021 with an interval covering chance.  
1160 The arm shows the same artefact as the cross-region reversals rather than confirming them. That verdict covers **elevation**, and slope, which behaves the same way; the arm’s thermal channels carry no verdict at all, because the 2022 event’s own predictors are not exported and the reconstruction used here is valid only for the year-invariant ones (Appendix A(o)). We keep the

1165 section because the structural properties below have no analogue elsewhere  
in the paper, and because this arm is what motivated the frame test.

The two arms are not disjoint samples: they share 38,789 of the 2022  
arm’s 38,790 cells, and elevation, slope and land cover are identical to the  
digit across all 73,098 grid cells, so only NDVI and the six thermal channels  
1170 carry new information between them. And the removal is the target’s own  
positive class, so in the 2022-to-2021 direction not one target positive is in  
the source training population while 38,789 of 38,819 target negatives are,  
and membership of the source training set alone separates the 2021 target’s  
classes at ROC-AUC 0.9996. **We therefore cannot bound what that**  
1175 **asymmetry does to a transfer estimate between the two arms.** The  
known part of the bias runs the safe way: the removed cells are high, with  
a median of 563 m against the population’s lower centre, so removing them  
strips high negatives from the 2022 arm and makes the observed reversal  
smaller rather than larger.

1180 *Appendix A.7. A(n). The two interventions, in full*

Appendix A(n) states both results; the four feature-set configurations,  
the per-region deltas and the pooled per-target shortfalls are here.

**(a) Pooled multi-region training** (Fig. A.1). At the point estimate,  
training on the pooled primary populations of the other four regions never  
1185 beats the best single-source transfer for any target, with shortfalls of 0.02 to  
0.22, and it stays 0.28 to 0.50 AUC below the within-region ceiling; for two  
targets it falls below the pairwise mean and below chance, though only Bejís  
is below chance with interval support, at 0.417 [0.369, 0.467]. Aggregation  
does not recover what single-source transfer loses.

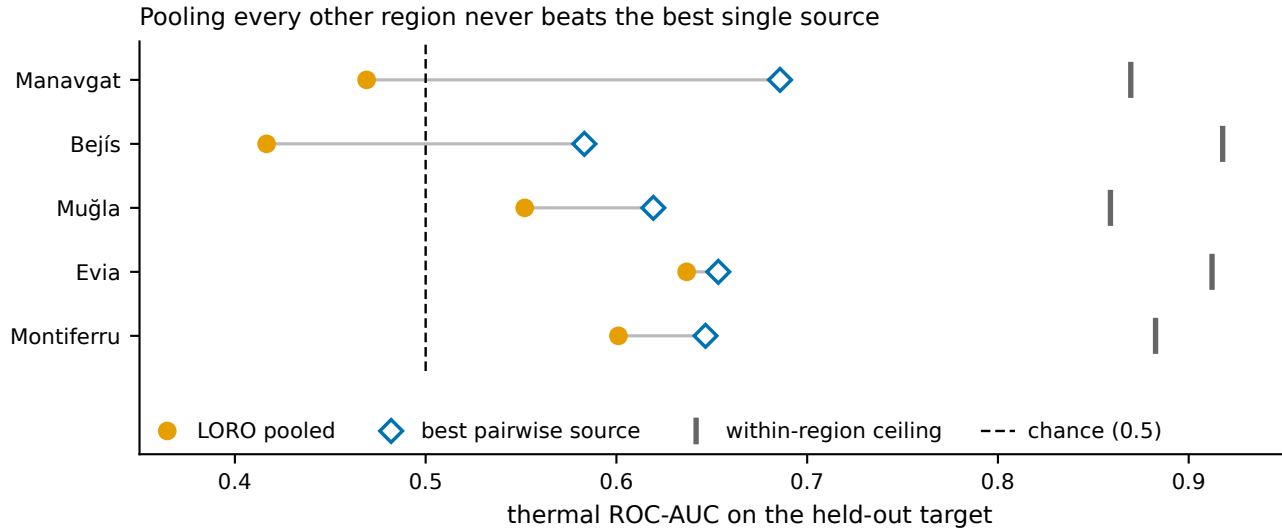


Figure A.1: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

1190 (b) **Removing the direction-reversing features.** The two predictors whose signed association reverses between regions **with bootstrap support on the frames as drawn** are **elevation\_mean** and **lst\_anomaly\_mean**. Section 4.4 withdraws that support under an equalised frame, so this selection rule is frame-dependent and the arm below measures what the removal costs under the original protocol, not a consequence of an established reversal. The per-region signed associations and the three pair-level reversals that meet that criterion are in Appendix B, Tables B2 and B3. Retraining without them costs **−0.081** of mean within-region AUC, supported in every region (per-region deltas  $-0.060$ ,  $-0.130$ ,  $-0.074$ ,  $-0.063$ ,  $-0.079$ , every interval entirely below zero), and changes mean transfer by **+0.014** [**−0.017**, **+0.045**], which spans zero.

Two qualifications belong with those numbers rather than after them. First, the debit is not the thermal block’s. Dropping elevation alone accounts for  $-0.061$  of it, from 0.888 to 0.827. Dropping the LST anomaly alone 1205 accounts for  $-0.013$ , from 0.888 to 0.875. Roughly three quarters of the cost is therefore the removal of a **baseline** terrain variable, not of a thermal channel. Second, both figures are post-selection. The two predictors were chosen because they reverse, using the same data on which the  $-0.081$  and the  $+0.014$  are then estimated. No correction for that selection is applied.

| Configuration                   | Mean within-region AUC | Mean transfer AUC |
|---------------------------------|------------------------|-------------------|
| full                            | 0.888                  | 0.541             |
| 1210 drop <b>elevation_mean</b> | 0.827                  | 0.546             |
| drop <b>lst_anomaly_mean</b>    | 0.875                  | 0.544             |
| drop both                       | 0.807                  | 0.556             |

Fig. A.2 shows the four configurations together. What this measures is a local cost with no compensating transfer gain. It does not measure an exchange, because the transfer side is a null on both arms. The thermal block’s own contribution to transfer is +0.004, with an interval spanning zero (Section 4.5). Removing the reversing predictors returns +0.014, with an interval spanning zero. Two nulls on the portability axis are not a price paid.

*Appendix A.8. A(o). The thermal sign and the anomaly difference, in full*

Section 4.4 states both results; the per-region values, the reciprocal stratification and the difference-instrument argument are here.

**A weaker instrument does support the anomaly result.** Table B3’s note commits this paper to a difference interval on the pair as the sharper test, and it is applied here. Bootstrapping the two regions independently under the collar and differencing, four pairs have opposite-sided point estimates **and** a difference interval excluding zero, all on `1st_anomaly_mean` and three of the four involving Evia — the largest being Bejís against Evia at  $-0.191 [-0.295, -0.083]$  (Appendix A(1), Table A6). The anomaly is the one channel the frame-and-terrain mechanism of this section cannot explain, being decorrelated from elevation. Three caveats keep it weak and all three are stated rather than buried: nine of ninety feature-by-pair differences clear zero against about 4.5 expected under the null with no multiplicity correction; the nine features are effectively two to three dimensions, as the collinearity paragraph below shows; and Evia’s own interval-support status turns on 0.003. **The honest statement is that no reversal meets this paper’s strict criterion under the collar, and that the LST anomaly differs be-**

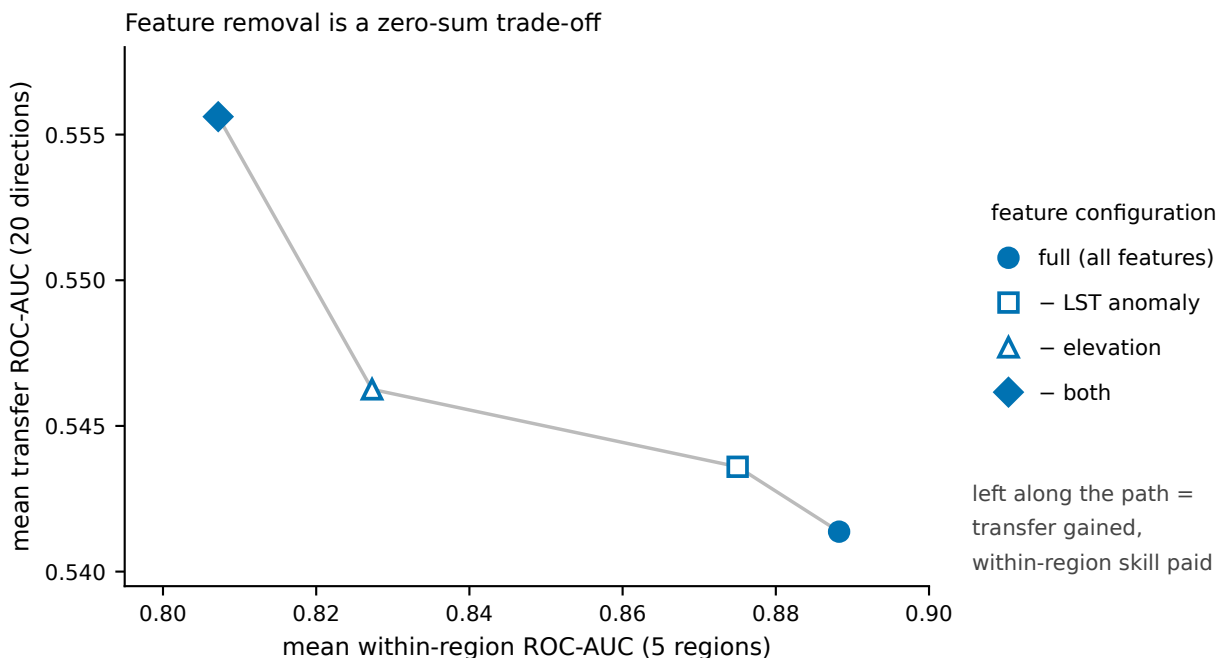


Figure A.2: **Removing the direction-reversing features costs locally and returns nothing measurable on transfer.** Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically ( $0.541 \rightarrow 0.556$ ) while within-region skill falls monotonically ( $0.888 \rightarrow 0.807$ ). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The within-region cost is 0.081 AUC and is supported in every region. The change in transfer is +0.014 AUC on an interval that spans zero, so no exchange between the two is demonstrated: a local cost is measured and no compensating transfer gain is.



case in the cohort, and it does not survive either. A fixed study area does not mean a fixed evaluation frame: its 2022 arm has **93.2 % of cells beyond 10 km of any burned cell, median 43.6 km**, a larger far field than any cross-region arm. Applying the same collar (`mugla_two_event_collar.csv`; the full-frame values reproduce Table B5 on the year-invariant channels only, for the reason given below):

**What this arm can and cannot evaluate.** The 2022 event's own step8a export is not in this tree. The released script (`code/verify_mugla_collar.py`) reconstructs the arm by taking the **2021** predictor file and substituting the 2022 burned mask, justified on the ground that elevation, slope and land cover are identical between the arms. That justification holds for those channels and **not for the state-dependent ones**, which the script nonetheless computes. Checked against the frozen step9g export, which uses the 2022 arm's own predictors, the reconstruction agrees on the year-invariant channels — elevation 0.297 against 0.296, slope 0.559 against 0.558 — and diverges on every seasonal one, by +0.136 on `lst_anomaly_mean`, +0.092 on `tvdi_difference_mean`, -0.078 on `ndvi_mean` and -0.025 on `current_tvdi_mean`. **So only elevation and slope can be given a collar verdict for this arm**, and the thermal channels cannot until the 2022 arm's own predictors are exported.

| Feature        | 2021, full           | 2022, full           | full      | 2021, collar         | 2022, collar         | collar |
|----------------|----------------------|----------------------|-----------|----------------------|----------------------|--------|
|                |                      |                      | ver-      |                      |                      | ver-   |
|                |                      |                      | dict      |                      |                      | dict   |
| elevation_mean | 0.529 [0.529, 0.692] | 0.297 [0.229, 0.363] | supported | 0.564 [0.525, 0.685] | 0.565 [0.450, 0.677] | none   |
|                |                      |                      | re-       |                      |                      |        |
|                |                      |                      | ver-      |                      |                      |        |
|                |                      |                      | sal       |                      |                      |        |
| slope_mean     | 0.637 [0.584, 0.689] | 0.559 [0.463, 0.643] | none      | 0.635 [0.578, 0.692] | 0.457 [0.356, 0.548] | none   |
| the —          | —                    | —                    | —         | —                    | —                    | not    |
| five           |                      |                      |           |                      |                      | evalu- |
| state-         |                      |                      |           |                      |                      | able   |
| dependent      |                      |                      |           |                      |                      | here   |
| chan-          |                      |                      |           |                      |                      |        |
| nels           |                      |                      |           |                      |                      |        |

On the two channels this arm can evaluate, the collar removes the reversal. Elevation’s 2022 arm crosses to the same side of 0.5 with an interval covering chance, while the 2021 arm is nearly frame-invariant, which isolates the effect to the 2022 arm’s far field rather than to the collar. Slope reverses on neither frame.

Two corrections to earlier statements belong here. First, an earlier version generalised the elevation row to "no bootstrap-supported sign reversal survives anywhere in this paper once evaluation frames are equalised". **That generalisation is withdrawn:** it was computed for one feature and the other five could not have supported it either way. Second, a subsequent version reported `current_lst_mean` and `current_tvdi_mean` as supported reversals appearing under the collar. **Those are withdrawn too** — they are the 2021 pre-fire surface scored against the 2022 scar, not the 2022 fire’s own state. The defensible statement is narrower than either: **on the channels this design can evaluate, the collar removes this arm’s reversal; on the rest the arm is silent.** Whatever it shows rests on eleven positive-

carrying blocks against this design's own floor of sixteen, so no verdict from it is strong.

1300 Two points bound how many independent reversals could have been counted in the first place. The channels whose reversal vanishes are the ones partly measuring terrain: current LST correlates with elevation at  $-0.695$  to  $-0.125$  across the regions and TVDI at  $-0.722$  to  $-0.298$ , so their proxy strength varies fivefold and a frame that shifts the elevation distribution shifts them  
1305 with it. And within the collar `fused_lst_mean` correlates with `current_lst_mean` at 0.99 to 1.00, `downscaled_lst_mean` at 0.97 to 0.99 and `current_tvdi_mean` at 0.87 to 0.98, so the four absolute channels are close to one axis and the two differenced channels correlate at 0.64 to 0.94: the "five of nine directions reverse" count of Appendix A(s) counts features, not independent  
1310 quantities, and **in effective dimensions it is closer to two.**

Two scope statements belong with the table. The full-frame values here are not numerically the published ones —  $\rho = +0.86$  over fourteen directions against the  $+0.84$  over sixteen in Section 4.6 — because the support test is itself bootstrap-dependent and at this setting three region pairs carry no  
1315 jointly supported feature rather than two; the argument rests on before-and-after under identical settings, not on the published figure. And **only these diagnostics were recomputed on the collar**: the other eighteen failed on the frames as drawn and were not rerun, so their correlations against the equalised transfer vector are unknown rather than shown to be null.

1320 **The thermal block's paired contribution to transfer, with its interval.** The two matrices were differenced direction by direction. The mean is  $+0.004$ , but the mean is not the informative statistic here. The individ-

ual paired contributions span **-0.148 to +0.133**, with **twelve positive and eight negative** (`baseline_vs_thermal_transfer.csv`). The spread  
1325 is thirty times the mean, and the sign is not a property of the block but of the pair it is asked to cross: the same six predictors that add +0.133 in one direction subtract 0.148 in another. A mean near zero here records cancellation, not consistent absence of effect. The directions are not independent, because each region appears in eight of the twenty. The interval therefore  
1330 depends on what is treated as the resampling unit. All four units the design permits give the same answer:

| Resampling unit                    | n  | 95 % interval on the mean paired contribution |
|------------------------------------|----|-----------------------------------------------|
| Directions, naive                  | 20 | $[-0.027, +0.034]$                            |
| Unordered pairs, cluster bootstrap | 10 | $[-0.028, +0.036]$                            |
| Unordered pairs, t on pair means   | 10 | $[-0.034, +0.042]$                            |
| Regions, leave-one-out jackknife   | 5  | $[-0.037, +0.046]$                            |

Every interval spans zero, and the point estimate is a small fraction of each width. The per-region jackknife shows how little the mean is anchored.  
1335 Holding out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives +0.0148, +0.0021, +0.0065, **-0.0081** and +0.0060. **Dropping Evia alone reverses the sign of the headline.** None of these four units propagates within-direction sampling variability. They resample between directions only.

**In precision terms the transfer is worse than the ROC figures**  
1340 **suggest.** ROC-AUC is the metric used throughout this paper, for comparability with the literature, but a susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity. Across the twenty

directions the thermal model’s PR-AUC averages **0.156 against a no-skill baseline of 0.136**. The mean of the twenty per-direction lifts is 1.16; the  
1345 ratio of the two means just quoted is 1.146. **Six of the twenty fall below their own no-skill baseline at the point estimate, and five of those six have intervals entirely below it.** The exception is Bejís to Manavgat, at 0.034 [0.029, 0.042] against a baseline of 0.038, whose interval covers the baseline and which is therefore a point-estimate case only. Only one direc-  
1350 tion, Evia to Manavgat, exceeds twice its baseline, at 0.094 against 0.038. A transferred model therefore ranks burned cells about a sixth better than random on average, and worse than random in five directions with interval support and a sixth at the point. This is a sharper statement than the ROC means support and it should be the one a practitioner reads.

1355 **The within-region increment survives the same correction, and is reported here because Section 4.2 establishes it on the frames this section calls incomparable.** Running the baseline arm on the collar as well (`collar_increment_and_cosine.csv`, `paper/code/verify_collar_increment.py`), the thermal increment at 5 km blocking is +0.041, +0.030, +0.091,  
1360 +0.134 and +0.090 across the five regions, **positive in all five**, with a mean of +0.077 against +0.086 on the frame as drawn. It does erode as the frame tightens further: at a 5 km collar the mean halves to +0.041, still positive in all five. These are point estimates; the intervals in Table 1 are computed on the frame as drawn. The paper’s one surviving predictor-level positive claim  
1365 therefore holds on the frame it argues is the correct one.

**The five areas of interest are not comparable frames.** Each region is a rectangle drawn around a fire, and the rectangles differ by an order of

magnitude in how much unburnt far field they enclose.

The share of modelled cells lying beyond 10 km of any burned cell is  
1370 60.1 % in Manavgat, 63.1 % in Bejís, 55.3 % in Muğla, 43.7 % in Evia and  
2.1 % in Montiferru, with median distances of 13.4, 13.5, 11.3, 8.0 and 2.7  
km (Appendix B, Table B6). Montiferru's frame is fire-scale; Manavgat's  
and Bejís's are roughly three-fifths far field. The far field is not a neutral  
addition. In Manavgat the median elevation of modelled cells rises from 472  
1375 m within 5 km of the fire to 955 m at 10 to 20 km and 1,273 m at 20 to 50  
km, against 512 m for the burned cells themselves, in Taurus terrain that no  
plausible spread model would place at risk.

#### *Appendix A.9. A(s). The contrast pair, in full*

The clearest single view needs no ranking at all (Fig. A.3). Manavgat  
1380 and Muğla lie in the same country and fire year, 306 km apart by centroid,  
and their burned cells occupy the most similar environmental envelope of any  
pair in the matrix; Bejís and Montiferru occupy the least similar.

Per-quantity values for both pairs, on both frames, are in Appendix B,  
Table B10.

1385 Two readings of this pair do not survive Section 4.4 and are not offered.  
On the frames as drawn five of nine feature-response directions point opposite  
ways, elevation among them; under the collar that figure moves to 0.561  
against 0.606, on the same side of 0.5. And transfer is below chance in both  
directions as drawn but above chance under the collar. **What survives**  
1390 **is the ordinal contrast:** the most similar pair is among the weakest in  
the matrix and the least similar among the stronger, while neither is the  
extreme — the weakest direction is Manavgat to Bejís at 0.417 and the

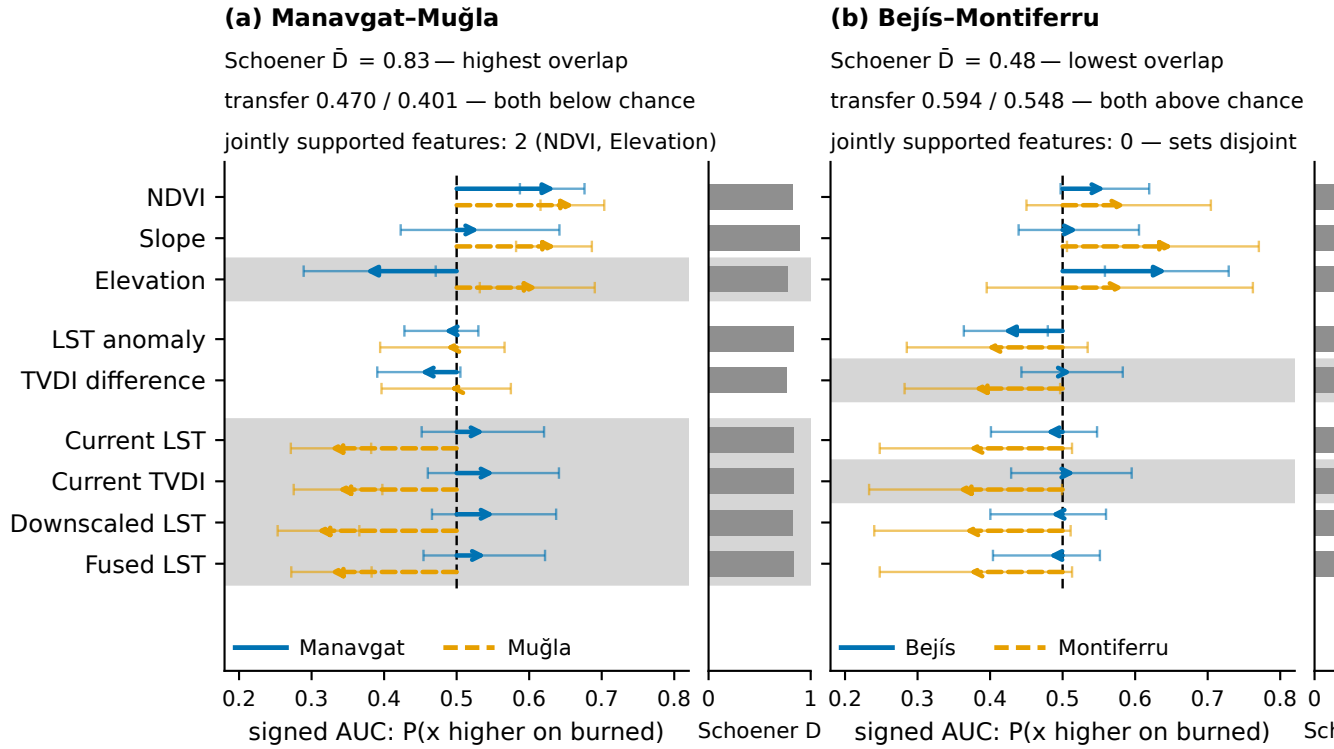


Figure A.3: **Burned-niche overlap does not determine transfer: the contrast pair.** Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block ( $\approx 5$  km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's  $D$  overlap. **(a)** Manavgat–Muğla, the pair with the *highest* burned-niche overlap ( $\bar{D} = 0.83$ ), transfers below chance in both directions on the frames as drawn (0.470 and 0.401); under the equalised frame of Section 4.4 it is 0.551 and 0.510, above chance but still among the weakest in the matrix. **(b)** Bejís–Montiferru, the pair with the *lowest* overlap ( $\bar{D} = 0.48$ ), transfers above chance in both directions (0.594 as drawn, 0.669 equalised, and 0.548 against 0.624). High niche overlap is therefore not sufficient for transfer, which is an ordinal claim on either frame. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which

strongest Muğla to Evia at 0.727. The claim this supports is that high envelope overlap does not buy transfer, not that it produces anti-prediction.

1395 The AoA shares are full-frame quantities.

Bejís and Montiferru sit at the opposite extreme. Their burned envelopes barely overlap, and they carry the most dissimilar values on every overlap measure. They transfer above chance in both directions. That half of the contrast rests on point estimates: neither direction carries a verdict at 5  
1400 km blocking, and the marginal applicability audit was never produced for Montiferru. The claim is one of *sufficiency*. Similarity does not guarantee transfer, and dissimilarity does not preclude it. The claim rests on two coexisting counterexamples, so it does not depend on the number of pairs available.

1405 *Appendix A.10.  $A(v)$ . The sensitivity arms, summarised*

Eight design choices were varied with everything else held fixed. They are the Evia AOI and its prevalence, the CORAL regularisation constant, the blocking scale, the closure date of the predictor window, the quality screening of the coarse thermal input, the contrast between the normalised  
1410 and the absolute dryness channels, the removal of the coordinate-informed channels, and the capacity of the classifier. None changes a conclusion above. Two bound how the results should be read, so they are carried into the main text here.

**The secondary population.** Section 3.5 defines a secondary popula-  
1415 tion of all valid cells, including cropland. The frozen export carries it for the within-region arm in Manavgat and Bejís only, so it is a two-region sensitivity rather than a cohort-wide one, and no transfer quantity is defined on it.

Across the two regions and the three block sizes the thermal increment runs +0.044 to +0.061 with every bootstrap interval above zero, against +0.046  
1420 to +0.067 on the primary population, and the paired difference between populations runs  $-0.008$  to  $+0.011$  with no consistent sign (`experiments/<region>/step10/within_robustness.json, by_population`). The increment is therefore not an artefact of excluding cropland, which is what this arm was run to test; it says nothing about the transfer results, where the  
1425 primary population is the only one available.

Coarsening the blocks from 1 km to 5 km moves the **paired thermal-minus-baseline delta** verdicts from ten positive, seven negative and three uncertain to six, four and ten. The above-chance verdicts on the thermal arm itself, a different quantity, move from twelve, six and two to nine, four  
1430 and seven. Support is removed from seven verdicts and added to none. The point estimates are unchanged, but that is an identity rather than a result. The blocking scale is the resampling unit, and it cannot move an estimate computed once over all target cells.

Manavgat’s whole downstream chain was then rebuilt from a quality-  
1435 screened MODIS input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to  $10.9\text{ }^{\circ}\text{C}$ . No signed univariate association moves by more than  $+0.0003$ . This closes the one processing-artefact candidate for that region’s behaviour. Appendix A reports all eight arms, including one that tests a claim of Section 1.2 and does not uphold it.

#### 1440 *Appendix A.11. $A(t)$ . Distance within a region*

One further arm measures how skill decays with distance inside a single region. A model is fitted on one half of a region and applied to the other, and

target cells are binned by their distance from the training cells (`distance_`  
`curve.md`, `paper/code/distance_curve.py`). Means are unweighted over  
1445 bins, whose positive counts range from 1 to 2,564, and the bins carry no  
intervals.

| Separation from training cells | Bins | Mean target AUC |
|--------------------------------|------|-----------------|
| 0 to 5 km                      | 18   | 0.692           |
| 5 to 10 km                     | 16   | 0.519           |
| 10 to 20 km                    | 11   | 0.499           |
| 20 to 40 km                    | 6    | 0.445           |
| 40 to 80 km                    | 3    | 0.541           |
| 80 to 160 km                   | 1    | 0.421           |
| cross-region, 306 to 2,802 km  | 20   | 0.541           |

By 10 to 20 km inside a single region the model is already at  
**chance**. That bounds how far a susceptibility surface of this kind can be  
1450 carried from the cells it was fitted on, and it is consistent with Section 4.3,  
where withholding a scar and replacing the model with a foreign one cost  
nothing distinguishable.

It does not reframe the paper’s negative result, and a reading  
that it does was considered and rejected. That reading argued that  
1455 because the twenty cross-region directions average 0.541, at or above the  
within-region plateau, they sit on the continuation of the curve and the fail-  
ure needs no regional mechanism — a rule fixed in advance in `positive_`  
`control.md`. Two objections defeat it, and `scar_control.md` records the re-  
framing as withdrawn in full. **The rule cannot fail**: once the curve reaches  
1460 the chance floor, any cross-region mean near 0.5 lies on its continuation by

construction, so the comparison could not have come out otherwise, and a test that cannot fail is not evidence. And **extrapolating an uninformative model does not produce a reliably reversed ranking**: Manavgat to Bejís is below chance with interval support on the frame as drawn and at  
1465 the 10 km collar, which is not what a model that has merely run out of skill returns.

Four limits bound even the descriptive reading. The two distance ranges do not overlap — within-region separations span 2 to 86 km, cross-region separations start at 306 km — so any comparison across the gap extrapolates  
1470 the curve. The far bins are thin, six at 20 to 40 km and one beyond 80 km, so their means should not be read closely and the apparent rise at 40 to 80 km is not evidence of anything. The near bins carry exactly the autocorrelation blocked validation exists to remove, so **0.692 is an upper bound on near-field skill rather than an estimate of it**. And this design separates  
1475 distance from crossing a study-area boundary, but not from the land cover, terrain and fire history that covary with it.

The arm therefore contributes a length scale for the within-region decay, not an attribution. The unit that fails to transfer is not established by this design, and Section 4.3 says so directly.

1480 *Appendix A.12.  $A(u)$ . What target labels cost*

Everything above measures a failure; this prices it. Label-free alignment does not close the residual gap, so the missing resource is information about the target that alignment cannot synthesise, and the direct way to supply it is target labels. A frozen labelled-budget diagnostic answers how many,  
1485 for three regions across all six ordered directions, using one 10-cell ( $\sim 5$  km)

spatial block as the unit of labelling effort and reading recovery against a matched target-only ceiling of 0.777 to 0.824.

Thirty-two labelled blocks recover **85 to 89 % of that ceiling in three of the six directions**, two of which started below chance; 51 to 57 % in two more, the directions where the conditional gap is widest; and 30 % in the sixth. That budget is 7 to 20 % of the target’s natural-vegetation population, which at the 2,700 to 3,000 labelled cells those blocks carry (S1.3) and this grid’s effective cell area of 0.199 to 0.208 km<sup>2</sup> (Appendix C) is roughly **540 to 620 km<sup>2</sup>**, so it is a real answer and not a cheap one. Two properties of the measurement bound it and are stated in the supplement rather than buried: at the top budget the selection pool is nearly exhausted, so the narrow upper-budget intervals reflect saturation rather than precision, and the labelled blocks are drawn from the event being predicted, which is not a resource available before that event burns. The protocol, the full curve and the limits are in Supplement S1.

Two limits belong with the number. At small budgets the same intervention damages the direction that already transfers best without any labels. And six directions in three regions cannot support a general label budget, so none is offered.

**Row D is a mean over four foreign models, and the spread behind it is large.** Decomposing it gives **32** scar-by-source combinations running from 0.374 to 0.724, with **ten below chance** and a mean spread of 0.171 across the four sources for a given scar (Appendix A, Table A3). The pooled 0.555 is therefore not a statement that any foreign model does as well as a same-region one, but that the *average* one does, and the variation it

averages over is exactly the pair-specific instability that the rest of Section 4 is about.

One confound must be stated. Holding out a region's only large scar also removes most of its positives: the source model retains 11 positives in  
1515 Evia, 88 in Manavgat and 97 in Montiferru, against about 2,000 in Muğla, which has four separate scars. Those four non-Muğla arms average 0.525 and the four Muğla arms 0.579. Three of the four are genuinely starved — the fourth, Montiferru's second component, retains 472 source positives — so row C mixes "the fire was withheld" with "almost all the positives  
1520 were withheld". Restricted to Muğla, C is 0.579 and D is 0.597, and the C-to-D comparison still shows nothing. **With eight scars, three of them starved, this design cannot establish a fire-specific residual, only bound it at about 0.18.**

What this establishes is bounded rather than positive. Rows C and D  
1525 both sit close to chance, so the comparison between them has little dynamic range, and neither the fire's identity nor the region boundary is shown to cost anything here. What is measurable is the change of evaluation geometry between rows A and B. Nor does it establish the fire event as the unit, because the held-out patch is defined by the labels, so its identity cannot  
1530 be separated from its location. The within-region half-split, which does see half of the target scar, returns 0.574 against leave-one-scar-out's 0.552, with overlapping intervals.

**The result is robust to how the held-out patch is defined.** Sweeping the two parameters that define it — minimum burned-component size  
1535 over 25, 50, 100 and 200 cells, and connectivity over the 4- and 8-neighbourhoods

— moves the row-C mean between **0.545 and 0.566** across all eight settings, against the 0.552 of the reported configuration, which the sweep reproduces to 0.001 (`scar_definition_sweep.csv`, `paper/code/scar_definition_sweep.py`). The number of qualifying scars falls from eleven to seven as the size threshold rises, which is what moves the mean; connectivity changes it by at most 0.001 at any threshold. The result is also flat across buffers of 2, 5 and 10 km.

**A weaker instrument does support the anomaly result.** Table B3’s note commits this paper to a difference interval on the pair as the sharper test. Applied under the collar, four pairs have opposite-sided point estimates **and** a difference interval excluding zero, all on `lst_anomaly_mean` and three of the four involving Evia (Appendix A(1), Table A6). Three caveats keep it weak: nine of ninety feature-by-pair differences clear zero against about 4.5 expected under the null with no multiplicity correction, the nine features are effectively two to three dimensions, and Evia’s own interval-support status turns on 0.003. **The honest statement is that no reversal meets this paper’s strict criterion under the collar, and that the LST anomaly differs between regions on the weaker difference instrument.** Elevation under the collar is above 0.5 in all five regions but individually supported in only two.

**The sign the five regions agree on is not the one the dryness framing predicts.** In every region a hotter pre-fire surface is associated with **less** burning, and the same holds for TVDI. It is neither a lapse-rate artefact nor greenness acting through fuel load: on mutual adjustment the surviving channel is LST, not NDVI. Two caveats bound it — in two re-

gions it is a residual spatial gradient that disappears when distance to the nearest burned cell is stratified inside the collar, and interval support is not uniform, so "all five agree" is a statement about point estimates. What we can defend is that the absolute thermal channels behave here as **static**  
1565 **land-surface descriptors** rather than as a dryness index, and that the two internally differenced channels carry no consistent cross-region direction at all. Compositing depth was not tested and remains an open alternative. Per-region values are in Appendix A(k); Section 5.2 states the consequence for the moisture-stress motivation of Section 1.2.

1570 *Appendix A.13. A(w). The frame test, elaborated*

Section 4.4 states these results; the paragraphs it condensed are here.

**The five areas of interest are not comparable frames.** Each is a rectangle drawn around a fire, and they differ by an order of magnitude in how much unburnt far field they enclose: the share of modelled cells beyond  
1575 10 km of any burned cell runs from **2.1 %** in Montiferru to **63.1 %** in Bejís (Appendix B, Table B6). That far field is not a neutral addition — in Manavgat the median elevation of modelled cells rises from 472 m within 5 km of the fire to 1,273 m at 20 to 50 km, against 512 m for the burned cells themselves.

1580 **Under an equalised frame the sign reversals of Appendix A(s) do not survive.** Restricting every region to cells within 10 km of any burned cell removes only far-field negatives; every burned cell is at distance zero and is retained at any radius, so the protection against choosing a flattering radius is the sweep below, not the retention of positives. On the equalised  
1585 frame all five regions agree in sign on elevation, on LST and on TVDI, and

both bootstrap-supported elevation reversals of Table B3 disappear. Under this paper’s own criterion from Section 3.10 — each region’s own 10-cell block-bootstrap interval must exclude 0.5 — **no between-region reversal remains bootstrap-supported**. The qualifier matters: this arm covers  
1590 the five regions, and the same-geography two-fire arm is treated separately in Appendix A(o), where only the year-invariant channels can be given a verdict at all. Two features straddle 0.5 at the point estimate and neither is supported: `lst_anomaly_mean` at 0.392 [0.324, 0.460] in Bejís against 0.584 [0.497, 0.669] in Evia, where Evia’s interval includes 0.5 by 0.003; and `tvdi_difference_mean`  
1595 at 0.509 [0.417, 0.597] in Muğla against 0.384 [0.288, 0.504] in Montiferru, where no region’s interval excludes 0.5. These are the two internally differenced channels. An earlier draft treated the first as uniquely informative because it carries no lapse-rate signal; that does not hold, since the second is equally decorrelated from elevation and behaves the same way.

1600 **A weaker instrument does support the anomaly result.** Table B3’s note commits this paper to a difference interval on the pair as the sharper test; applied under the collar, four pairs have opposite-sided point estimates **and** a difference interval excluding zero, all on `lst_anomaly_mean` and three of the four involving Evia. Three caveats keep it weak: nine of  
1605 ninety feature-by-pair differences clear zero against about 4.5 expected under the null with no multiplicity correction, the nine features are effectively two to three dimensions, and Evia’s support status turns on 0.003. **The honest statement is that no reversal meets this paper’s strict criterion under the collar, and that the LST anomaly differs between regions on the weaker instrument** (Appendix A(l)).  
1610

The sign the five regions now agree on is not the one the dryness  
 framing predicts. In every region a hotter pre-fire surface is associated  
 with less burning, and the same holds for TVDI. It is neither a lapse-rate  
 artefact nor greenness acting through fuel load: on mutual adjustment the  
 surviving channel is LST, not NDVI. Two caveats bound it — in two regions  
 it is a residual spatial gradient that disappears when distance to the nearest  
 burned cell is stratified inside the collar, and interval support is not uniform,  
 so "all five agree" is a statement about point estimates. The absolute thermal  
 channels therefore behave here as **static land-surface descriptors** rather  
 than as a dryness index, and the two internally differenced channels carry no  
 consistent cross-region direction at all. Compositing depth was not tested  
 and remains an open alternative. Per-region values are in Appendix A(k);  
 Section 5.2 states the consequence for the motivation of Section 1.2.

The same test determines what the diagnostics of Section 4.6  
 can establish. The only two candidates there with intervals excluding zero  
 measure agreement in the sign of each predictor's association between source  
 and target, counted over features interval-supported in both. Those are built  
 out of exactly the signed AUCs this section has shown to be frame artefacts,  
 and in Section 4.6 they are correlated against transfer measured on the same  
 unequal frames. Recomputing both sides under the 10 km collar gives Table 3  
 of the body (Section 4.4), which is not repeated here.

The two fail differently and both fail. Once frames are equalised every  
 region pair agrees in sign on every jointly supported feature, so the agreement  
 fraction has no variance left. The supported cosine keeps a trace of variance  
 — unanimous signs fix directions but not magnitudes — and simply stops

tracking transfer. This is not a marginal shift: features supported in both regions *rise* from 1.20 to 3.40 per direction, so the diagnostics are better determined and unanimous. The disagreements they were reading were the far fields.

1640 Two scope statements belong with Table 3 and are given in Appendix A(o): the full-frame values here are not numerically the published ones, because the support test is itself bootstrap-dependent; and only these diagnostics were recomputed on the collar, so the other eighteen are unknown against the equalised transfer vector rather than shown to be null.

1645 This settles what Contribution 3 can claim. It is not that conditional similarity orders transfer where marginal similarity fails; it is that **no diagnostic tested here has been shown to order transfer**. The eighteen marginal, niche and regime candidates do not order it on the frames as drawn, and the two conditional variants that do stop doing so once the frames are  
1650 comparable, because what they were reading is how the study rectangles were drawn. The practitioner’s position is therefore worse than the as-drawn analysis suggests, not better: there is no screen, and the apparent exception is an artefact. Section 4.6 reports its numbers as computed under the pre-registered protocol, which is what a reader following that protocol would  
1655 obtain.

The same-geography arm of Appendix A(m) does not survive either, and it is the most extreme case in the cohort. A fixed study area is not a fixed evaluation frame: its 2022 arm has 93.2 % of cells beyond 10 km of any burned cell against 55.3 % for 2021. Under the same collar the  
1660 2021 elevation figure barely moves while the 2022 figure crosses to the same

side of 0.5 with an interval covering chance, so the elevation reversal does not survive equalisation. **This does not generalise across features:** on the same arm two absolute thermal channels become supported reversals under the collar that were not reversals as drawn, so equalisation moves this arm's reversal rather than removing it. The seven-feature table is in Appendix A(o); values in Appendix A(m).

**The within-region increment survives the same correction,** remaining positive in all five regions on the equalised frame with a mean of +0.077 against +0.086 as drawn, and halving but staying positive in all five at a 5 km collar. The paper's one surviving predictor-level positive claim therefore holds on the frame it argues is the correct one. Per-region values are in Appendix A(o).

**The data-provenance correction.** Muğla's 500 m modelling dataset at the canonical path had been overwritten by a quality-screening rebuild after the frozen tables were computed, so this section's scripts had read a version of that one region differing from Table 1 and B9 in `downscaled_1st_mean` and `fused_1st_mean`. The two files agree on every other column; the proof that the difference is the data and not the fitting is that both give a within-region baseline of 0.6980 at 10-cell blocking while the thermal arm gives 0.7834 against the frozen 0.7773. Two regions' files carry the rebuild's timestamp, Manavgat's and Muğla's, so both were replaced; the distinction that matters is whether the replacement agrees with what was published. Manavgat's does, reproducing its Table 1 row to the printed precision at 0.748 and 0.797, as does the rebuild's own reported increment interval. Muğla's does not. Bejís, Evia and Montiferru were not touched. Every

arm of Section 4.4 was therefore re-run with Muğla read from the surviving frozen copy and the other four regions as they stand, which is the configuration Table 1 and B9 were computed under. The correction moves forty of the hundred per-direction transfer values by up to 0.022, the largest being Muğla  
1690 to Montiferru at  $0.509 \rightarrow 0.531$ . It leaves every headline quantity of Table 4 unchanged to within 0.0012, the largest being the full-to-10 km paired delta at  $+0.0021$  against  $+0.0033$ : the equalised mean 0.6163 against 0.6166, its paired delta  $+0.0231$  against  $+0.0234$ , and the above-chance counts identical in four rows of five, the 5 km row moving from 18 to 19 as one direction crosses  
1695 0.5. The signed-AUC results are unaffected in substance — only Muğla’s two channels move, by at most 0.008, and the same two features straddle 0.5 on each frame. Sources `aoi_frame_transfer_frozen_mugla.csv`, `aoi_frame_auc_frozen_mugla.csv`, `frozen_mugla_recheck.json`; code `code/frozen_mugla_verify_aoi_transfer.py`.

1700 **The transfer matrix moves as well.** Restricting source and target to the same collar gives Table 4 of the body (Section 4.4), which is not repeated here.

The reference arm reproduces the frozen matrix, at 0.541 against Table B9’s 0.541 and 14 of 20 exactly, so this is measuring the same quantity.  
1705 **The baseline control moves with it and must be restated on this frame:** the static baseline transfers at 0.593 against the thermal model’s 0.616, a paired difference of  $+0.023$  rather than the  $+0.004$  of the frame as drawn. The control still holds in kind — the static predictor class is not the portable one either — but the gap between them is about six times larger once  
1710 frames are comparable, and Sections 1, 5 and 6 quote only the as-drawn pair.

Table 4’s above- and below-chance counts are point counts. Under the same 10-cell block bootstrap used for Table B9, at 1000 replicates, the full frame gives nine directions above chance and four below with interval support, and the collar frame fifteen above and one below, so the headline movement is six  
1715 to one at the point estimate and **four to one with interval support** (`aoi_frame_transfer.csv`, which carries the per-direction bounds). The largest movers are Bejís to Evia, 0.383 to 0.602, and Bejís to Manavgat, 0.440 to 0.601.

**What this settles, and what it leaves standing.** Five quantities re-  
1720 ported in the sections that follow are properties of the frames rather than of the predictor-burning relationship, and are identified as such where they appear: the count of six anti-predictive directions, which becomes one; the sign reversal of elevation, LST and TVDI as a mechanism; the sign-agreement diagnostic of Section 4.6; the same-geography arm of Appendix A(m); and the  
1725 paired thermal contribution, which is +0.004 as drawn and +0.023 equalised. What the correction leaves standing is the central negative result, and its size must be stated on a matched comparison. Setting the equalised transfer mean of 0.616 against a within-region reference of about 0.87 would compare a 10 km-collar number with a full-rectangle one at 1 km blocking, which is the  
1730 most generous reference in the paper and the very figure Section 4.3 argues is an upper bound. Recomputing the within-region reference on the same frame and at the 5 km blocking this design defends (`matched_frame_gap.csv`, `paper/code/verify_matched_gap.py`):

| Frame          | Within-region (5 km blocking) | Mean transfer | Gap          |
|----------------|-------------------------------|---------------|--------------|
| full rectangle | 0.798                         | 0.540         | 0.258        |
| 10 km collar   | <b>0.772</b>                  | <b>0.616</b>  | <b>0.155</b> |
| 5 km collar    | 0.737                         | 0.608         | 0.129        |

1735 Paired by target region, the collar shortfall is **+0.155** [**+0.094**, **+0.217**]  
 (Student  $t$  over the five target regions; per-region +0.086 Montiferru, +0.127  
 Manavgat, +0.161 Muğla, +0.196 Bejís, +0.206 Evia), which reproduces the  
 point estimate above and excludes zero. The shortfall therefore survives  
 on every matched row, but it is **0.155 at the collar, not the 0.25 the**  
 1740 **unmatched comparison implies**, and it shrinks as the frame approaches  
 the fire — which is where a susceptibility surface is actually used.

This is Section 4.3’s effect acting between regions rather than within one,  
 on frames whose fire-adjacent share ranges from 37 % to 98 %. Appendix  
 C.5(x) records the frame as a limitation of this cohort rather than of the  
 1745 method.

#### *Appendix A.14. $A(x)$ . The transfer matrix and adaptation, elaborated*

Section 4.5 states these results; the paragraphs it condensed are here.

**Raw transfer is heterogeneous and includes anti-predictive di-**  
**rections.** Raw target AUC spans 0.326 to 0.686 (Fig. A.4). Twelve of 20  
 1750 directions are above chance with CI support. Six are *below* chance with CI  
 support: both directions of Manavgat to Bejís and of Manavgat to Muğla,  
 plus Bejís to Evia and Evia to Bejís — of which only Manavgat to Bejís sur-  
 vives frame equalisation. Two intervals span 0.5. Even the best raw transfer,  
 Evia to Manavgat at 0.686, stays far below that target’s own within-region  
 1755 thermal performance of 0.870. Across all directions the raw deficit against

the within-region reference is 0.184 to 0.592 AUC. That reference is a region-level blocked estimate, and Section 4.3 shows it is not matched to a transfer evaluation.

Those counts belong to the 2-cell blocking of Table B9. At the more  
1760 conservative 10-cell blocking the same points give 9 above, 4 below and 7  
uncertain, and no direction changes side of the chance line (Appendix A(v)).  
Four of the six below-chance directions keep their support there. Bejís to  
Manavgat and Manavgat to Muğla lose it and carry no verdict. The quali-  
tative statement is unchanged. The counts should not be read as exact.

1765 **In precision terms the transfer is worse than the ROC fig-  
ures suggest.** A susceptibility surface is used as a ranked area budget,  
so precision-recall is the operational quantity. Across the twenty directions  
the thermal model’s PR-AUC averages **0.156 against a no-skill baseline  
of 0.136. Six of the twenty fall below their own no-skill baseline**  
1770 **at the point estimate, five of them with intervals entirely below it;**  
the exception is Bejís to Manavgat, whose interval covers its baseline. Only  
one direction exceeds twice its baseline. These are frame-as-drawn quantities  
and the PR arm was not recomputed on the collar. Per-direction values are  
in Appendix B, Table B4.

1775 **The static baseline does not transfer either.** The same twenty  
directions were run with the terrain, fuel and greenness baseline alone, giving  
a mean target AUC of **0.537** against **0.541** for the thermal model. Static  
attributes of a place are the class Dimarco et al. transfer successfully and the  
class this paper’s framing treats as portable; here that class is itself barely  
1780 above chance. The transfer failure below is therefore not specific to the

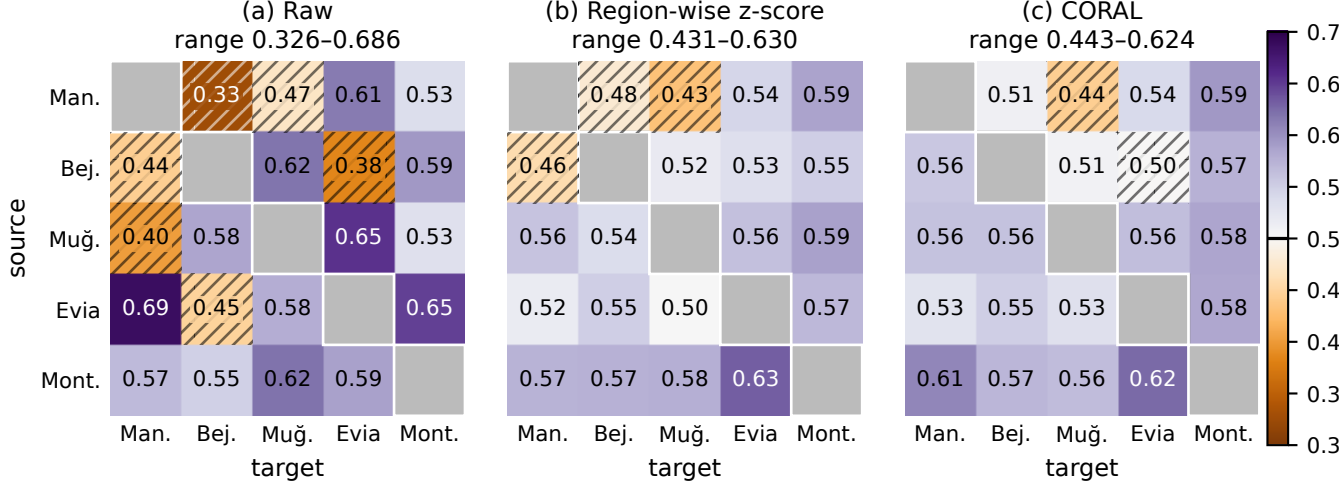


Figure A.4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. **(a)** Raw transfer, no adaptation. **(b)** After region-wise z-score standardisation. **(c)** After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table B9.

dynamic block: a baseline that does not travel, plus pre-fire thermal state, gives a model that does not travel.

**The thermal block’s paired contribution to transfer, with its interval.** Differencing the two matrices direction by direction gives a mean of **+0.004**, but the mean is not the informative statistic. The individual contributions span **−0.148 to +0.133**, twelve positive and eight negative, so the spread is thirty times the mean and the sign belongs to the pair rather than to the block; a mean near zero records cancellation, not consistent absence of effect. The directions are not independent, since each region appears in eight of the twenty, so the interval depends on the resampling unit — **all four units the design permits give the same answer**, from  $[-0.027, +0.034]$  treating directions as independent to  $[-0.037, +0.046]$  jackknifing regions, and none propagates within-direction sampling variability (Appendix A(o)). The leave-one-region-out jackknife shows how little the mean is anchored: **dropping Evia alone reverses its sign.**

**On the frames as drawn, six directions are below chance with interval support**, the sharpest at 0.326  $[0.305, 0.349]$ . Section 4.4 has shown most of that count is a property of the frames: equalising them leaves **one**, Manavgat to Bejís at 0.417  $[0.349, 0.488]$ , supported at the 10 km collar though its interval covers chance at 5 km. That direction still needs a mechanism acting on the direction of the relationship, because no account of merely lost skill produces a reliably reversed ranking, and Section 4.6 pursues it. Everything below this point is computed on the frames as drawn and should be read against Section 4.4. Per-split and per-scar detail is in Appendix A(i); per-direction values for all twenty directions, raw and under

both adaptations, are in Appendix B, Table B9.

**Label-blind adaptation compresses the matrix toward chance rather than repairing it.** Under region-wise z-scoring the twenty directions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half  
1810 the raw spread, with no adapted direction exceeding 0.631 against the unmatched within-region references of 0.859 to 0.918. Adaptation raises the failing directions and degrades most of those that already transferred. Taking the better of the two adaptations per direction, 14 of the 20 end closer to chance than they began and 6 end further from it; five of those six involve  
1815 Montiferru, the smallest and last-added region, and move upward, while the sixth is Manavgat→Muğla moving downward from 0.470 to 0.443. The 14 to 6 split should be read at the precision of limitation (ix) in Appendix C.5, since Bejís→Manavgat is counted as compressed on a margin of 0.001.

The per-direction decomposition is in Appendix A(j), Table A5; two fig-  
1820 ures from it matter here and are used below.

**Against the right reference, adaptation is not failing** (Fig. A.5). The three controls above give an achievable reference for a model applied to a fire it has not seen: 0.574 for the half-split, 0.552 for the leave-one-scar-out. CORAL, committed to in advance, averages **0.552** across the twenty direc-  
1825 tions against 0.541 raw, which is exactly that reference. Taking whichever of the two adaptations scores higher per direction gives 0.556, but that selection uses the target labels the protocol forbids, so it is an oracle upper bound rather than an achievable result and is reported as one. What it does is regress the matrix towards that value: fourteen of twenty directions move  
1830 closer to chance, which harms the directions that already worked and helps

the ones that did not. The verdict that follows is narrower than "alignment fails". Alignment cannot exceed what a model can achieve on an unseen fire, and it does not.

In the six directions where raw transfer was below chance, the best label-free method recovers at most 34 % of the gap to the within-region reference, so the remaining unrecovered fraction is at least 0.66 everywhere. The half-split control above shows that the larger part of that remainder is already incurred inside the region, so it should not be read as a measure of concept shift. On the four-AOI twelve-direction subset for which the decomposition is defined (Appendix A(j)), seven directions show *negative* recovery, meaning adaptation moves the score away from the reference; in six of those raw transfer was already above chance and adaptation destroyed that advantage. Label-free alignment therefore does not act as a repair mechanism.

*Appendix A.15. A(y). The diagnostics, elaborated*

Section 4.6 states these results; the paragraphs it condensed are here.

Twenty candidate diagnostics from five families were each rank-correlated with the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap.

Table 5 of the body (Section 4.6) groups them by family and is not repeated here.

Geographic separation does not order the matrix on either construction: over all twenty directions the Spearman correlation between centroid separation and transfer is  $-0.32$ , and on the twelve-direction common subset  $-0.24$ , both spanning zero. The two nearest directions are among the worst on the frames as drawn, while the 2,802 km pair returns 0.326 and 0.444, so

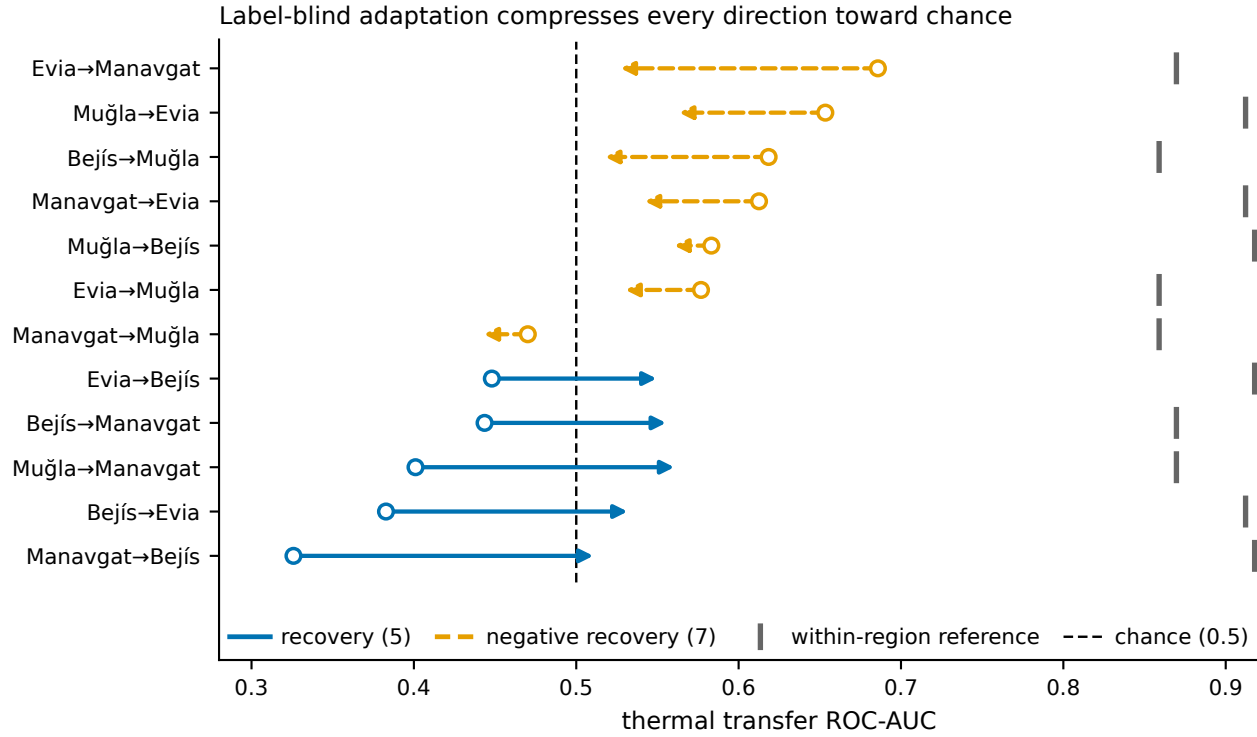


Figure A.5: **Label-blind adaptation regresses transfer onto the level a model reaches on an unseen scar, rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of  $-0.862$ . No arrow reaches its within-region tick.

the transfer mean of 0.541 is not the value at any one separation.

**Only two diagnostics have intervals excluding zero, and both are conditional:** the sign-agreement fraction over interval-supported features at  $\rho = +0.84$   $[+0.58, +0.88]$ , and its cosine variant at  $+0.81$ . No marginal  
1860 measure was shown to order the matrix, including area-of-applicability dissimilarity, climatic distance and geographic distance, and neither were the niche-overlap and regime families. The learned domain classifier is at ceiling, separating source from target at  $\text{AUC} \geq 0.96$  for every pair; it always succeeds, which is why it carries no ordering information.

1865 Four limits are stated with the result rather than after it. **Size:** the index’s tie structure caps the achievable Spearman at  $+0.861$ , so the observed  $+0.840$  sits on that ceiling, and its exact one-sided permutation p of 0.0060 is the smallest that structure can produce against a Bonferroni threshold of 0.0026 over nineteen computed variants — **no outcome of this diagnostic**

1870 **could have cleared family-wise correction on ten effective pairs.** **Labels:** signed associations need burned labels in both regions, so the family that appears to work is not available before deployment while the family that fails is. **Selection,** the sharpest of the three: the two rows that clear zero are the *supported-feature* variants, whose subset is chosen by whether two regions’  
1875 bootstrap intervals happen to be disjoint — a data-dependent selection made on the same data, with no correction. Their unselected counterparts over all nine features are  $\rho = +0.50$   $[-0.17, +0.83]$  and  $+0.18$   $[-0.40, +0.72]$ , both spanning zero, so the result lives in the selection step and is reported as such.

The families sit on unequal samples — twelve directions for the marginal,  
1880 applicability, climatic and geographic rows, sixteen for the supported-conditional

rows, twenty for the rest — so every row was recomputed on the common twelve. Published values reproduce to  $4.8 \times 10^{-5}$ , the conditional rows still lead at +0.87 [+0.65, +0.88] and +0.85 [+0.43, +0.88], and every marginal row still spans zero. The ordering is not an artefact of unequal samples.

1885     **A fourth limit, established in Section 4.4, removes the result entirely.** The index is built from signed associations that Section 4.4 shows to be frame artefacts, and was correlated against transfer measured on the same frames. Recomputed on an equalised frame it is unanimous, with no variance left to correlate. Everything in this section is what the original  
1890 protocol yields; the conclusion that survives is that **no diagnostic tested here was shown to order transfer once the frames are comparable.**

*Appendix A.16.  $A(z)$ . The four evaluations, elaborated*

Section 4.3 states these results; the paragraphs it condensed are here.

**Where the skill is lost, on a matched comparison.** Four evaluations  
1895 are reported; the last three are scored on **identical cells**, so they differ only in what the model was trained on, and the first shows why an unmatched comparison misleads. The held-out unit is a burned connected component of at least 50 cells together with all cells within 2 km of it. What makes that a harder problem than a whole region is the composition of its negatives, not  
1900 its burned fraction, and that can be shown rather than argued. Taking the same fitted model and the same out-of-fold predictions and scoring them on a random sample of region cells drawn at the scar area’s own burned fraction gives **0.782 against the region-wide 0.782**: matching the prevalence changes nothing, at  $-0.000$  [ $-0.003$ ,  $+0.002$ ] over the nine scars. Scoring  
1905 the same predictions on the scar area itself gives 0.627, a fall of **+0.155**

[+0.093, +0.217]. The whole effect is the negative pool. Every negative in a scar collar is fire-adjacent, sharing the terrain, land cover and synoptic conditions of the positives, whereas a region's negatives include its easy far field. The burned fraction, 34 to 87 % against 3.8 to 28.7 % for a region, is  
1910 a symptom of that construction, and ROC-AUC is in any case invariant to class balance at fixed class-conditional distributions.

Table 2 of the body (Section 4.3) carries the four evaluations and is not repeated here; its rows are labelled A to D and referred to by those letters below.

1915 Table 2's four rows are means over the **same eight scars**. A ninth burned component, Bejís, is excluded **from it**: it is that region's only component of any size, so holding it out leaves no usable source model and there is no row C for it. It is *not* excluded from the arms that need no leave-one-scar-out model, so the prevalence control two paragraphs above and Tables A3 and  
1920 A4 are computed on nine scars. That is why their means, 0.782 and 0.627, differ from Table 2's 0.776 and 0.634. Rows A, B and D are given there on the eight so that the differences are paired, and the intervals are Student  $t$  over those eight, which is the resampling unit for this arm rather than the spatial-block bootstrap used elsewhere in the paper. Eight is a small number  
1925 and the intervals are wide accordingly. Four of the eight scars are in Muğla and two in Montiferru, so they are not independent; row A in particular is a region-level quantity repeated across the scars of a region, and its interval is pseudo-replicated and should not be read as coverage. The two differences are computed per scar and paired, which is what they are reported for.

1930 **The same model, scored two ways on the same region, differs by**

**0.143 AUC.** Rows A and B are one model. The only change is which cells it is scored on: the whole region, or the burn scar and its 2 km collar. That change alone costs **0.143** [**+0.077**, **+0.208**] of the 0.225 [+0.157, +0.293] fall from A to C, about two thirds, and it is the size of the entire increment  
1935 this literature usually reports. Both intervals are Student  $t$  over the eight scars, so the "two thirds" is a ratio of two estimates and is compatible with anything from roughly a third to nine tenths; the point of the comparison is the size of the numerator, not the precision of the fraction. A region-wide blocked figure is therefore an upper bound on what the same model achieves  
1940 where the fire actually is, and the difference is not small enough to ignore. Any comparison of a scar-level result against a region-level reference inherits that, and the region-level reference is what a paper of this kind normally reports.

**Neither withholding the fire nor moving 2,800 km has a measurable cost.** B minus C, the fire-specific residual on identical cells, is **+0.082** [**-0.011**, **+0.175**], which spans zero. C minus D, the effect of replacing a same-region model with one fitted between 306 and 2,802 km away, is **-0.003** [**-0.075**, **+0.069**], which also spans zero and is an order of magnitude smaller. The second is the cleaner result: on the same cells, a model  
1950 fitted in another country does as well as one fitted in the same region with the scar withheld, and it survives restriction to Muğla, the only region where holding out one scar still leaves the source model properly trained, at 0.579 against 0.597.

**Row D averages four foreign models over a large spread, 32 scar-**  
1955 **by-source combinations running 0.374 to 0.724 with ten below chance, so**

the pooled 0.555 says the *average* foreign model matches a same-region one, not that any particular one does. One confound must be stated: holding out a region's only large scar also removes most of its positives, so row C mixes "the fire was withheld" with "almost all the positives were withheld";

1960 restricted to Muğla, where four scars remain, C is 0.579 and D is 0.597 and the comparison still shows nothing. **With eight scars, three of them starved, this design cannot establish a fire-specific residual, only bound it at about 0.18.** Rows C and D both sit close to chance, so the comparison between them has little dynamic range: what is measurable is the

1965 change of evaluation geometry between rows A and B. **It is not shown to be zero on an unseen fire, and eight scars cannot show that; what this design establishes is that it is not established there.** Nor does it establish the fire event as the unit, because the held-out patch is defined by the labels and its identity cannot be separated from its location. Sweeping

1970 the two parameters that define that patch — minimum burned-component size over 25 to 200 cells and connectivity over the 4- and 8-neighbourhoods — moves the row-C mean only between 0.545 and 0.566, and it is flat across buffers of 2, 5 and 10 km. Per-scar and per-source values, the full sweep and the half-split control are in Appendix A(i).

1975 **The increment declines with the holdout, and is not established once the fire is withheld.** Contribution 1 is about the paired thermal-minus-baseline difference, so the same evaluations were run on it.

| Evaluation                              | Thermal increment | Interval                    |
|-----------------------------------------|-------------------|-----------------------------|
| Blocked cross-validation,<br>per region | +0.056 to +0.153  | every interval above zero   |
| Within-region half-split                | +0.027            | positive in 13 of 18 splits |
| Leave-one-scar-out                      | <b>+0.022</b>     | <b>[−0.032, +0.077]</b>     |
| Cross-region, twenty<br>directions      | +0.004            | [−0.028, +0.036]            |

The point estimate falls monotonically as the holdout hardens and the last  
1980 two intervals span zero, so **the within-region increment of +0.056 to +0.153 is substantially a property of interleaved holdout**. Per-scar values are in Appendix A(i).

*Appendix A.17. A(aa). The six further arms, summarised*

Six further arms bear on the findings above without changing them; each  
1985 is reported in full in Appendix A. **The contrast pair** (A(s)): the most environmentally similar pair in the matrix is among the weakest in transfer and the least similar among the stronger, so high envelope overlap does not buy transfer — an ordinal claim, not one about the endpoints. **Interventions** (A(n)): pooling four regions into one training set does not recover  
1990 what single-source transfer loses, and removing the two reversing predictors costs −0.081 of within-region AUC while returning +0.014 [−0.017, +0.045] on transfer, so a local cost is measured and no compensating transfer gain is. **Sensitivity** (A(a)–A(h)): eight design choices were varied with everything else held fixed; none changes a conclusion above, except arm (f), which tests  
1995 a claim of Section 1.2 and does not uphold it. **The same geography, a second fire** (A(m)): the one arm that held place fixed shows an elevation

reversal on the frame as drawn, which Section 4.4 withdraws. **Distance** (A(t)): within a single region the model is already at chance by 10 to 20 km from its training cells, which bounds how far a surface of this kind can be carried but cannot be turned into an attribution, for the reason given there.

**The price of labels** (A(u)): thirty-two labelled 5 km blocks recover 85 to 89 % of the target’s matched ceiling in three of six directions and 30 to 57 % in the rest, which is 7 to 20 % of the target’s natural-vegetation population and not a cheap answer. Two limits bound it: at the top budget the selection pool is nearly exhausted, so the narrow upper-budget intervals reflect saturation rather than precision, and **the labelled blocks are drawn from the event being predicted, which is not a resource available before that event burns.**

## Appendix B. Appendix B. All twenty transferability diagnostics

Section 4.6 summarises this table by family. The full ranking is given here because the negative result is the point: a reader should be able to see every candidate that was tried, not only the families’ best members, and to check that no diagnostic was dropped after it failed.

**Table B1. All transferability diagnostics versus raw thermal transfer (20 ordered directions).** Spearman  $\rho$  with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with  $n = 12$  exist only for the four-AOI subset, because those diagnostics were never produced for Montiferru. The supported-features conditional rows use the 16 directions (8 pairs) with at least one CI-supported feature.

| Diagnostic                                    | Family                    | Exp. | n   | dir | Spearman $\rho$<br>[95% CI] | CI excludes 0 |
|-----------------------------------------------|---------------------------|------|-----|-----|-----------------------------|---------------|
| <b>Agreement fraction, supported features</b> | <b>P(y x) conditional</b> | +    | 16  |     | <b>+0.84</b> [+0.58, +0.88] | <b>yes</b>    |
| <b>Cosine, supported features</b>             | <b>P(y x) conditional</b> | +    | 16  |     | <b>+0.81</b> [+0.33, +0.88] | <b>yes</b>    |
| Cosine, all 9 features                        | P(y x) conditional        | +    | 20  |     | +0.50 [−0.17, +0.83]        | no            |
| Vector                                        | P(y x)                    | +    | 20  |     | +0.27 [−0.36, +0.77]        | no            |
| Spearman, all 9                               | conditional               |      |     |     |                             |               |
| Agreement count, all 9                        | P(y x) conditional        | +    | 20  |     | +0.18 [−0.40, +0.72]        | no            |
| Schoener’s D, 1-D mean                        | P(x y=1) niche            | +    | 20  |     | +0.24 [−0.45, +0.74]        | no            |
| Warren’s I, 1-D mean                          | P(x y=1) niche            | +    | 20  |     | +0.22 [−0.42, +0.73]        | no            |
| Schoener’s D, PCA-2D                          | P(x y=1) niche            | +    | 20  |     | +0.10 [−0.51, +0.68]        | no            |
| Warren’s I, PCA-2D                            | P(x y=1) niche            | +    | 20  |     | −0.07 [−0.66, +0.49]        | no            |
| Mahalanobis, burned centroids                 | P(x y=1) niche            | −    | 20  |     | −0.23 [−0.75, +0.44]        | no            |
| Domain-classifier AUC                         | P(x) marginal             | −    | 20  |     | −0.32 [−0.78, +0.33]        | no            |
| Predictor-space mean dissimilarity            | P(x) marginal             | −    | 12  |     | −0.10 [−0.54, +0.43]        | no            |
| Predictor-space p95 dissimilarity             | P(x) marginal             | −    | 103 | 12  | −0.08 [−0.59, +0.49]        | no            |
| Fraction inside weighted AoA                  | P(x) marginal             | +    | 12  |     | +0.22 [−0.48, +0.59]        | no            |
| Fraction inside unweighted                    | P(x) marginal             | +    | 12  |     | +0.08 [−0.89, +0.63]        | no            |

*Table note.* A null row means the diagnostic was **not shown to order transfer** on this design, not that it was shown incapable of ordering it. With ten effective region pairs the power is low, and the intervals are wide enough to admit moderate true correlations in either direction. The last row is retained rather than deleted because it was computed: on two directions the statistic has no meaningful value, and reporting that is more honest than dropping the variant.

**Reading the two rows that clear zero.** Both are supported-feature variants, where the predictor subset is chosen by whether two regions' bootstrap intervals happen to be disjoint. That is a data-dependent selection made on the same data, uncorrected. Their all-nine-feature counterparts are in the table and both span zero. The result lives in the selection step, and Section 4.6 says so.

**Equal-sample check.** The families sit on unequal samples: marginal, applicability, climatic and geographic rows on twelve directions, the supported-conditional rows on sixteen, the rest on twenty. Recomputing every row on the common twelve directions reproduces the published values to  $4.8 \times 10^{-5}$  and leaves the ordering unchanged — the conditional rows still lead at +0.87 [+0.65, +0.88] and +0.85 [+0.43, +0.88], every marginal row still spans zero. Source: `paper/diagnostics_common_subset.md`.

*Appendix B.1. B2. The signed univariate associations the reversal claim rests on*

Appendix A(n) drops two predictors because their signed association with burning reverses between regions with bootstrap support on the frames as drawn. Section 4.4 shows that both supported elevation reversals disappear

once the frames are equalised, so this table is the evidence for the feature-removal arm and for the narrowed mechanism claim inside Contribution 1, not for a general reversal mechanism. It was computed for the frozen analysis and is reproduced here because the claim is otherwise asserted rather than shown.

**Table B2. Signed univariate AUC of each predictor against burned, by region.** Primary natural-vegetation population; 10-cell ( $\sim 5$  km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is never folded to  $\max(\text{AUC}, 1 - \text{AUC})$ , so a value below 0.5 means lower values rank burned and is a direction rather than weakness. **Bold** marks a region whose own interval excludes 0.5.

| Feature              | Manavgat                    | Bejís                       | Muğla                       | Evia                        | Montiferru                  |
|----------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| elevation_mean       | <b>0.374</b> [0.289, 0.471] | <b>0.643</b> [0.558, 0.729] | <b>0.611</b> [0.532, 0.690] | 0.541 [0.448, 0.626]        | 0.584 [0.395, 0.762]        |
| slope_mean           | 0.531 [0.423, 0.642]        | 0.521 [0.439, 0.605]        | <b>0.637</b> [0.582, 0.686] | 0.487 [0.418, 0.554]        | <b>0.652</b> [0.506, 0.771] |
| ndvi_mean            | <b>0.636</b> [0.587, 0.676] | 0.559 [0.497, 0.619]        | <b>0.662</b> [0.616, 0.704] | <b>0.639</b> [0.575, 0.701] | 0.586 [0.450, 0.704]        |
| lst_anomaly_mean     | 0.482 [0.428, 0.530]        | <b>0.418</b> [0.364, 0.480] | 0.485 [0.395, 0.566]        | <b>0.640</b> [0.567, 0.710] | 0.395 [0.285, 0.535]        |
| current_lst_mean     | 0.538 [0.452, 0.621]        | 0.477 [0.401, 0.547]        | <b>0.325</b> [0.271, 0.382] | <b>0.377</b> [0.301, 0.456] | 0.370 [0.248, 0.513]        |
| current_tvdi_mean    | 0.552 [0.460, 0.641]        | 0.517 [0.429, 0.595]        | <b>0.336</b> [0.275, 0.398] | <b>0.362</b> [0.285, 0.442] | <b>0.356</b> [0.233, 0.499] |
| tvdi_difference_mean | 0.449 [0.391, 0.505]        | 0.512 [0.443, 0.583]        | 0.490 [0.396, 0.575]        | 0.519 [0.444, 0.589]        | <b>0.378</b> [0.282, 0.497] |
| downscaled_lst_mean  | 0.552 [0.466, 0.637]        | 0.484 [0.400, 0.560]        | <b>0.307</b> [0.253, 0.366] | <b>0.377</b> [0.297, 0.459] | 0.365 [0.240, 0.511]        |
| fused_lst_mean       | 0.540 [0.454, 0.622]        | 0.481 [0.404, 0.551]        | <b>0.325</b> [0.272, 0.383] | <b>0.376</b> [0.300, 0.456] | 0.370 [0.248, 0.513]        |

**What counts as a reversal.** A pair of regions is called a reversal

only when their signed associations point to opposite sides of 0.5 **and each**  
 2060 **region’s own interval excludes 0.5**. That is stricter than requiring the  
 two regions’ intervals to be disjoint, and the difference matters: for `current_`  
`lst_mean` between Manavgat and Muğla the two intervals are disjoint, but  
 Manavgat’s own interval, [0.452, 0.621], includes 0.5, so Manavgat has no  
 established direction to reverse from. That pair is a point reversal, not a  
 2065 supported one.

**Table B3. The cross-region reversals that meet the stricter criterion.** Difference intervals are from the same paired bootstrap.

| Feature                       | Region A | AUC   | Region B | AUC   | Difference [95 %<br>CI]    |
|-------------------------------|----------|-------|----------|-------|----------------------------|
| <code>elevation_mean</code>   | Manavgat | 0.374 | Bejís    | 0.643 | +0.269 [+0.138,<br>+0.397] |
| <code>elevation_mean</code>   | Manavgat | 0.374 | Muğla    | 0.611 | +0.235 [+0.102,<br>+0.360] |
| <code>lst_anomaly_mean</code> | Bejís    | 0.418 | Evia     | 0.640 | +0.221 [+0.123,<br>+0.313] |

Three pair-level reversals across **two** features, which is why Section 3.12  
 2070 removes exactly those two. Twenty-nine further pairs reverse at the point  
 estimate only, spread across eight of the nine features, and they are not  
 counted. The conservative criterion costs the paper findings rather than  
 manufacturing them: a difference interval on the pair, which is the instrument  
 Appendix A(m) uses, would support more reversals than the three listed here.

2075 *Appendix B.2. B4. The transfer matrix in precision-recall terms*

ROC-AUC is reported throughout the main text for comparability with the susceptibility literature. A susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity, and at target prevalences of 3.8 to 28.7 % the two can differ sharply. Read from the same  
2080 frozen step9b exports as Table B9.

**Table B4. Thermal transfer, PR-AUC against the no-skill baseline.** The baseline is the target's burned prevalence. Lift is PR-AUC divided by that baseline; a lift of 1 is no better than random ranking. Ordered by lift.

| Direction                         | ROC-AUC      | PR-AUC       | No-skill     | Lift        |
|-----------------------------------|--------------|--------------|--------------|-------------|
| Evia $\rightarrow$ Manavgat       | 0.686        | 0.094        | 0.038        | <b>2.45</b> |
| Bejís $\rightarrow$ Montiferru    | 0.594        | 0.289        | 0.212        | 1.36        |
| Evia $\rightarrow$ Montiferru     | 0.647        | 0.283        | 0.212        | 1.34        |
| Bejís $\rightarrow$ Muğla         | 0.619        | 0.093        | 0.070        | 1.33        |
| Muğla $\rightarrow$ Evia          | 0.653        | 0.379        | 0.287        | 1.32        |
| Montiferru $\rightarrow$ Bejís    | 0.548        | 0.093        | 0.072        | 1.28        |
| Montiferru $\rightarrow$ Muğla    | 0.619        | 0.089        | 0.070        | 1.27        |
| Evia $\rightarrow$ Muğla          | 0.577        | 0.086        | 0.070        | 1.23        |
| Muğla $\rightarrow$ Bejís         | 0.583        | 0.088        | 0.072        | 1.22        |
| Manavgat $\rightarrow$ Evia       | 0.613        | 0.343        | 0.287        | 1.20        |
| Manavgat $\rightarrow$ Montiferru | 0.533        | 0.243        | 0.212        | 1.15        |
| Montiferru $\rightarrow$ Manavgat | 0.567        | 0.043        | 0.038        | 1.11        |
| Montiferru $\rightarrow$ Evia     | 0.586        | 0.317        | 0.287        | 1.11        |
| Muğla $\rightarrow$ Montiferru    | 0.531        | 0.214        | 0.212        | 1.01        |
| Bejís $\rightarrow$ Manavgat      | 0.444        | 0.034        | 0.038        | <b>0.90</b> |
| Manavgat $\rightarrow$ Muğla      | 0.470        | 0.063        | 0.070        | <b>0.90</b> |
| Evia $\rightarrow$ Bejís          | 0.448        | 0.059        | 0.072        | <b>0.82</b> |
| Bejís $\rightarrow$ Evia          | 0.383        | 0.226        | 0.287        | <b>0.79</b> |
| Muğla $\rightarrow$ Manavgat      | 0.401        | 0.029        | 0.038        | <b>0.75</b> |
| Manavgat $\rightarrow$ Bejís      | 0.326        | 0.049        | 0.072        | <b>0.67</b> |
| <b>Mean</b>                       | <b>0.541</b> | <b>0.156</b> | <b>0.136</b> | <b>1.16</b> |

Six directions fall below their own no-skill baseline, and only one exceeds twice it. The six are the same six that are below chance on ROC-AUC, which is what a reversed ranking predicts in either metric. Section 4.4 shows that this count is largely a property of the evaluation frames rather than of

2090 a reversed predictor-burning relationship.

*Appendix B.3. B5. The same-geography event pair, in full*

Appendix A(m) reports this arm and Section 4.4 withdraws its **elevation** reversal as a frame artefact; Appendix A(o) explains why the thermal channels of this arm carry no verdict either way. The per-feature values are  
2095 kept here because the arm is what motivated the frame test, and because its structural asymmetries have no analogue in the twenty-direction matrix.

**Table B5. Signed univariate feature-burned AUC, Muğla 2021 versus 2022.** Raw AUC against burned, never folded to  $\max(\text{AUC}, 1 - \text{AUC})$ ; 10-cell ( $\approx 5$  km) spatial-block bootstrap, 1,000 replicates, seed 42  
2100 (Section 3.15). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm.** Table 1’s note sets sixteen as the floor this design supports at that blocking, so the 2022 intervals here fall below the paper’s own standard and are read as indicative, exactly as the 20-  
2105 cell row of Table 1 is. The 2022 arm is additionally a single compact scar, so its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

| Feature               | 2021 AUC [95 %<br>CI]           | 2022 AUC [95 %<br>CI]           | Reversal                        |
|-----------------------|---------------------------------|---------------------------------|---------------------------------|
| <b>elevation_mean</b> | <b>0.611 [0.532,<br/>0.690]</b> | <b>0.296 [0.230,<br/>0.355]</b> | <b>bootstrap-<br/>supported</b> |
| current_lst_mean      | 0.325 [0.271,<br>0.382]         | 0.515 [0.434,<br>0.580]         | point only                      |
| current_tvdi_mean     | 0.336 [0.275,<br>0.398]         | 0.594 [0.475,<br>0.674]         | point only                      |
| downscaled_lst_mean   | 0.307 [0.253,<br>0.366]         | 0.508 [0.435,<br>0.571]         | point only                      |
| fused_lst_mean        | 0.325 [0.272,<br>0.383]         | 0.519 [0.436,<br>0.583]         | point only                      |
| ndvi_mean             | 0.662 [0.616,<br>0.704]         | 0.707 [0.624,<br>0.777]         | none                            |
| slope_mean            | 0.637 [0.582,<br>0.686]         | 0.558 [0.468,<br>0.634]         | none                            |
| lst_anomaly_mean      | 0.485 [0.395,<br>0.566]         | 0.380 [0.249,<br>0.502]         | none                            |
| tvdi_difference_mean  | 0.490 [0.396,<br>0.575]         | 0.397 [0.265,<br>0.514]         | none                            |

2110 *Appendix B.4. B6. The evaluation frames, and the signed associations they produce*

Section 4.4 states these results; the per-region values are here.

**Table B6. Evaluation-frame geometry of the five study regions.**

2115 Primary natural-vegetation population. Distance is Euclidean to the nearest burned cell on the 500 m grid, at 0.45 km per cell. Source `aoi_frame_auc.csv`; recomputable by `paper/code/verify_aoi_frame.py`.

| Region     | cells  | burned | median distance to<br>burned | share beyond 10 km |
|------------|--------|--------|------------------------------|--------------------|
| Manavgat   | 20,511 | 784    | 13.4 km                      | <b>60.1 %</b>      |
| Bejís      | 15,190 | 1,100  | 13.5 km                      | <b>63.1 %</b>      |
| Muğla      | 41,730 | 2,911  | 11.3 km                      | 55.3 %             |
| Evia       | 9,298  | 2,664  | 8.0 km                       | 43.7 %             |
| Montiferru | 2,544  | 539    | 2.7 km                       | <b>2.1 %</b>       |

**Table B7. Signed univariate AUC, frame as drawn against a 10 km collar.** Point estimates; the intervals that decide the reversal question are given in the text below and in `collar_frame_bootstrap.csv` (10-cell blocks, 1000 replicates, seed 42). Signed and never folded to  $\max(\text{AUC}, 1 - \text{AUC})$ , so a value below 0.5 is a direction, not weakness. The collar drops no burned cells in any region.

| Signed<br>univariate<br>AUC            | Manavgat     | Bejís | Muğla | Evia  | Montiferru | straddles 0.5 |
|----------------------------------------|--------------|-------|-------|-------|------------|---------------|
| elevation, full<br>frame (Table<br>B2) | <b>0.374</b> | 0.643 | 0.611 | 0.541 | 0.584      | <b>yes</b>    |
| elevation, 10<br>km collar             | 0.561        | 0.614 | 0.606 | 0.648 | 0.581      | no            |
| current LST,<br>full frame             | <b>0.538</b> | 0.477 | 0.325 | 0.377 | 0.370      | <b>yes</b>    |
| current LST,<br>10 km collar           | 0.386        | 0.405 | 0.332 | 0.286 | 0.376      | no            |
| current<br>TVDI, full<br>frame         | <b>0.552</b> | 0.517 | 0.336 | 0.362 | 0.356      | <b>yes</b>    |
| current<br>TVDI, 10 km<br>collar       | 0.392        | 0.454 | 0.342 | 0.250 | 0.361      | no            |

**Table B8. Region summary: populations and gate outcomes.** Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled population, `burnable_tree_shrub_grass` and `valid_for_modeling == True`, which is the population every model in this paper was fitted and scored on.

| Region                                   | Total cells | Valid cells | Burned | Prevalence<br>(all<br>valid) | TSG cells | Burned in TSG | TSG prevalence | Burned<br>natural-<br>veg<br>frac-<br>tion | Grade verdict |
|------------------------------------------|-------------|-------------|--------|------------------------------|-----------|---------------|----------------|--------------------------------------------|---------------|
| Manavgat<br>2021                         | 24,450      | 24,087      | 796    | 0.033                        | 20,511    | 784           | 0.038          | 0.984                                      | pass          |
| Bejís<br>2022                            | 15,759      | 15,759      | 1,103  | 0.070                        | 15,190    | 1,100         | 0.072          | 0.991                                      | pass          |
| Mugla<br>2021                            | 73,098      | 73,045      | 3,026  | 0.041                        | 41,730    | 2,911         | 0.070          | 0.958                                      | pass          |
| North<br>Evia<br>2021<br>(ex-<br>tended) | 22,925      | 22,906      | 2,788  | 0.122                        | 9,298     | 2,664         | 0.287          | 0.945                                      | pass          |
| Monte<br>León<br>2021                    | 3,234       | 3,173       | 697    | 0.220                        | 2,544     | 539           | 0.212          | 0.723                                      | pass          |

**Table B9. Cross-region transfer matrix, thermal model, TSG population.** Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ( $\lambda = 10^{-5}$ ).

| Direction           | Raw                  | Region-wise z-score  | CORAL                |
|---------------------|----------------------|----------------------|----------------------|
| Manavgat→Bejís      | 0.326 [0.305, 0.349] | 0.477 [0.451, 0.502] | 0.511 [0.484, 0.534] |
| Bejís→Manavgat      | 0.444 [0.408, 0.480] | 0.457 [0.420, 0.497] | 0.555 [0.528, 0.583] |
| Manavgat→Muğla      | 0.470 [0.451, 0.488] | 0.431 [0.411, 0.449] | 0.443 [0.423, 0.461] |
| Muğla→Manavgat      | 0.401 [0.378, 0.426] | 0.559 [0.531, 0.587] | 0.560 [0.535, 0.587] |
| Manavgat→Evia       | 0.613 [0.593, 0.631] | 0.542 [0.520, 0.565] | 0.539 [0.518, 0.561] |
| Evia→Manavgat       | 0.686 [0.653, 0.716] | 0.516 [0.489, 0.544] | 0.527 [0.500, 0.553] |
| Bejís→Muğla         | 0.618 [0.601, 0.635] | 0.518 [0.501, 0.535] | 0.507 [0.489, 0.524] |
| Muğla→Bejís         | 0.583 [0.561, 0.607] | 0.535 [0.512, 0.557] | 0.560 [0.538, 0.581] |
| Bejís→Evia          | 0.383 [0.363, 0.402] | 0.532 [0.509, 0.551] | 0.499 [0.479, 0.518] |
| Evia→Bejís          | 0.448 [0.426, 0.470] | 0.549 [0.524, 0.575] | 0.549 [0.525, 0.573] |
| Muğla→Evia          | 0.653 [0.636, 0.671] | 0.561 [0.543, 0.580] | 0.563 [0.545, 0.582] |
| Evia→Muğla          | 0.577 [0.560, 0.593] | 0.501 [0.485, 0.518] | 0.530 [0.515, 0.546] |
| Montiferru→Manavgat | 0.567 [0.539, 0.594] | 0.573 [0.538, 0.609] | 0.606 [0.574, 0.639] |
| Manavgat→Montiferru | 0.533 [0.488, 0.580] | 0.586 [0.540, 0.629] | 0.592 [0.550, 0.631] |
| Montiferru→Bejís    | 0.548 [0.521, 0.578] | 0.574 [0.552, 0.596] | 0.569 [0.548, 0.591] |
| Bejís→Montiferru    | 0.594 [0.560, 0.631] | 0.550 [0.500, 0.601] | 0.574 [0.530, 0.621] |
| Montiferru→Muğla    | 0.619 [0.604, 0.634] | 0.576 [0.562, 0.589] | 0.565 [0.550, 0.579] |
| Muğla→Montiferru    | 0.531 [0.495, 0.568] | 0.587 [0.549, 0.624] | 0.584 [0.547, 0.623] |
| Montiferru→Evia     | 0.586 [0.565, 0.606] | 0.630 [0.611, 0.649] | 0.624 [0.605, 0.641] |
| Evia→Montiferru     | 0.647 [0.608, 0.682] | 0.568 [0.528, 0.609] | 0.581 [0.539, 0.623] |

**Table B10. The most and least environmentally similar pairs, on both frames.** Schoener’s  $D$  is computed over burned cells only and is therefore collar-invariant. Transfer values are the two ordered directions of each pair; ranks are out of the twenty directions on the equalised frame. As-drawn transfer is read from Table B9, collar transfer from `aoi_frame_transfer.csv`.

|                                           | Manavgat–Muğla         | Bejís–Montiferru      |
|-------------------------------------------|------------------------|-----------------------|
| Schoener’s $D$ , mean 1-D                 | <b>0.826</b> (highest) | <b>0.479</b> (lowest) |
| per-feature $D$                           | 0.77 to 0.89           | 0.23 to 0.77          |
| transfer, frames as drawn                 | 0.470, 0.401           | 0.594, 0.548          |
| transfer, 10 km collar                    | 0.551, 0.510           | 0.669, 0.624          |
| rank of 20 on the collar, from the bottom | 5th, 2nd               | 15th, 11th            |
| target cells inside the AoA               | 0.875, 0.531           | —                     |

## Appendix C. Appendix C. Protocol detail

2145 These protocols are given here in full rather than in Methods, because each is a specification a reader needs only when checking the corresponding result, and none is needed to follow the argument. Sections 3.2, 3.3, 3.11 and 3.15 state in summary what each does.

2150 *Appendix C.1. C.1 Burned-area label, the reconstructed analysis grid, and the admissibility gate*

Labels come from MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1) retrieved through Google Earth Engine [11], whose omission and commission characteristics [4] bound every model here. The product is exported onto the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which  
2155 duplicates each native  $\sim 500$  m observation across the 30 m pixels beneath it.

The analysis grid is therefore **reconstructed rather than native**: the 30 m grid is partitioned into non-overlapping blocks of  $\text{round}(500 / 30) = 17 \times 17$  pixels, giving a nominal cell edge of 510 m, and each cell is identified

2160 by its integer block indices `row_500m` and `col_500m`. This approximates the  
MODIS sinusoidal cell rather than reproducing it, being anchored to the  
Landsat/EPSG:4326 reference grid, and marginal blocks are truncated. The  
cell is square in degrees but not on the ground, about 510 m north to south  
and 390 to 407 m east to west, so its area is 0.199 to 0.208 km<sup>2</sup> against a  
2165 MODIS cell's 0.250 km<sup>2</sup>; every block-size label in this paper is therefore the  
north-south dimension. The companion paper develops what follows from  
that.

A cell's representative burn date is the **mode** of the positive sub-pixel  
day-of-year values in the block, tested against the region's label window. Be-  
2170 cause the exported raster carries no out-of-window positives, in this dataset  
a single in-window positive sub-pixel makes the cell burned. The fraction of  
positive sub-pixels agreeing with the modal date is recorded as `burn_date_  
pixel_agreement_fraction` but no agreement threshold is imposed. The  
label never affects a cell's eligibility for modelling: unburned, all-no-data  
2175 and out-of-window cells all remain as the negative class.

Two safeguards are recorded rather than assumed. Cells that burned  
inside a region's own predictor window are removed from its analysis universe  
rather than counted as unburned, which ran for Muğla, Evia and Montiferru,  
is recorded as not run for Manavgat and has no recorded status for Bejís.  
2180 Burning in earlier years is screened for no region in the five-region cohort;  
the only historical exclusion in the study removes the 2021 Muğla scar from  
the 2022 event-relative experiment of Section 3.15.

Before any modelling each region passes a gate that answers one ques-  
tion: of the cells labelled burned, what fraction is dominated by natural

2185 vegetation? A region is admitted as a wildfire candidate when that fraction reaches 0.50 and at least 30 burned cells are present, and is rejected as a cropland-dominated control when the cropland fraction reaches 0.50 instead. The gate uses ESA WorldCover classes aggregated to the same reconstructed cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which 2190 MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

### *Appendix C.2. C.2 Transferability diagnostics versus transfer*

Twenty candidate diagnostics from five families are each rank-correlated against the same target quantity, the raw thermal transfer AUC over the 2195 twenty ordered directions, under one common pair-based bootstrap. The families are marginal predictor-distribution measures  $P(x)$ , burned-niche overlap  $P(x|y=1)$ , fire-regime label-pattern structure  $P(y)$ , and conditional feature-response direction  $P(y|x)$ .

The **marginal** family includes area-of-applicability-style dissimilarity in 2200 scaled, importance-weighted predictor space [23, 24, 18], climatic and geographic distance, and a learned domain classifier trained to separate source from target cells. The **niche-overlap** family includes Schoener’s D [36] and Warren’s I [44] over the burned cells of each region. The **regime** family includes burned-patch component counts and effective component counts. The 2205 **conditional** family is the sign-agreement index: the fraction of predictors whose signed univariate association points the same way in both regions, computed over all predictors and over the subset whose associations are individually interval-supported.

Two properties of this design are stated in advance because they bound

2210 what it can show. The candidate set was fixed in a project analysis log before the correlations were computed; that log is not a formal pre-registration and no independent timestamped registration exists. And with five regions the effective sample is ten unordered pairs, so nineteen computed variants are evaluated at that power with no family-wise error control claimed; where a  
2215 diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

*Appendix C.3. C.3 Same-geography event-to-event comparison (Muğla 2021 versus 2022)*

Muğla is the one region for which a second fire event is analysed on the  
2220 identical AOI and analysis grid, which allows the direction of feature-label associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor window closing the day before ignition and a 49-day label window opening on it, matching the 2021 durations exactly.

2225 **This is not a clean temporal-transfer design and is not presented as one.** The 2022 event ignites about six weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no difference can be attributed to the year alone. The registered claim is same-geography event-to-event. The pair is deliberately kept out of the twenty-  
2230 direction matrix, and that exclusion is enforced in released code rather than merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion applies to the 2022 arm alone, masking every cell that burned in 2021 so the previous year’s scar does not enter the following year’s analysis: it removes

2235 3,073 cells, of which 2,941 belong to the primary population, taking it from  
41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of  
38,790 cells and three byte-identical static predictors, and in the 2022-to-2021  
direction every target positive lies outside the source training population.  
Appendix A(m) states what follows for how these two numbers may be read.

2240 **Table C1. Study regions, areas of interest and temporal win-  
dows.** Bounding boxes are in EPSG:4326, as registered. Window lengths in  
brackets are inclusive day counts. The baseline years are the four window-  
symmetric years preceding each predictor window.

| Region                      | Bounding box<br>(lon min, lat min,<br>lon max, lat<br>max) | Predictor window                   | Label window                       | Baseline years            |
|-----------------------------|------------------------------------------------------------|------------------------------------|------------------------------------|---------------------------|
| Manavgat 2021<br>(Türkiye)  | 31.05, 36.72,<br>31.85, 37.35                              | 2021-06-01 to<br>2021-07-27 (57 d) | 2021-07-28 to<br>2021-08-31 (35 d) | 2017, 2018, 2019,<br>2020 |
| Bejís 2022<br>(Spain)       | -1.05, 39.68,<br>-0.35, 40.15                              | 2022-06-15 to<br>2022-08-14 (61 d) | 2022-08-15 to<br>2022-09-30 (47 d) | 2018, 2019, 2020,<br>2021 |
| Muğla 2021<br>(Türkiye)     | 27.1, 36.6, 28.9,<br>37.45                                 | 2021-06-01 to<br>2021-07-28 (58 d) | 2021-07-29 to<br>2021-09-15 (49 d) | 2017, 2018, 2019,<br>2020 |
| North Evia 2021<br>(Greece) | 23.05, 38.55,<br>23.85, 39.15                              | 2021-06-05 to<br>2021-08-02 (59 d) | 2021-08-03 to<br>2021-09-30 (59 d) | 2017, 2018, 2019,<br>2020 |
| Montiferru 2021<br>(Italy)  | 8.45, 40.05, 8.75,<br>40.27                                | 2021-05-25 to<br>2021-07-23 (60 d) | 2021-07-24 to<br>2021-08-31 (39 d) | 2017, 2018, 2019,<br>2020 |

2245 *Appendix C.4. C.4 Predictor provenance, compositing and the downscaler*

Ten predictors are used, four baseline and six thermal, all summarised  
per cell over the predictor window, through the chain shown in Fig. C.1.  
All optical and thermal predictors come from Landsat 8 Collection 2 Level-2  
(LANDSAT/LC08/C02/T1\_L2), quality-screened per pixel from the QA\_PIXEL  
2250 band with the water bit deliberately preserved; the coarse-resolution thermal  
input is MODIS/061/MOD11A1.

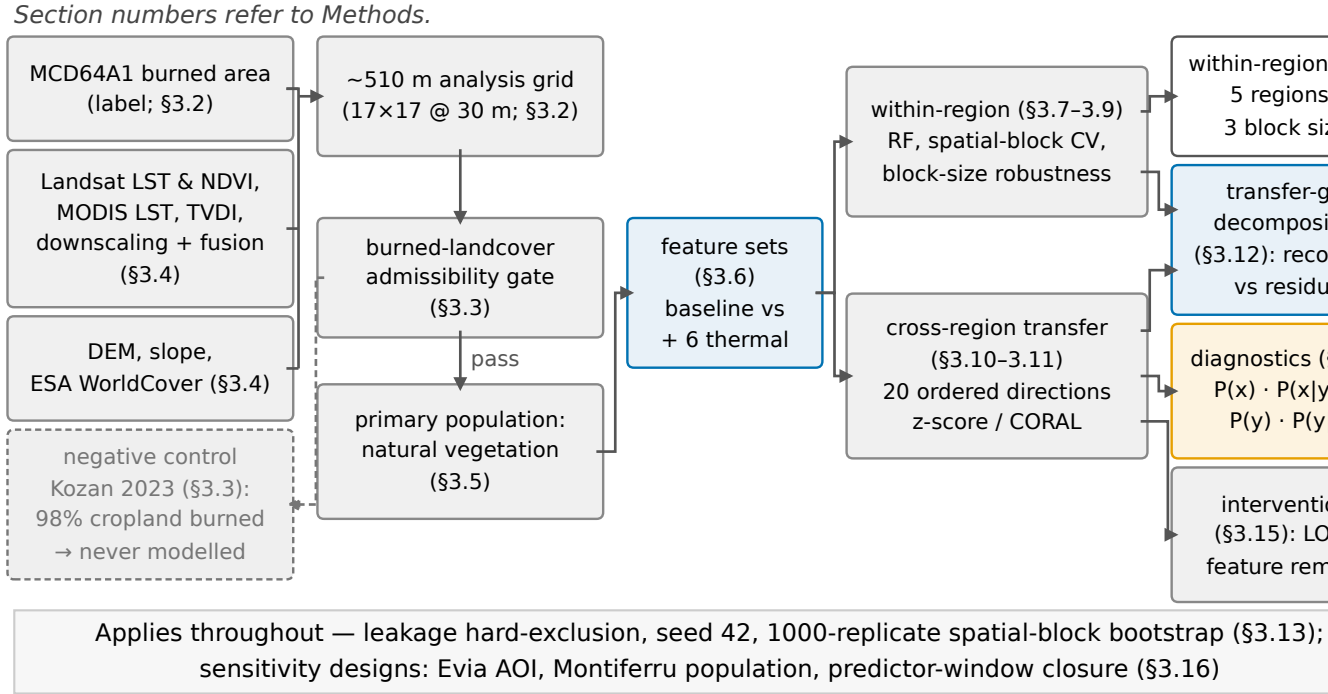


Figure C.1: **Processing pipeline and evaluation programme.** Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed  $\sim 510$  m analysis grid ( $17 \times 17$  pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

- **NDVI**, the predictor-window median of Landsat surface reflectance, is the baseline’s one time-varying member.
- **Elevation** and **slope** come from the Copernicus DEM GLO-30.
- 2255 • **Land cover** is ESA WorldCover v200 [48], entering as the dominant class code.
- **Current LST**, the predictor-window median of Landsat surface temperature in °C.
- 2260 • **LST anomaly**, a z-score of the current-window LST median against the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C or observations are too few.
- 2265 • **TVDI** and **TVDI difference**. The Temperature-Vegetation Dryness Index [35] is computed in the LST-NDVI feature space, with wet and dry edges taken as the 2nd and 98th LST percentiles within each of twenty NDVI bins and the index clamped to [0, 1]. The edges are percentiles of the LST values a given scene contains, so they are fitted per AOI and per window, and a TVDI of 0.5 denotes a different physical moisture state in each region. The companion paper reports what that scene dependence does and does not explain. **TVDI difference** 2270 **ence** is the raw anomaly of that index against the same four window-symmetric baseline years used for the LST anomaly: `tvdi_difference = current_tvdi - mean(baseline_tvdi)`, in index units rather than standard deviations, which is why it is reported alongside the z-scored channel rather than in place of it.

2275 • **Downscaled LST and fused LST.** A MODIS-to-Landsat downscal-  
 ing model is trained on the predictor window and applied to the full 30  
 m grid. The fused product equals observed Landsat LST wherever that  
 is valid, and the downscaled surface only where it is not. Gap-filling  
 therefore never replaces or blends a valid observation. The gap-filled  
 2280 share is 0.11 % to 9.70 % by region. The downscaler’s own inputs  
 include coordinates, which is the one route by which a coordinate-  
 derived surface re-enters a feature set from which Section 3.14 excludes  
 coordinates. Appendix A(g) reports the increment without these two  
 channels.

2285 *Appendix C.5. C.5 Limitations, in full*

Section 5.8 states the four that bind the conclusions; all eleven are here.

- (i) **No meteorological covariates** enter the models, so we cannot say  
 how local skill and portability behave for a mixed thermal-plus-weather pre-  
 dictor set.
- 2290 (ii) **The same-geography comparison covers one region only**, and  
 even there year and seasonal phase are confounded. That confound cannot  
 be resolved in this study area, and the reason is specific: the two events sit  
 42 days apart in median burn day-of-year, neither year contains a second  
 event at the other’s phase, and a calendar-matched arm would carry nine  
 2295 burned cells against this design’s gate minimum of thirty. The positive-block  
 count for the 2022 arm is separately below the floor this design sets itself,  
 at eleven against sixteen (Appendix A(m)). Its 331 burned cells also leave  
 the thermal reversals unresolved at interval level, and the pair holds place  
 fixed but not population. Those two arms are additionally the only transfer

2300 directions here computed by us rather than read from the pipeline author’s frozen export, with his unmodified code and the same pinned environment.

(iii) **All labels derive from a single burned-area product**, MCD64A1 [10], whose omission and commission characteristics [4] bound every model evaluated here. No second burned-area product covers 2021 and 2022 at this  
2305 resolution; the companion paper reports what an independent active-fire observation says about the omission concern.

(iv) **Evia remains the most imbalance-atypical population** even after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072 in three of the others.

2310 (v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires multi-year labels.

(vi) **Cross-region point estimates carry an implementation tolerance** of roughly  $\pm 0.02$  to 0.03 across scikit-learn versions. All reported  
2315 numbers are fixed to one verified version, but exact reproduction elsewhere requires the archived environment.

(vii) **The diagnostic correlations rest on an effective sample of ten region pairs**. Both the successes and the failures of Section 4.6 should be read at that power.

2320 (viii) **Manavgat’s atypical transfer behaviour remains unexplained**.

It is where the conditional diagnosis bites hardest and where feature removal recovers most. Three candidates have now been tested and none survives: its meteorology, which was not extreme; the quality screening of its coarse thermal input, which propagates widely but moves no signed association by

2325 more than  $+0.0003$  (Appendix A(v), Appendix A(e)); and the evaluation  
frame, which explains its elevation figure but not its transfer behaviour (Sec-  
tion 4.4). With one fire season per region the remaining candidates are not  
separable in this design.

(ix) **The interval-support counts are less stable than the point**  
2330 **estimates behind them.** Several verdicts sit within a thousandth of their  
reference value, and at 1 km blocking the published split of ten positive, seven  
negative and three uncertain turns on a lower bound of  $-0.00045$ . The point  
estimates and the sign pattern are stable; the counts are not. Every sentence  
in this paper that leans on an exact count of supported directions should be  
2335 read at that precision. (x) **The five areas of interest are not comparable**  
**frames, and this cohort cannot fully repair it** (Section 4.4). We report  
the equalised arm alongside the frame-as-drawn arm rather than replacing  
one with the other, because the collar radius is itself a choice and 5 km and  
10 km do not agree exactly (0.608 against 0.616). The deeper limitation  
2340 is that the frames were fixed upstream of this work, in `repo/`, so we can  
restrict them but not extend them; a region whose rectangle is already fire-  
scale, Montiferru, cannot be given a far field for symmetry. Any future cohort  
should fix the frame by an explicit accessible-area rule [3] before any predictor  
is computed, and we treat that as the main design lesson of this paper.

2345 (xi) **One classifier family.** The headline numbers use a random forest  
with unlimited depth, the configuration most able to encode local structure  
and least able to extrapolate. Appendix A(h) shows the transfer result is not  
an artefact of that choice: three further estimators, including a penalised  
linear one, all land between 0.510 and 0.556 and all place fourteen of twenty

2350 directions above chance. Those are point estimates without intervals, so the ordering among them is not claimed as a result. Other model families were not tried, and a different inductive bias might behave differently, but within this family the negative result is a property of the predictors rather than of an unregularised estimator.

2355 *Appendix C.6. C.6 Leakage control and reproducibility, in full*

An explicit forbidden-column set is enforced at every model fit. Coordinates (`lon`, `lat`, `row`, `col` and their normalised forms), every burn-date and label-provenance column, and the agreement fraction are excluded from all feature sets, and the enforcement runs as an assertion rather than a convention. 2360 The natural-vegetation mask is used only to define the population, never as a predictor. Spatial blocking prevents a cell from sharing a fold with its own neighbours.

**Reproducibility.** All randomness uses seed 42 and the bootstrap uses 1000 replicates throughout. The transfer and adaptation analysis runs in an environment separate from the upstream pipeline’s. 2365 Every within-region model was therefore refitted there and compared against the frozen upstream output, and the independently implemented adaptation was compared against the pipeline’s own. The within-region comparisons agree exactly and the twenty transfer directions to within  $1.6 \times 10^{-7}$ . All numbers 2370 here were produced under scikit-learn 1.9.0 or verified against it; the implementation tolerance that applies if the version is not fixed is stated in Appendix C.5(vi), and the companion paper reports the version sensitivity and the reproduction check in full.

**Sensitivity analyses.** Every headline result is repeated across two anal-

2375 ysis populations, three spatial-block sizes, the CORAL sweep, both feature  
sets and four classifier capacities; where a conclusion depends on one of  
those choices the dependence is reported rather than resolved by choosing  
the favourable setting (Appendix A).

2380 *Appendix C.7. C.7 Transfer-gap decomposition and the concept-shift diagnostic, in full*

For each direction the gap between the target’s own within-region skill  
and the raw transfer result is split in two. One part is what the best label-  
free adaptation recovers, and the other is what it does not. The recovered  
fraction is (adapted – raw) / (within – raw), signed and unclipped, with  
2385 its interval from the same paired bootstrap; it bounds what covariate-level  
correction can achieve. Section 4.3 shows the remainder should not be read  
as a conditional residual, because much of it is incurred inside a single region  
(Appendix A(j)).

The mechanism is diagnosed by **signed univariate association**. For  
2390 each numeric predictor the raw ROC-AUC of that predictor against **burned**  
is computed in each region and never folded to  $\max(\text{AUC}, 1 - \text{AUC})$ , so a  
value below 0.5 is read as a direction rather than as weakness. A reversal is  
called bootstrap-supported only when the two regions’ point estimates fall on  
opposite sides of 0.5 **and each region’s own interval excludes 0.5**, under  
2395 the same 10-cell spatial-block bootstrap. That is stricter than requiring the  
two regions’ intervals to be disjoint: a feature whose intervals are disjoint  
but one of which straddles 0.5 has not been shown to point anywhere in  
that region, so it is recorded as a point reversal only. Appendix B states  
the rule again beside the counts, and `conditional_similarity_transfer`.

2400 json carries it as machine-readable metadata.

## Declarations

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**Competing interests.** The authors declare no competing interests.

2415 **Ethics approval.** Not applicable. This study involved no human participants, animal subjects, or personally identifiable data.

**Data availability.** All satellite inputs are public. Burned-area labels are MODIS MCD64A1 Collection 6.1 and land cover is ESA WorldCover, both retrieved through Google Earth Engine; no proprietary or restricted 2420 data were used. The frozen numeric outputs behind every table and figure, together with the analysis code that reads them, are available at <https://github.com/dryuemco/thermal-twin>. The upstream processing pipeline that produced those outputs is a separate release, <https://github.com/>

emrehann17/satellite-thermal-digital-twin, and the commit of record  
2425 for every number reported here is 48b56e7. No digital object identifier is  
minted and no archival deposit exists.

**Code availability.** The analysis code, its configuration and the scripts  
that regenerate each reported artefact are in the first repository above. Three  
components of the upstream pipeline fall outside the released commit, and a  
2430 reader cannot re-run them: the reproduction-check driver and its validation  
package, on which Section 3.14 rests; the few-shot commit 19d825b, cited for  
provenance in the supplementary material; and the ERA5-Land diagnostic,  
whose manifest records commit a07ea33. Every other quantity in the paper  
can be regenerated from the released code and the frozen outputs.

2435 **Author contributions.** Stated in CRediT terms. **Emrehan Metin:**  
Software, Data curation, Investigation, Validation, Writing – review and  
editing. **Yunus Emre Cogurcu:** Conceptualization, Methodology, For-  
mal analysis, Investigation, Writing – original draft, Writing – review and  
editing, Supervision.

## 2440 References

- [1] Alkan Akıncı, H., Akıncı, H., 2023. Machine learning based forest fire  
susceptibility assessment of Manavgat district (Antalya), Turkey. *Earth  
Science Informatics* 16, 397–414. doi:10.1007/s12145-023-00953-5.
- [2] Archibald, S., Lehmann, C.E.R., Gómez-Dans, J.L., Bradstock, R.A.,  
2445 2013. Defining pyromes and global syndromes of fire regimes. *Proceed-  
ings of the National Academy of Sciences* 110, 6442–6447. doi:10.1073/  
pnas.1211466110.

- [3] Barve, N., Barve, V., Jiménez-Valverde, A., Lira-Noriega, A., Mather, S.P., Peterson, A.T., Soberón, J., Villalobos, F., 2011. The crucial role of the accessible area in ecological niche modeling and species distribution modeling. *Ecological Modelling* 222, 1810–1819. doi:10.1016/j.ecolmodel.2011.02.011.
- [4] Boschetti, L., Roy, D.P., Giglio, L., Huang, H., Zubkova, M., Humber, M.L., 2019. Global validation of the Collection 6 MODIS burned area product. *Remote Sensing of Environment* 235, 111490. doi:10.1016/j.rse.2019.111490.
- [5] Breiman, L., 2001. Random forests. *Machine Learning* 45, 5–32. doi:10.1023/A:1010933404324.
- [6] de Bruin, S., Brus, D.J., Heuvelink, G.B.M., van Ebbenhorst Tengbergen, T., Wadoux, A.M.J.C., 2022. Dealing with clustered samples for assessing map accuracy by cross-validation. *Ecological Informatics* 69, 101665. doi:10.1016/j.ecoinf.2022.101665.
- [7] Chuvieco, E., Cocero, D., Riaño, D., Martín, P., Martínez-Vega, J., de la Riva, J., Pérez, F., 2004. Combining NDVI and surface temperature for the estimation of live fuel moisture content in forest fire danger rating. *Remote Sensing of Environment* 92, 322–331. doi:10.1016/j.rse.2004.01.019.
- [8] Dimarco, N.A., Faraji, I., Wahbi, M., Maatouk, M., Boulaassal, H., Yazidi Aalaoui, O., El Kharki, O., 2026. Generalizing human-driven wildfire ignition models across Mediterranean regions using harmonized

remote-sensing and machine-learning data. *Geomatics* 6, 13. doi:10.3390/geomatics6010013.

- 2475 [9] Gelabert, P.J., Jiménez-Ruano, A., Ochoa, C., Alcasena, F., Sjöström, J., Marrs, C., Ribeiro, L.M., Palaiologou, P., Bentu Martínez, C., Chuvieco, E., Vega-García, C., Rodrigues, M., 2025. Assessing human-caused wildfire ignition likelihood across Europe. *Natural Hazards and Earth System Sciences* 25, 4713–4729. doi:10.5194/nhess-25-4713-2025.
- 2480 [10] Giglio, L., Boschetti, L., Roy, D.P., Humber, M.L., Justice, C.O., 2018. The Collection 6 MODIS burned area mapping algorithm and product. *Remote Sensing of Environment* 217, 72–85. doi:10.1016/j.rse.2018.08.005.
- 2485 [11] Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment* 202, 18–27. doi:10.1016/j.rse.2017.06.031.
- 2490 [12] Huang, Y., Liu, N., Chen, W., Tan, W., Deng, Y., Wang, Z., Chlus, A., Shen, J., Townsend, P.A., 2026. Enhancing transferability of foliar trait retrieval models: A comparative analysis of transfer learning strategies and domain shift characterization. *Remote Sensing of Environment* 337, 115336. doi:10.1016/j.rse.2026.115336.
- [13] Iban, M.C., Sekertekin, A., 2022. Machine learning based wildfire susceptibility mapping using remotely sensed fire data and GIS: A case

- study of Adana and Mersin provinces, Turkey. *Ecological Informatics* 69, 101647. doi:10.1016/j.ecoinf.2022.101647.
- [14] Jodhani, K., Bhatiya, L., Sirvi, L., Chanda, A., Thakkar, S., Rathnayake, U., 2026. Artificial intelligence and remote sensing frameworks for wildfire monitoring and risk analysis across multiple ecosystems: A review. *AIMS Environmental Science* 13, 56–98. doi:10.3934/environsci.2026004.
- [15] Kondylatos, S., Prapas, I., Camps-Valls, G., Papoutsis, I., 2023. Meso-geos: A multi-purpose dataset for data-driven wildfire modeling in the Mediterranean, in: *Advances in Neural Information Processing Systems* 36 (NeurIPS 2023) Datasets and Benchmarks Track, pp. 50661–50676. doi:10.52202/075280–2204.
- [16] Liu, C., Mostafavi, A., 2025. WildfireGenome: Interpretable machine learning reveals local drivers of wildfire risk and their cross-county variation. *arXiv preprint arXiv:2511.11589*. doi:10.48550/arXiv.2511.11589, arXiv:2511.11589.
- [17] Lobo, J.M., Jiménez-Valverde, A., Real, R., 2008. AUC: a misleading measure of the performance of predictive distribution models. *Global Ecology and Biogeography* 17, 145–151. doi:10.1111/j.1466-8238.2007.00358.x.
- [18] Ludwig, M., Moreno-Martinez, A., Hölzel, N., Pebesma, E., Meyer, H., 2023. Assessing and improving the transferability of current global spa-

tial prediction models. *Global Ecology and Biogeography* 32, 356–368.  
doi:10.1111/geb.13635.

- [19] Maffei, C., Alfieri, S.M., Menenti, M., 2018. Relating spatiotemporal patterns of forest fires burned area and duration to diurnal land surface temperature anomalies. *Remote Sensing* 10, 1777. doi:10.3390/rs10111777.
- [20] Maffei, C., Lindenbergh, R., Menenti, M., 2021. Combining multispectral and thermal remote sensing to predict forest fire characteristics. *ISPRS Journal of Photogrammetry and Remote Sensing* 181, 400–412. doi:10.1016/j.isprsjprs.2021.09.016.
- [21] Maffei, C., Menenti, M., 2019. Predicting forest fires burned area and rate of spread from pre-fire multispectral satellite measurements. *ISPRS Journal of Photogrammetry and Remote Sensing* 158, 263–278. doi:10.1016/j.isprsjprs.2019.10.013.
- [22] Marino, E., Yáñez, L., Guijarro, M., Madrigal, J., Senra, F., Rodríguez, S., Tomé, J.L., 2024. Transferability of empirical models derived from satellite imagery for live fuel moisture content estimation and fire risk prediction. *Fire* 7, 276. doi:10.3390/fire7080276.
- [23] Meyer, H., Pebesma, E., 2021. Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution* 12, 1620–1633. doi:10.1111/2041-210X.13650.
- [24] Meyer, H., Pebesma, E., 2022. Machine learning-based global maps of

ecological variables and the challenge of assessing them. *Nature Communications* 13, 2208. doi:10.1038/s41467-022-29838-9.

- 2540 [25] Meyer, H., Reudenbach, C., Hengl, T., Katurji, M., Nauss, T., 2018. Improving performance of spatio-temporal machine learning models using forward feature selection and target-oriented validation. *Environmental Modelling & Software* 101, 1–9. doi:10.1016/j.envsoft.2017.12.001.
- [26] Milà, C., Mateu, J., Pebesma, E., Meyer, H., 2022. Nearest neighbour distance matching leave-one-out cross-validation for map validation. *Methods in Ecology and Evolution* 13, 1304–1316. doi:10.1111/2041-210X.13851.
- 2550 [27] Moreno-Torres, J.G., Raeder, T., Alaiz-Rodríguez, R., Chawla, N.V., Herrera, F., 2012. A unifying view on dataset shift in classification. *Pattern Recognition* 45, 521–530. doi:10.1016/j.patcog.2011.06.019.
- [28] Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D.G., Piles, M., Rodríguez-Fernández, N.J., Zsoter, E., Buontempo, C., Thépaut, J.N., 2021. ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data* 13, 4349–4383. doi:10.5194/essd-13-4349-2021.
- 2555 [29] Pausas, J.G., Keeley, J.E., 2021. Wildfires and global change. *Frontiers in Ecology and the Environment* 19, 387–395. doi:10.1002/fee.2359.
- [30] Persello, C., Bruzzone, L., 2012. Active learning for domain adaptation in the supervised classification of remote sensing images. *IEEE Transac-*
- 2560

tions on Geoscience and Remote Sensing 50, 4468–4483. doi:10.1109/TGRS.2012.2192740.

- [31] Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., Lyapustin, A., Gourlet-Fleury, S., Pélissier, R., 2020. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. Nature Communications 11, 4540. doi:10.1038/s41467-020-18321-y.
- [32] Podschwit, H., Jolly, W., Alvarado, E., Verma, S., Ponce, B., Markos, A., Aliaga-Nestares, V., Rodriguez-Zimmermann, D., 2022. Reliability of cross-regional applications of global fire danger models: A Peruvian case study. Fire Ecology 18, 25. doi:10.1186/s42408-022-00150-7.
- [33] Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillerá-Aroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller, W., Warton, D.I., Wintle, B.A., Hartig, F., Dormann, C.F., 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40, 913–929. doi:10.1111/ecog.02881.
- [34] Rousseau, J.S., Betts, M.G., 2022. Factors influencing transferability in species distribution models. Ecography 2022, e06060. doi:10.1111/ecog.06060.
- [35] Sandholt, I., Rasmussen, K., Andersen, J., 2002. A simple interpretation of the surface temperature/vegetation index space for assessment of

- surface moisture status. *Remote Sensing of Environment* 79, 213–224. doi:10.1016/S0034-4257(01)00274-7.
- 2585 [36] Schoener, T.W., 1968. The Anolis lizards of Bimini: Resource partitioning in a complex fauna. *Ecology* 49, 704–726. doi:10.2307/1935534.
- [37] Sun, B., Feng, J., Saenko, K., 2016. Return of frustratingly easy domain adaptation, in: *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence (AAAI-16)*, AAAI Press, Phoenix, Arizona, USA. pp. 2058–2065. doi:10.1609/aaai.v30i1.10306.
- 2590 [38] Tuia, D., Persello, C., Bruzzone, L., 2016. Domain adaptation for the classification of remote sensing data: An overview of recent advances. *IEEE Geoscience and Remote Sensing Magazine* 4, 41–57. doi:10.1109/MGRS.2016.2548504.
- 2595 [39] Valavi, R., Elith, J., Lahoz-Monfort, J.J., Guillerá-Arroita, G., 2019. blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution* 10, 225–232. doi:10.1111/2041-210X.13107.
- 2600 [40] VanDerWal, J., Shoo, L.P., Graham, C., Williams, S.E., 2009. Selecting pseudo-absence data for presence-only distribution modeling: How far should you stray from what you know? *Ecological Modelling* 220, 589–594. doi:10.1016/j.ecolmodel.2008.11.010.
- [41] Vesik, P.A., Morris, W.K., Neal, W.C., Mokany, K., Pollock, L.J., 2021.

- 2605      Transferability of trait-based species distribution models. *Ecography* 44,  
134–147. doi:10.1111/ecog.05179.
- [42] Vibhandik, P., Sawant, S., Joshi, A., Bidwe, R., 2026. A systematic  
literature review on forest fire susceptibility mapping using geo-spatial  
technology and future research directions. *Discover Artificial Intelligence*  
2610      6, 921. doi:10.1007/s44163-026-00921-0.
- [43] Wadoux, A.M.J.C., Heuvelink, G.B.M., de Bruin, S., Brus, D.J., 2021.  
Spatial cross-validation is not the right way to evaluate map accuracy.  
*Ecological Modelling* 457, 109692. doi:10.1016/j.ecolmodel.2021.  
109692.
- 2615      [44] Warren, D.L., Glor, R.E., Turelli, M., 2008. Environmental niche equiv-  
alency versus conservatism: Quantitative approaches to niche evolution.  
*Evolution* 62, 2868–2883. doi:10.1111/j.1558-5646.2008.00482.x.
- [45] Xu, Y., Dai, Y., Chang, L., Wang, Q., Dong, Y., 2026. Does  
your wildfire prediction model actually work, or just score well?  
2620      arXiv preprint arXiv:2605.18911. doi:10.48550/arXiv.2605.18911,  
arXiv:2605.18911.
- [46] Yates, K.L., Bouchet, P.J., Caley, M.J., Mengersen, K., Randin, C.F.,  
Parnell, S., et al., 2018. Outstanding challenges in the transferability of  
ecological models. *Trends in Ecology & Evolution* 33, 790–802. doi:10.  
2625      1016/j.tree.2018.08.001.
- [47] Yebra, M., Dennison, P.E., Chuvieco, E., Riaño, D., Zylstra, P., Hunt,  
E.R., Danson, F.M., Qi, Y., Jurdao, S., 2013. A global review of remote

- sensing of live fuel moisture content for fire danger assessment: Moving towards operational products. *Remote Sensing of Environment* 136, 455–468. doi:10.1016/j.rse.2013.05.029.
- [48] Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W., Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz, S., Lesiv, M., Herold, M., Tsendbazar, N.E., Xu, P., Ramoino, F., Arino, O., 2022. ESA WorldCover 10 m 2021 v200. Zenodo. doi:10.5281/zenodo.7254221.